Protein Interaction as an Operator

From Potts Models to Transformers

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Preface

Protein science, statistical physics, graphical modeling, and modern machine learning repeatedly use the same words for different mathematical objects. A coupling may mean a term in a joint Hamiltonian, a conditional dependence parameter, a message map, or an experimentally measured epistatic effect. A graph may encode model factorization, computational communication, an operator-valued response, or physical contact. Softmax may normalize configurations, residue identities, source positions, memories, or latent edge states. The formulas can look nearly identical while their scientific claims remain incompatible.

This book develops a typed mathematical language for navigating those translations. Its central object is a reference-centered operator-valued graph: every sequence position carries a categorical contrast space, and every directed edge carries a map between such spaces. This construction places Potts couplings, protein-language-model responses, attention routes, latent memories, spectral summaries, and biological interventions in common coordinates without declaring them equivalent.

The purpose is not to attach physical language to every neural computation. It is to identify exactly what additional assumptions turn a score into an energy coordinate, a collection of conditionals into a joint law, a route into an interaction, a scalar network into an operator graph, and a predictive association into biological evidence. Where those assumptions fail, the book treats the failure as information rather than as a defect to conceal.

The material grew from a theory paper and retains theorem–proof precision, but its natural scope is broader. The translation dictionary, gauge geometry, static–dynamic decomposition, higher-order hierarchy, graph probability, model correspondences, spectral analysis, realizability constraints, and falsification program are parts of one framework. Presenting them as a book allows each layer to keep its proper meaning instead of forcing every distinction into the narrow arc of a single research claim.

How to read this book

Part I establishes the cross-disciplinary vocabulary, the hierarchy of claims, and the scales that connect sequence, structure, function, evolution, and observation. Part II is the theorem–proof core: it builds the operator geometry and develops static, dynamic, higher-order, and probability-exact edge objects. Part III first compares Potts models, Gaussian models, Hopfield systems, RBMs, and Transformers in common coordinates. It then separates phylogeny from the residue-position graph, declares the observations and estimands, audits the three generations of AlphaFold, and only then introduces graph spectra. The AlphaFold comparison is an architectural reading of published systems; its claims concern retained evidence, relational state, and output law. Part IV separates architectural capacity from conditional information limits, then closes with falsification and biological interpretation. The detailed MSA Pairformer, graph-network, and diffusion derivations remain inside their protein-facing chapters. The optional appendix isolates the population-selection bridge.

Readers interested primarily in protein language models can begin with Chapters 1, 4, 7, 9, and 10. Readers coming from statistical physics may prefer Chapters 2–6, 8, and 11. Readers designing sparse or interpretable architectures should read Chapters 3, 4, 12, 13, and 14 before treating an internal route as an identified interaction.

The front-matter object map gives a lighter first route through the notation. On a first reading, the opening prose, boxed equations, worked cases, and codas carry the conceptual argument. Readers following the protein-machine-learning route can skip the optional population appendix. The corresponding chapters preserve the exact operator-limit and stochastic-transport derivations.

Objects and Notation

The notation in this book is easiest to read as a sequence of typed transformations. Data do not begin as an interaction graph, and a spectrum is not computed until an edge object has been declared. The main path is

observations $\longrightarrow$ estimator $\longrightarrow$ typed edge operator $\longrightarrow$ declared compression $\longrightarrow$ spectrum or experiment.

Every arrow can lose information. The book repeatedly asks what survives each arrow and what new assumption would be required to reverse its interpretation.

The recurring objects

Symbol	Type	Question answered
$X = (X_1, \dots, X_L)$	sequence in $[q]^L$	Which residue identity occurs at each position?
$X^{(n)} \in [q]^L$	alignment row n of depth N	Which sampled homologues, with what weights, enter an estimate?
p_i	reference distribution on the alphabet at site i	Relative to which background are categorical contrasts centered?
$\mathcal{H}_i^0$	$(q-1)$-dimensional centered categorical fiber	Which source and target perturbations count as genuine contrasts?
P_i	projection onto $\mathcal{H}_i^0$	Which constant and one-body directions are removed from a table?
$G_{i\leftarrow j}$	reference-centered $q \times q$ kernel	Which source-category contrasts are coupled to which target-category contrasts?
$\mathcal{K}_{i\leftarrow j}$	map $\mathcal{H}_j^0 \to \mathcal{H}_i^0$ induced by $G_{i\leftarrow j}$	How does a centered contrast at j act on centered predictions at i?
$\mathbb{K}$	block operator on $\bigoplus_i \mathcal{H}_i^0$	How are all local maps assembled without scalarizing them?
$G_{i\leftarrow j}(C)$	categorical kernel conditional on background C	Does the effective relation change with the surrounding sequence?
$\overline{G}, \delta G$	mean operator and context-dependent residual	What is the best static approximation, and what does it leave unexplained?
Q_{ij}	pair probability law with declared marginals	Does a finite normalized pair law realize the local score geometry?
A, L_G	scalar shadow and a declared graph Laplacian	Which positional organization remains after categorical information is compressed?
ϵ_{ij}	projected experimental epistasis table	Does a matched intervention support the predicted categorical relation?

The same letter can appear with a superscript identifying its origin, such as G^{P} for a Potts

estimate or G^{pLM} for a language-model response. An arrow $j \to i$ uses j as source and i as target. Static pair energies may eventually identify the two directions, but directed responses are kept separate until reciprocity has been tested.

The distinction between G and $\mathcal{K}$ is representational, not a second scientific claim. In the declared categorical bases, $G_{i \leftarrow j}$ is the matrix kernel of $\mathcal{K}_{i \leftarrow j}$. The calligraphic symbol emphasizes the typed input and output; the plain capital emphasizes the entries that are estimated, projected, plotted, or compressed.

How the notation carries type

The typography follows a small grammar. It does not make every symbol globally unique, but it narrows the possibilities before an equation is read in detail.

Form	Usual role	Examples
Lower-case italic	indices, states, scalar functions, or vectors	$i, j, a, b, t;\ f, u, v$
Plain capital	finite table, matrix, random object, or model state	T, J, G, A, P, X
Calligraphic capital	space, set, typed map, or functional	$\mathcal{H}, \mathcal{U}, \mathcal{K}, \mathcal{E}$
Script capital	latent architecture space or propagator	$\mathscr{Z}, \mathscr{S}$
Blackboard/fraktur capital	assembled system or expectation operator	$\mathbb{K}, \mathbb{J}, \mathfrak{G}, \mathfrak{E}, \mathfrak{R}, \mathbb{E}$
Arrow and indices	source, target, and level of action	$G_{i \leftarrow j}$ maps source j toward target i

Decorations have stable meanings throughout the main construction:

$$\widehat{G}:\ \text{sample estimate}, \qquad \overline{G}:\ \text{declared background average},$$
$$\delta G:\ G - \overline{G}, \qquad (\cdot)^{*}:\ \text{population target or adjoint, as stated locally.}$$

Superscripts such as P, pLM, stat, and dyn name origin or role; they do not change the domain and codomain unless the text explicitly says so. Conditional expectations over one coordinate or over the context are written $\mathbb{E}_u$ and $\mathbb{E}_C$. The encoder–transport–readout factors assembled in the architecture chapters are $\mathfrak{E}_C$, $\mathbb{J}_C^{\mathrm{lat}}$, and $\mathfrak{R}_C$; they are transport operators, not expectations.

Several overloaded letters remain because they are standard in the literatures being compared. Their syntax disambiguates them. L is sequence length in $[q]^L$ and an index bound, whereas L_G or a subscripted $L_{\mathrm{sym}}, L_{\mathrm{rw}}$ is a graph Laplacian and L^2 is a function space. N is alignment depth with row index n; in the interaction-hierarchy theorem a bare N is the neighbor set of a site and $\mathcal{N}(i)$ is a graph neighborhood, while the architecture chapters let a locally declared N count layers, blocks, or recurrent steps. P_i is categorical centering, bare P in the routing chapters is a row-stochastic matrix, $\mathbb{P} = \bigoplus_i P_i$ is the assembled projection, and $P(\cdot)$ denotes a probability. Q_{ij} is a pair law, while Q_i is the Euclidean form of a categorical projector; nuisance residualizers are written R rather than adding another meaning to Q. Attention follows the query–key convention: uppercase $Q_i^{(h)}, K_j^{(h)}$ in the audit formalism and lowercase q, k, v inside a quoted architecture, with the head superscript and the argument keeping them separate from Q_{ij} and $\mathcal{K}$. Bare G can denote a generic graph in a quoted graph-network formalism, while an indexed $G_{i \leftarrow j}$ is always a categorical kernel in this book. Bare A is the scalar shadow; in the function-space hierarchy theorem A and B are

position subsets with projections Π_A, Π_B, and the ANOVA primer reuses A, B as scalar coefficients. The letter h is a Potts field $h_i(a)$ and, with a head superscript, an attention-head index; ψ_{ij} is a cavity factor, while an unindexed ψ is a spectral generating function whose scale parameter is s. In the MSA Pairformer case study, q indexes the query row; the alphabet size does not appear there. An unsubscripted ϵ in the diffusion appendix is Gaussian noise, while ϵ_{ij} is always the projected epistasis table.

Three spaces that should not be merged

The book moves among three geometries:

Geometry	Native object	Typical operation
Observation space	MSA rows, positions, and residue categories	weighting, residualization, covariance estimation
Categorical operator space	centered contrasts and maps between local fibers	gauge projection, reference transport, response audit
Position-graph space	one scalar or transported vector state per residue position	Laplacian spectrum, heat filtering, wavelets

An eigenvector across MSA rows, a categorical constant mode, and an eigenvector across residue positions are therefore different objects even when all three are called modes.

A first and a second reading

The main argument can be followed from the opening prose of each chapter, the boxed equations, the worked cases, and the codas. The theorem and proof blocks establish exactness but need not be read on the first pass. Long derivations labeled as mathematical interludes answer a narrower question and can likewise be deferred without losing the book's conceptual sequence.

On a second reading, each proposed correspondence can be checked with the same ledger:

$$\text{sample space}, \quad \text{normalization axis}, \quad \text{available context},$$
$$\text{gauge or symmetry}, \quad \text{state dynamics}, \quad \text{estimator}, \quad \text{validation endpoint}.$$

These are not seven hurdles that every useful model must clear. They are seven coordinates that prevent a valid statement in one space from being silently promoted into a stronger statement in another.

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Part I

Language and Foundations

Chapter 1

One Edge, Several Meanings

Consider two positions, i and j, in a protein family. A structural biologist may draw a line between them because their atoms lie within a chosen distance. A statistical physicist may fit a Potts model and attach a $q \times q$ coupling table to the same pair. A Transformer may send a large attention weight from j to i. A model audit may find that substituting the residue at j changes the predicted distribution at i. A double-mutant experiment may report a nonadditive assay endpoint. Each result is naturally drawn as an edge. The pictures look alike, and that visual agreement invites a stronger conclusion: perhaps all five procedures have found the same interaction.

They have not—at least not yet. The structural line is defined by geometry. The Potts table is a component of a probability model for an ensemble of homologous sequences. Attention is a route inside a computation. The substitution response is a directed input–output property of a fitted function. Experimental epistasis is a statement about an intervention and a measured biological endpoint. Agreement among these objects can be scientifically important. Shared indices (i, j) provide the comparison point; the carried objects and validation criteria determine the meaning.

This book begins with that apparently minor distinction because much of the conceptual difficulty in protein machine learning is concentrated inside it. Words such as *interaction*, *energy*, *memory*, *direct*, and *graph* travel easily between statistical physics and neural computation. The equations often travel with them, while their assumptions must be carried explicitly.

1.1 A residue pair enters five descriptions

Let $X = (X_1, \ldots, X_L)$ denote a protein sequence over an alphabet of size q. Suppose that a multiple-sequence alignment, a trained protein language model, a three-dimensional structure, and a mutational assay are all available for one family. For the pair (i, j) they provide five candidate edge objects:

The entries differ along several axes at once. Some are scalar and some are categorical tables. Some are symmetric and some directed. Some are fixed across a family and some depend on the input sequence. Some belong to a joint probability law; others belong to an algorithm or an assay. Even their zeros mean different things. A zero Potts block may express conditional factorization within a fitted model, a zero route may mean skipped computation, and a missing structural contact may only mean that one distance threshold was not crossed.

The first principle of the book is therefore simple:

$$\boxed{\text{shared endpoints do not imply a shared edge semantics.}} \tag{1.1}$$

This principle sets the standard for unification: a successful comparison must preserve the relevant

Description	Edge object	Question answered
Structure	contact indicator or geometric relation	Are the two sites close under a declared structural criterion?
Potts model	coupling table $J_{ij}(a, b)$	Which pair term helps a fitted family distribution explain sequence variation?
Attention	route coefficient $\alpha_{ij}^{(h,\ell)}(x)$	How much source-normalized weight does one head assign to this computational route?
Model response	categorical map $G_{i\leftarrow j}(C)$	How does a substitution at j change the model prediction at i in background C?
Experiment	epistasis or another intervention response	Does the joint perturbation differ from the declared additive biological null?

Figure 1.1: The same endpoints support five different edge semantics. Agreement must be established after the carried object and validation question have been declared.

types. If two edge objects are to be compared, their state spaces, normalization, conditioning information, invariances, dynamics, and validation criteria must remain visible during the comparison.

The same caution applies to familiar formulas and familiar adjectives. Softmax can normalize residue identities, source positions, memories, or complete configurations. The normalization axis determines which probability law the shared algebra defines. Likewise, an edge may be called *direct* because it is a one-step computational route, a term in a declared factorization, a spatial contact, or an experimental effect that survives a lower-order null. Chapter 2 separates these meanings systematically. Here it is enough to retain one reading rule: before interpreting a formula or an edge, name the space in which it lives and the alternative explanations that have actually been excluded.

1.2 From an edge weight to an edge operator

For categorical sequences, a scalar edge weight is a compression of a richer object. If residue b at site j is replaced by b', the effect at site i is a change across all possible target identities a. A pair relation is therefore naturally represented by a table or operator

$$G_{i\leftarrow j}(a, b), \tag{1.2}$$

mapping source substitutions to target contrasts.

This operator contains information that every scalar compression discards. Two edges can have the same Frobenius norm while favoring different amino-acid exchanges. They can have the same support while acting in different singular directions. They can generate the same contact score but make opposite predictions for a particular mutation. A graph of scalar strengths is useful, but it is a shadow of the typed relation carried by the edge.

Raw categorical tables introduce another problem. A constant term, a function of a alone, and a function of b alone can be moved between one-body and pair terms without changing the modeled law. The pair object is therefore an equivalence class until a reference geometry is declared. The reference-centered operator developed later in the book removes those lower-order directions and retains the genuinely bivariate action. Gauge fixing supplies the identified coordinates that make edge operators comparable across models and references.

1.3 One family graph, many sequence responses

The contrast between a Potts model and a protein language model supplies the running mathematical problem. In an additive pairwise Potts model, a fixed gauged block describes the response of site i to site j throughout the family model. A nonlinear protein language model generally produces a different effective response in each sequence background C:

$$G_{i\leftarrow j}(C) = \overline{G}_{i\leftarrow j} + \delta G_{i\leftarrow j}(C). \tag{1.3}$$

The mean $\overline{G}_{i\leftarrow j}$ is the best static approximation in a declared reference geometry. The residual $\delta G_{i\leftarrow j}(C)$ measures context dependence. Under a fixed product reference, its variance can be identified with higher-order functional components that contain the audited source and take values in the target contrast fiber. The target category is an output coordinate, not an assumed input to the masked model. Under correlated empirical backgrounds, additional cross terms appear and the interpretation must change.

This decomposition separates two questions that are often merged. Does the model contain a stable family-like pair response? How much of its behavior depends on the surrounding sequence? A model can answer both strongly, one strongly, or neither. Calling the entire response a "learned Potts coupling" erases the distinction.

Directionality adds another layer. The response from j to i and the response from i to j are separately measurable. They define a joint pair energy only if they satisfy a reciprocity condition after reference transport. A directed response graph is therefore a valid primary object. Symmetrization becomes a later operation, justified after its asymmetry has been measured.

1.4 The coordinate system to come

The rest of the book builds a mathematical coordinate system in which these comparisons can be made without flattening them. Each model will be described by the variables carried at its nodes, the support and type of its edges, the context available to an edge, the represented interaction order, the updated state, the normalization or conserved quantity, the estimation procedure, and the external endpoint used for validation.

Figure 1.2 places the main spaces on one page. A sequence family supports two parallel constructions: a declared probability law, reached through a gauge choice and an estimator, and a trained model with its internal routes and memories. The two constructions meet only after a reference-centered audit places both edge objects in common typed coordinates. Every arrow in the map adds an assumption, and the downward arrows out of the operator graph either compress type into a scalar shadow or confront the prediction with an intervention.

The construction proceeds from local meaning to global claims. We first establish a translation language across physics, probability, and computation. We then define reference-centered categorical fibers and the operators between them. Static and dynamic responses lead to higher-order decompositions, probability-exact edge geometry, and compatibility conditions for joint energies. Model correspondences, phylogenetic transport, observation ledgers, AlphaFold, and spectra then place those objects in successively broader settings. The final part treats architectural capacity, conditional information limits, and biological evidence.

The ambition is to make differences between protein graphs mathematically explicit enough that genuine agreement becomes meaningful. The edge between i and j begins a sequence of questions: an edge in which space, carrying what object, conditioned on what information, invariant under which changes, and validated against which world? Section 14.2 returns to this pair and carries all five candidate edges through the complete audit.

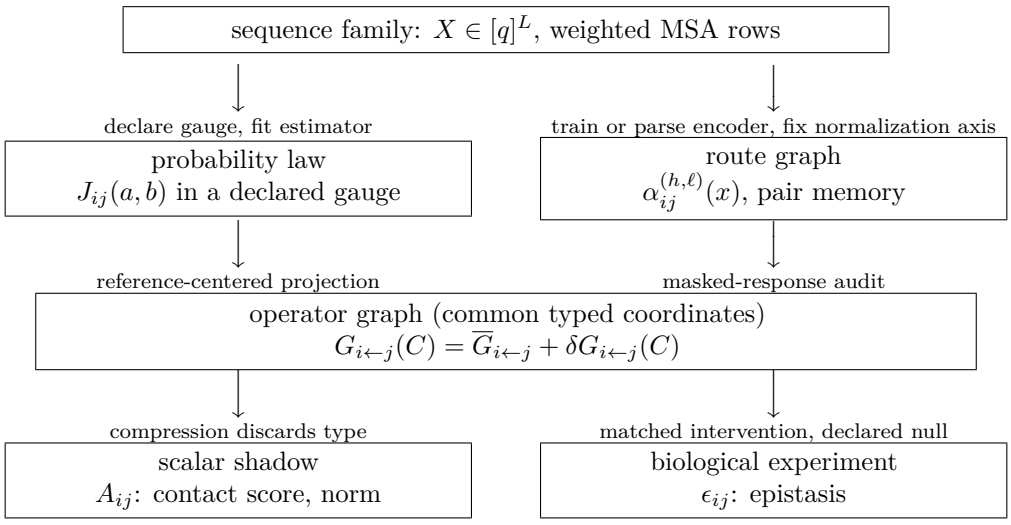


Figure 1.2: The spaces of this book on one page. Shared indices (i, j) appear in every box; the carried object differs. Only the operator-graph level holds both the fitted law and the audited model in common typed coordinates, and every arrow is a declared transformation whose lost or added assumptions the corresponding chapter makes explicit.

Chapter 2

When Equations Change Their Meaning

Mathematical notation creates a powerful illusion of continuity. Write an exponential, divide by a sum, and a probability simplex appears. Write a quadratic form and one is tempted to call it an energy. Place coefficients between indexed objects and a graph seems to have been defined. These moves are algebraically legitimate. The difficulty begins when the semantic content of the symbols is allowed to travel farther than the derivation.

Cross-disciplinary work depends on such travel. Statistical physics, graphical modeling, and machine learning would remain isolated if every familiar operation had to be renamed in each field. The aim is to preserve enough type information for analogies and equivalence claims to remain distinguishable.

This chapter develops that discipline. The central rule is that an equation is interpreted only after its random variable, sample space, conditioning information, invariance class, and validation claim have been named. The same numerical kernel can then connect several theories without collapsing them into one.

2.1 The sample space comes before the probability

A probability vector is incomplete until we know what its coordinates enumerate. Consider

$$\mathrm{softmax}_u(s)_u = \frac{e^{s_u}}{\sum_v e^{s_v}}. \tag{2.1}$$

The expression guarantees positivity and unit mass. The meaning of that mass comes from whether u denotes a residue identity, a source position, a stored memory, an edge type, or an entire protein configuration. Each choice produces a different random object.

For a Potts conditional, $u = a \in [q]$ and the simplex describes the possible identities of one residue given the remaining sequence. For attention, $u = j \in [L]$ and the simplex distributes one unit of routing mass over source positions for a fixed target and head. In a modern Hopfield update, $u = \mu \in [M]$ indexes memories that compete for retrieval. In a latent relation model, $u = z$ may index the possible states of one candidate edge. A globally normalized Gibbs law is different again: its index ranges over complete configurations, often an exponentially large set.

These distributions can share software and calculus while retaining different normalization constraints. Source-softmax makes positions compete. Edge-state softmax resolves uncertainty about one relation and acquires a degree budget only when edges are coupled. A site conditional is

normalized for each context; compatibility with one joint distribution is a further property to be established.

The normalization axis is consequently part of the mathematical type. Omitting it is comparable to omitting the units from a physical measurement: the remaining number can still be manipulated, but its comparisons become unsafe.

2.2 The exact content of the Gibbs analogy

Once a finite state space $\mathcal{Z}$ has been declared, softmax possesses an exact variational meaning. Let r_z be a score and let $E_z = -r_z$. At temperature $\tau > 0$,

$$p_z = \frac{e^{-E_z/\tau}}{\sum_{z'} e^{-E_{z'}/\tau}}. \tag{2.2}$$

The associated free energy satisfies

$$-\tau \log \sum_z e^{-E_z/\tau} = \min_{q \in \Delta(\mathcal{Z})} \left\{ \sum_z q_z E_z - \tau H(q) \right\}, \qquad H(q) = -\sum_z q_z \log q_z. \tag{2.3}$$

This is more than a metaphor. The distribution exactly balances expected energy against entropy. As τ decreases it concentrates on score maxima; as τ increases it spreads mass across alternatives. Adding the same constant to every E_z changes the free energy by that constant but leaves p unchanged, so only energy differences are observable within this coordinate.

The identity gives the exact geometry of a declared categorical choice. A physical reading adds further structure: calibrated energy units, a thermodynamic temperature, an equilibrium ensemble, and compatibility with a global Hamiltonian. Learned logits and routing temperatures supply these features only when the corresponding model and calibration establish them.

This distinction is especially sharp for graphs. Suppose each candidate edge e has a latent state z_e and an independent softmax posterior $q_e(z_e \mid x)$. The resulting law is

$$Q(G \mid x) = \prod_{e \in E_c} q_e(z_e \mid x). \tag{2.4}$$

Every factor has a local Gibbs representation. The product is a factorized graph distribution; shared degree budgets, reciprocal constraints, loop energies, connectivity, geometric exclusion, and chemical consistency require additional coupled factors or a restricted graph space.

A coupled graph ensemble instead has one normalization over admissible graphs:

$$P(G \mid x) = \frac{\exp[-\mathcal{H}_{\text{graph}}(G; x)/\tau]}{\sum_{G'} \exp[-\mathcal{H}_{\text{graph}}(G'; x)/\tau]}. \tag{2.5}$$

The two laws coincide only when the graph energy separates over edges and the admissible set adds no global coupling. Thus “each edge is Gibbs distributed” and “the graph is a Gibbs system” are different statements.

2.3 Four uses of interaction

The word *interaction* can now be separated by the object whose nonadditivity is at issue.

Computational interaction. Changing one node can alter another through a route, message, attention coefficient, residual path, or nonzero network response. This is a property of an implemented function and its execution graph.

Statistical interaction. A joint or conditional function contains a component outside the span of the declared lower-order terms. The statement depends on a reference measure, a projection or factorization, and an interaction order.

Physical interaction. A term belongs to a justified Hamiltonian or effective energy under an explicit coarse graining, symmetry class, and regime of validity. Statistical fit alone does not supply those assumptions.

Biological interaction. A joint perturbation produces a reproducible effect on structure, a declared assay endpoint, or a population growth coordinate relative to a declared null. This is an experimental statement and inherits the scope of the assay and demographic model.

The four meanings can support one another without forming a chain of automatic implications. A computational route can help predict a statistical coupling. A fitted coupling can correlate with contact geometry. A contact can constrain a mutational assay or population growth rate. Each transition is an empirical result with its own failure modes.

2.4 Four graphs on the same protein

The corresponding graphs must also be typed.

The *route graph* records where a computation can read or send a message. In ordinary self-attention its support is usually the complete directed graph; what varies is the weight and the effective value-path operator. The *statistical support graph* records nonzero pair components in a declared probability or conditional model. The *operator graph* records the full categorical action attached to each directed pair. The *physical or biological graph* records relations defined by structural, energetic, kinetic, or intervention criteria.

These distinctions matter even when all graphs predict the same contact benchmark. Contact prediction compresses a rich pair object to a scalar label. Agreement at that level leaves substitution specificity, reciprocity, background dependence, and mediation as separate comparison questions. Agreement after scalar compression is evidence, but it is evidence about the compressed observable.

Topology adds information that no isolated edge contains. Support determines which effects are represented as direct and which require paths. Degree, components, cycles, bottlenecks, and spectral modes describe collective organization. Yet topology alone remains incomplete for categorical systems. Two graphs with identical support can carry different edge operators and make different mutational predictions. This is why the position graph will later appear as a shadow of a richer operator-valued object.

Static and dynamic graphs occupy different roles as well. A family-level interaction graph held fixed while homologous sequences are modeled resembles a quenched structure. A route graph that changes with sequence, layer, and head is an adaptive computational state. It is sometimes useful to compare the latter with an annealed graph, but input dependence alone does not produce an equilibrium measure over graph configurations.

2.5 A claim is a path with missing arrows

Cross-disciplinary overstatement often takes the form of a skipped arrow. A route is observed and called an interaction; a conditional score is observed and called a Hamiltonian; a contact correlation is observed and called a mechanism. A more durable argument makes the missing arrows explicit:

$$\begin{aligned}\text{projected bivariate component} &\xRightarrow[\text{fixed context}]{\text{endpoint exclusion}} \text{conditional response} \\ \xRightarrow[\text{one joint law}]{\text{compatibility}} \text{distributional interaction} &\xRightarrow[\text{matched endpoint}]{\text{independent intervention}} \text{biological interaction.}\end{aligned} \tag{2.6}$$

Each arrow names an additional assumption or experiment. The ladder preserves the value of every established level: a computational route can be useful before conditional semantics are known, and a statistical interaction can remain predictive while its microscopic interpretation is open.

This also changes how negative results are read. Failure of reciprocity may reject a joint Potts interpretation while revealing directed model behavior. Failure of background invariance may reject a static pair description while quantifying higher-order context. Failure to predict contacts may reject one structural interpretation without erasing an operator's role in sequence prediction.

2.6 The translation ledger

Before identifying two constructions, we will record the following coordinates:

$$\boxed{\begin{array}{c}\text{object and sample space,} \quad \text{normalization axis,} \quad \text{conditioning information,} \\ \text{gauge or invariance class,} \quad \text{state dynamics,} \quad \text{estimator and validation endpoint.}\end{array}} \tag{2.7}$$

The ledger is deliberately asymmetric across fields. Translating a Hamiltonian into a score may discard physical assumptions; translating a score into a Hamiltonian requires adding them. A hidden feature can function as a memory without being a metastable thermodynamic state. A sparse execution graph can save computation without identifying conditional independence. A gauge-fixed pair table can isolate statistical nonadditivity without establishing biological causality.

This asymmetry is the price of honest unification. The remaining sections first fix the elementary sequence and MSA objects, then broaden the translation ledger through four statistical-mechanics lenses and through the scales of protein science. Part II begins with the categorical edge itself: what remains of a pair table after constants and one-body effects have been removed, and how that remainder acts as an operator between reference-centered state spaces.

2.7 From one protein sequence to an alignment

A protein sequence records the amino-acid residues of a polymer chain in their order from one terminus to the other. A *residue* is one amino-acid unit in that chain; a *site* or *position* is the coordinate at which its identity is recorded. The *query* is the particular sequence whose structure, function, or model response is under study. A *homologue* is a sequence related to the query by common ancestry. Sequence similarity is evidence used to find homologues; homology itself is a historical relation.

A multiple-sequence alignment (MSA) places homologous sequences in rows and proposes corresponding ancestral positions in columns. The following toy alignment uses the query as its coordinate system:

row	1	2	3	4	5	6	7	8
query	M	K	T	A	E	L	V	G
homologue 1	M	K	S	A	E	L	I	G
homologue 2	M	R	T	A	D	L	V	G
homologue 3	M	R	S	–	D	L	I	G

Columns 1, 6, and 8 are conserved in this sample. Columns 2 and 5 covary: the observed rows pair K with E and R with D. The dash in column 4 is an alignment gap, representing an insertion, deletion, missing observation, or an alignment decision according to the data pipeline. It is not an amino acid. Row order in an alignment file has no biological meaning; relationships among rows are supplied by common ancestry, taxonomy, sampling, or an inferred tree.

The model alphabet must therefore be declared. The twenty standard amino acids may be augmented with gap, unknown, ambiguous, or masked symbols. In a query-anchored analysis, indices i and j usually denote retained query-aligned columns. They need not equal the raw residue numbers in every homologue, and poorly aligned or insertion-rich regions can make the correspondence uncertain.

Conservation is a one-column statement: the marginal distribution $p_i(a)$ is concentrated on a small set of residue identities. Covariation is a two-column statement: the joint distribution $p_{ij}(a, b)$ differs from $p_i(a)p_j(b)$. The toy alignment displays both, yet its four rows are far too few to establish either quantity reliably. More generally, observed covariation can contain several mechanisms at once:

Source	How the pattern arises	What must be checked
Direct structural or functional constraint	substitutions at two sites compensate to preserve folding, binding, catalysis, or another selected property	matched structure, function, or mutational evidence
Indirect dependence	i and j are both coupled to a third site k, or are joined by a longer path	conditioning, inverse modeling, and controlled synthetic cases
Shared ancestry	related rows inherit both residue states from common ancestors	a tree-based or clade-aware sampling null
Mixture and data construction	subfamilies, taxonomic imbalance, gaps, alignment choices, or paired-MSA assignment change the empirical joint table	stratification and pipeline-matched nulls

Finite depth and database redundancy add uncertainty to every row of this ledger. Sequence weighting can reduce the influence of nearly duplicated observations, while the ancestral process and alignment uncertainty remain part of the data-generating law. An MSA is therefore an observed three-axis object—row, aligned position, and residue category—from which a position-level interaction operator may be estimated. The estimator and its null determine which part of the observed covariation the resulting edge can claim to represent.

With these three axes fixed, the next step is to distinguish operations performed on the resulting many-variable system. A global energy law, a factor-graph cavity message, a dynamical response, and a cascade on a support graph act on different state spaces even when each is described with the word “interaction.” The four statistical-mechanics lenses that follow provide exact reference cases for those operations before the argument crosses from sequence data to physical and evolutionary scales.

2.8 Three solvable lenses on interaction

Three lines of statistical mechanics are especially useful for the distinctions developed in this book: Bernard Derrida's random-energy landscapes, Marc Mézard's cavity inference, and Haim Sompolinsky's dynamical response theory. They expose three different operations on a many-variable system.

Lens	Native state space	Native operation	Coordinate clarified in this book
Derrida	complete configurations	Gibbs sampling from a random energy table	interaction order and background-dependent response
Mézard	variables and factors of a joint law	deletion of one factor followed by message passing	coupling, marginal, message, and susceptibility
Sompolinsky	trajectories of a high-dimensional network	propagation of an infinitesimal perturbation	parameter graph, local Jacobian, and time-ordered response

The three cases also organize three common translation failures. A global energy law need not have a useful low-order graph. A message on an edge is derived from a probability model but is not its pair potential. A fixed parameter matrix can produce a response operator that changes with state and time. Each case will reappear where its native mathematics becomes useful: random energy in the interaction-order chapter, dynamical mean field in the static–dynamic chapter, and cavity equations in the probability chapter.

This placement follows the objects rather than the chronology. It turns a historical lineage from equilibrium statistical mechanics through nonequilibrium dynamics, optimization, neuroscience, and artificial intelligence into a sequence of exact audits. Protein-facing claims enter only after the corresponding state space, operator, estimator, and validation endpoint have been declared.

The three lenses distinguish operations on a many-variable system. Protein science adds a second issue: one claim may also cross sequence, conformational, functional, evolutionary, and observational scales. The next section makes those maps explicit before the operator formalism begins.

2.9 Protein interaction across scales

The five edges in Chapter 1 arise at different resolutions of the same protein system. A residue identity belongs to sequence space, an atomic contact belongs to a conformational description, a family coupling belongs to an evolutionary sample, a model response belongs to a fitted computation, and epistasis belongs to an intervention with a measured endpoint. Their shared residue indices make comparison possible. Moving between them requires a declared coarse graining or estimator.

Let $x \in [q]^L$ be a sequence and let r collect atomic coordinates, solvent, protonation, and any other physical variables retained by a molecular model. A microscopic energy first defines a conditional conformational partition function and free energy,

$$Z_{\mathrm{conf}}(x) = \int e^{-\beta E_{\mathrm{atom}}(x,r)}\, d\mu_x(r), \qquad F_{\mathrm{conf}}(x) = -\beta^{-1} \log Z_{\mathrm{conf}}(x). \tag{2.8}$$

The integral removes physical degrees of freedom. Its value depends on the ensemble, measure, temperature, chemical state, and environmental conditions encoded by (μ_x, β). A single predicted

structure is one representative coordinate configuration; it does not by itself specify the conformational integral in Eq. 2.8.

Function first enters through a typed measurement. Let e index an assay protocol, let c collect the environment and sequence background held fixed by that protocol, and let $\mathsf{P}_{x,c}$ denote the molecular or cellular state law presented to the measurement. The experimental endpoint is

$$Y_e(x, c) = \mathcal{A}_e(\mathsf{P}_{x,c}, c) \in \mathcal{Y}_e. \tag{2.9}$$

The codomain $\mathcal{Y}_e$ may contain a binding constant, catalytic rate, abundance, fluorescence, growth-curve statistic, or a vector of jointly measured quantities. Assay identity, units, time window, and preprocessing are part of the definition. Structure can mediate Y_e through $\mathsf{P}_{x,c}$; expression, localization, kinetics, partners, and cellular context can change the endpoint while a representative folded geometry remains similar.

Population selection requires a further genotype-to-demography map. The map may use measured traits, latent ecological variables, or a full population state; its output has its own units and dynamics. The main text leaves that map unspecified and uses P_{evo} for the resulting evolutionary law. The Malthusian-rate construction is given in the optional population-selection appendix.

Evolution acts on populations and histories rather than on one isolated structure. A mutation process, population model, and genealogy together induce a law on the observed leaves of an alignment,

$$(E_{\mathrm{atom}}, \mu, \{\mathcal{A}_e\}, \mathrm{population}, T) \longmapsto P_{\mathrm{evo}}(X_{\mathcal{L}}), \qquad X_{\mathcal{L}} \in [q]^{N \times L}. \tag{2.10}$$

Chapter 8 develops the tree component of this construction. Selection, drift, mutation, recombination, sampling, and shared ancestry can all shape the alignment. A sequence-level marginal $P_{\mathrm{evo}}^{\mathrm{seq}}$ consequently combines several mechanisms. Its negative-log coordinate

$$H_{\mathrm{seq}}(x) = -\log P_{\mathrm{evo}}^{\mathrm{seq}}(x) \tag{2.11}$$

is a statistical sequence score. Connecting it quantitatively to $F_{\mathrm{conf}}(x)$ requires the intervening assay, demographic, and evolutionary maps to be specified.

Population selection is a downstream model choice. The assay endpoint Y_e and the induced sequence law P_{evo} can be used without assigning either one a universal fitness coordinate. A specific mutation–selection operator, its equilibrium eigenproblem, and reversible Gibbs-like special cases are collected in Appendix A. The main text only carries the boundary condition: interpreting an assay value or a model score as population fitness requires an explicit genotype-to-demography map and its units.

The complete path used by this book is therefore

$$\boxed{\begin{array}{c}\text{atomic law} \longrightarrow \text{conformational ensemble} \longrightarrow \text{structure, kinetics, and function} \\ \longrightarrow \text{population and phylogenetic law} \longrightarrow \text{observed MSA} \longrightarrow \text{estimated sequence operator} \\ \longrightarrow \text{declared compression or spectrum} \longrightarrow \text{matched experiment.}\end{array}} \tag{2.12}$$

Each arrow changes the state space or the available information. A structural contact can explain part of an evolutionary constraint; a distal functional relation can also survive without a short geometric edge. The operator framework begins near the middle of this chain, where sequence data or a trained model are converted into a typed categorical relation. Physical and functional claims are recovered only by supplying the missing upstream derivation or downstream endpoint.

The word *entropy* also changes type along this path. Four recurring quantities are:

Quantity	Distribution or state space	Question answered
Conformational entropy	molecular configurations conditional on sequence and environment	how physical probability mass is distributed across conformations
Sequence entropy	$P_{\text{evo}}^{\text{seq}}$ on $[q]^L$	how dispersed the induced sequence ensemble is
Phylogenetic path or leaf uncertainty	ancestral histories or $X_{\mathcal{L}}$ on a tree	how much evolutionary history or leaf state remains uncertain
Predictive entropy	one model output simplex under a declared context	how uncertain the fitted predictor is about its normalized variable

These quantities may influence one another through Eq. 2.12. Their numerical equality would require an additional theorem tying the corresponding probability laws together.

There is likewise no universal monotonicity law for entropy along evolution. Mutation can broaden a sequence distribution, while purifying selection can concentrate it; drift, bottlenecks, recombination, migration, environmental change, and biased sampling alter the result in different directions. A claim of entropy production requires a declared trajectory law, time direction, boundary conditions, and measure of irreversibility. The entropy of an observed MSA is a property of its induced and sampled distribution, not a clock that must increase along every lineage.

A cross-scale claim can now be read through six proof obligations:

$$\boxed{\begin{array}{c}\text{state space} \to \text{law} \to \text{observable} \\ \to \text{reference or small parameter} \to \text{controlled remainder} \to \text{validated effective claim.}\end{array}} \tag{2.13}$$

Later chapters use this ledger for phylogenetic nulls, sequence Hamiltonians, spectral modes, and biological interventions. In each case the effective object is licensed only to the level reached by its declared reference, controlled approximation, and matched validation endpoint.

Part II begins at the estimation end of this chain: with one categorical pair table and the contrasts it can identify.

Part II

Operator Geometry of Sequence Interactions

Chapter 3

Reference-Centered Operator-Valued Graphs

A pair table appears to be one of the simplest objects in sequence modeling. Choose two positions, list the possible residue identities along the rows and columns, and attach a number to every combination. The result is finite, visible, and easy to store. Yet before it can represent an interaction, a basic ambiguity must be resolved: which part of the table belongs specifically to the pair?

Suppose a table $T(a, b)$ is changed by

$$T(a, b) \longmapsto T(a, b) + u(a) + v(b) + c. \tag{3.1}$$

In a conditional model, the added row term may be absorbed into a local field at the target; in a joint energy, row and column terms can be redistributed among one-body potentials. Different raw tables can therefore encode the same observable distribution or the same categorical response. Taking the matrix at face value confuses a parameterization with an identified pair object.

The remedy is often called gauge fixing, but the phrase can make the operation sound like a convention chosen for numerical convenience. The construction in this chapter is stronger. A reference distribution at each endpoint defines which directions count as constants and one-body effects. Double projection removes exactly those directions. What remains is the canonical bivariate component relative to that reference.

The geometric interpretation is the key. Position i carries a $(q-1)$-dimensional space of centered categorical contrasts behind its q labels. A pair interaction is a map from the source contrast space to the target contrast space. The resulting protein graph is a bundle of local categorical spaces connected by typed linear operators; scalar edge weights are compressed shadows of these maps.

This shift resolves several problems at once. It explains why raw Potts tables are not directly comparable across gauges, why reference changes require a transport law, why scalar norms lose substitution direction, and why directed model responses can be compared with static couplings without pretending that their original parameters are the same. The formal definitions below make these claims exact.

A three-state table, taken apart

Begin with the smallest example that retains more than one categorical contrast. Let both endpoints have three states and use uniform references

$$p_i = p_j = \frac{1}{3}(1, 1, 1)^{\mathsf{T}}. \tag{3.2}$$

Write $D_i = D_j = \frac{1}{3}I$ and $P_i = P_j = I - \frac{1}{3}\mathbf{1}\mathbf{1}^\mathsf{T}$. The projector P subtracts the mean of a three-component categorical score. Applying it on both sides of a table removes the constant, row-only, and column-only contributions. Consider the raw pair table

$$T = \begin{pmatrix} 10 & 5 & 3 \\ 5 & 5 & 2 \\ 6 & 5 & 4 \end{pmatrix}. \tag{3.3}$$

Its grand mean is $c = 5$. Its row means are $(6, 4, 5)$, so the centered target-only term is

$$u = (1, -1, 0)^\mathsf{T}. \tag{3.4}$$

Its column means are $(7, 5, 3)$, giving the centered source-only term

$$v = (2, 0, -2)^\mathsf{T}. \tag{3.5}$$

Subtracting these three lower-order pieces leaves

$$G = T - c\mathbf{1}\mathbf{1}^\mathsf{T} - u\mathbf{1}^\mathsf{T} - \mathbf{1}v^\mathsf{T} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{3.6}$$

Every row and column of G sums to zero. Equivalently,

$$p_i^\mathsf{T} G = 0, \qquad G p_j = 0. \tag{3.7}$$

The large entry $T_{11} = 10$ is therefore not itself the interaction. It contains the grand baseline, a target propensity, a source propensity, and the bivariate residual. In this example those contributions are $5 + 1 + 2 + 2$.

Now alter the raw table by adding any new row and column effects,

$$T'(a, b) = T(a, b) + \widetilde{u}(a) + \widetilde{v}(b) + \widetilde{c}. \tag{3.8}$$

The numerical entries and their ordinary Frobenius norm can change substantially, but double projection returns the same G. The identified interaction is therefore the quotient class of T, while Eq. 3.6 is its representative in the uniform reference.

The edge becomes an operator when it acts on a source contrast. For this example define $\mathcal{K}_{i \leftarrow j} f = G D_j f$. Take

$$f = (1, -1, 0)^\mathsf{T}, \qquad p_j^\mathsf{T} f = 0. \tag{3.9}$$

Then

$$\mathcal{K}_{i \leftarrow j} f = G D_j f = \frac{1}{3} G f = \begin{pmatrix} 1 \\ -2/3 \\ -1/3 \end{pmatrix}, \tag{3.10}$$

whose target mean is again zero. The source contrast does not merely activate an edge of strength $\|G\|$; it produces a particular target contrast. Another source direction produces another response.

For this example, define the whitened edge matrix $\widetilde{G} = D_i^{1/2} G D_j^{1/2} = G/3$ and $\|\mathcal{K}\|_{\mathrm{HS}} = \|\widetilde{G}\|_F$. Its two nonzero singular values are 1 and $1/3$, and its Hilbert–Schmidt norm is

$$\|\mathcal{K}_{i \leftarrow j}\|_{\mathrm{HS}} = \frac{\sqrt{10}}{3}. \tag{3.11}$$

The scalar norm retains their combined magnitude but discards both singular directions. This one calculation contains the sequence of transformations used throughout the book:

$$\boxed{\text{raw table} \longrightarrow \text{gauge class} \longrightarrow \text{reference representative} \longrightarrow \text{fiber operator} \longrightarrow \text{scalar shadow.}} \tag{3.12}$$

Only the first four objects retain the typed substitution action. The final scalar is useful only after the loss has been declared.

This finite calculation will serve as a running case. We now restore the node, context, dynamics, estimator, and validation coordinates suppressed by the table calculation.

A machine-checked certificate accompanies this example (`lean/OperatorGraph/ThreeState.lean`): the decomposition of Eq. 3.3 into grand mean, target-only, source-only, and bivariate components, and the fiber-operator action of Eq. 3.10, are formalized in Lean 4, kernel-checked. The spectral and shadow properties of the gauged block G are certified separately in `lean/OperatorGraph/Shadow.lean`.

3.1 A finite-dimensional categorical geometry primer

The preceding table calculation uses ordinary finite-dimensional linear algebra. A vector $f \in \mathbb{R}^q$ assigns one numerical score to each residue category. A reference law p assigns weights to those scores: $p^{\mathsf{T}} f$ is their mean and $\sum_a p(a) f(a)^2$ is their squared norm. The notation $L^2(p)$ records this weighted geometry. Removing the mean leaves $q-1$ independent contrasts.

Three constructions below organize these contrasts. A *fiber* is the contrast space at one position. A *direct sum* collects one such vector per position. A *tensor product* represents joint contrasts across several positions. Whitening changes weighted coordinates into Euclidean ones, where singular vectors identify the source and target directions of an edge map.

The quotient language also has a concrete meaning. Adding a target-only term, a source-only term, or a constant to a pair table leaves its bivariate component unchanged. All tables related by these additions form one equivalence class. A chosen reference selects its centered representative. The definitions and formulas below make these operations explicit.

The common object must record more than a matrix attached to each pair. In particular, changing the context available to an edge or the state updated by that edge changes the model even when the displayed matrix is unchanged.

The formal tuple below is a second-reading map of the complete framework. On a first pass, track the node, its centered contrast space, the map carried by an edge, the context available to that map, and the endpoint used to validate it.

Definition 3.1 (Typed interaction system)**.** Let $(\mathsf{C}, \Sigma_{\mathsf{C}})$ be a measurable context space. A typed interaction system is a tuple

$$\mathfrak{G} = \big(V, \{(\Omega_i, p_i, \mathcal{H}_i^0)\}_{i\in V}, (\mathsf{C}, \Sigma_{\mathsf{C}}), E, h, \mathcal{K}, \mathcal{O}, \mathcal{D}, \mathcal{N}, \mathcal{E}_{\mathrm{st}}, \mathcal{V}_{\mathrm{d}}\big)\,. \tag{3.13}$$

Here V is the node set; (Ω_i, p_i) is the reference probability space at node i; $\mathcal{H}_i^0 \subseteq L_0^2(p_i)$ is the chosen centered feature fiber; $c \in \mathsf{C}$ is a context realization; the measurable map $E : \mathsf{C} \to 2^{V\times V}$ gives the support; $h : \mathsf{C} \to \bigoplus_i \mathcal{H}_i^0$ gives the one-body field; and

$$\mathcal{K}(c) = \big\{\mathcal{K}_{i\leftarrow j}(c) : \mathcal{H}_j^0 \longrightarrow \mathcal{H}_i^0\big\}_{(i,j)\in E(c)}$$

is the edge-map family. The higher-order coordinate is the measurable family

$$\mathcal{O}(c) = \left\{ \mathcal{K}_{i\leftarrow A}(c) : \bigotimes_{j\in A} \mathcal{H}_j^0 \longrightarrow \mathcal{H}_i^0 : i \in V,\ A \subseteq V\setminus\{i\},\ |A| \geq 2 \right\},$$

with absent maps represented by zero. The coordinate $\mathcal{D}$ records the represented latent and persistent state together with its input injection, update or transport, and output readout; $\mathcal{N}$ records the normalization or Lyapunov functional, $\mathcal{E}_{\mathrm{st}}$ the estimator and nuisance model, and $\mathcal{V}_{\mathrm{id}}$ the independent validation endpoint. A background or audit law is a probability measure on $(\mathsf{C}, \Sigma_{\mathsf{C}})$ and is declared separately from the type itself.

A model audit is a typed extraction map from model parameters, data, and interventions to $\mathfrak{G}$. Two models may agree in h and $\mathcal{K}$ while differing in $\mathcal{D}$, $\mathcal{N}$, or $\mathcal{E}_{\mathrm{st}}$; their agreement is coordinate-wise. A PSSM, for example, has a nonzero field h and zero off-diagonal edge operator. A Gaussian model can use centered linear features in $L^2_0(p_i)$, whereas the protein models below use the full centered categorical fiber.

3.2 Relational computation and the Graph Network interface

The Graph Network (GN) formalism isolates a reusable relational computation without prescribing the scientific semantics of its edges [39]. For a graph $G = (\mathbf{u}, V, E)$ with directed edge records (e_k, r_k, s_k), one GN block performs

$$e'_k = \phi^e(e_k, v_{r_k}, v_{s_k}, \mathbf{u}), \tag{3.14}$$

$$\bar{e}'_i = \rho^{e\to v}\big(\{e'_k : r_k = i\}\big), \qquad v'_i = \phi^v(v_i, \bar{e}'_i, \mathbf{u}), \tag{3.15}$$

$$\bar{e}' = \rho^{e\to u}(E'), \qquad \bar{v}' = \rho^{v\to u}(V'), \qquad \mathbf{u}' = \phi^u(\mathbf{u}, \bar{e}', \bar{v}'). \tag{3.16}$$

The learnable maps ϕ specify relation, node, and global updates. The reducers ρ are invariant to an arbitrary enumeration of their input sets, making the complete block equivariant to node relabeling when the graph and features are relabeled together. Parameter sharing permits the same local rule to act on new graph sizes and new combinations of entities. This creates an architectural basis for combinatorial generalization. Out-of-distribution generalization remains an empirical property of the trained system.

The typed system in Eq. 3.13 refines this interface. Its support $E(c)$ records the admissible senders and receivers; $\mathcal{K}_{i\leftarrow j}(c)$ types the edge update as a map between declared fibers; and $\mathcal{D}$ records the aggregation and node-state update. The remaining coordinates answer questions that a generic GN block intentionally leaves open: whether an edge is a route or an identified pair component, whether the updates arise from one joint probability law, which parameter redundancies preserve observables, how the edge object is estimated, and what endpoint validates it. Message-passing networks, Potts fields, and Transformers can therefore share a relational computation diagram while occupying different probabilistic and inferential coordinates. The unification is an interface with typed translations; semantic identity requires agreement across the complete coordinate system.

3.3 Node fibers and reference geometry

The word *fiber* carries little mystery here. At every residue position, take all numerical functions of residue identity and remove the constant function. The remaining $(q-1)$ directions encode comparisons such as "charged rather than hydrophobic" or any other centered categorical contrast. Calling these local spaces fibers keeps two indices separate: position identifies the node, while a vector inside the fiber identifies a pattern across residue categories.

Let positions be indexed by $[L] = \{1, \dots, L\}$ and categorical states by $[q] = \{1, \dots, q\}$. Fix a full-support reference distribution $p_i \in \Delta_{q-1}$ at each position and write $D_i = \mathrm{diag}(p_i)$. The node

fiber at position i is

$$\mathcal{H}_i = L^2(p_i), \qquad \langle f, g\rangle_{p_i} = f^{\mathsf{T}} D_i g, \tag{3.17}$$

with centered subspace

$$\mathcal{H}_i^0 = \{f : p_i^{\mathsf{T}} f = 0\}. \tag{3.18}$$

The total categorical tangent space is the Hilbert direct sum

$$\mathcal{H}^0 = \bigoplus_{i=1}^{L} \mathcal{H}_i^0. \tag{3.19}$$

The position index is the graph base, while amino-acid contrasts form the fiber over each node. This distinction prevents a position graph from being confused with a flat matrix of categorical coordinates.

Define the weighted zero-mean projector

$$P_i = I - \mathbf{1} p_i^{\mathsf{T}}. \tag{3.20}$$

It satisfies $P_i^2 = P_i$ and $P_i^{\mathsf{T}} D_i = D_i P_i$. In whitened coordinates its Euclidean form is

$$Q_i = D_i^{1/2} P_i D_i^{-1/2} = I - \sqrt{p_i}\sqrt{p_i}^{\mathsf{T}}. \tag{3.21}$$

Thus the removed constant direction depends on the declared categorical reference. A uniform reference, a family marginal, and a model marginal define different geometries; every audit keeps this choice explicit. Operationally, P_i takes any residue score vector and subtracts its p_i-weighted mean. It computes a centered contrast. Interaction estimation, biological choice of reference, and physical calibration enter at later stages.

3.3.1 Choosing the endpoint anchor and background law

A response audit contains two distributional choices. The endpoint anchors (p_i, p_j) define the categorical contrasts, block norm, and singular directions. The background law μ_{ij} weights the contexts used for the static–dynamic decomposition. These choices answer different questions: an endpoint anchor fixes the coordinate system, while a background law fixes the population over which persistence and variation are measured. Table 3.1 gives an operational default for common audit objectives.

The anchor is fixed before evaluating interaction scores and estimated from the training partition when data determine it. Small probabilities require a declared smoothing floor. Cross-family averages first transport every block to one common anchor. A primary anchor, one scientifically plausible alternative, and the resulting change in norms or rankings form the minimum sensitivity report. Product-reference and empirical-background dynamic fractions are reported together with the geometry gap in Eq. 4.25; the first supports interaction-order identification, and the second describes variation on the evaluated sequence population.

3.4 The canonical bivariate component

For any fixed pair table $T_{ij} \in \mathbb{R}^{q\times q}$, define

$$\boxed{G_{ij} = P_i T_{ij} P_j^{\mathsf{T}}.} \tag{3.22}$$

Audit objective	Primary endpoint anchor	Background law and required control
Exact algebraic calibration	Uniform or designed full-support reference	Designed product law; verify the exact identities under at least one additional full-support anchor.
Within-family static comparison	Smoothed family marginals estimated from the training MSA	Clade-held-out empirical contexts; report sequence weighting and a phylogeny-aware null.
Model-centered response audit	Detached model marginals from a fixed checkpoint and prompt population	Declared natural or mutational contexts; repeat across checkpoints when model drift is part of the question.
Cross-family or assay comparison	One common pooled, assay-defined, or design-defined anchor	Family-balanced or held-out assay contexts; report support overlap and transport sensitivity before pooling blocks or rankings.

Table 3.1: Reference choices follow the estimand. The anchor fixes endpoint geometry; the background law fixes the evaluation population.

Read this operation row by row and column by column: the left projection removes target-only variation, the right projection removes source-only variation, and the surviving block changes only when both endpoints vary. The theorem below establishes that this remainder is unique in the chosen reference geometry. Uniqueness is a statement about the decomposition of the table, not about how the table was generated.

Theorem 3.2 (Canonical categorical decomposition)**.** *Equip pair tables with*

$$\langle A, B\rangle_{p_i,p_j} = \sum_{a,b} p_i(a)p_j(b)A(a,b)B(a,b). \tag{3.23}$$

Every table has a unique orthogonal decomposition

$$T_{ij}(a,b) = c + u(a) + v(b) + G_{ij}(a,b), \tag{3.24}$$

where $p_i^{\mathsf{T}}u = 0$, $p_j^{\mathsf{T}}v = 0$, *and*

$$p_i^{\mathsf{T}}G_{ij} = 0, \qquad G_{ij}p_j = 0. \tag{3.25}$$

The bivariate component is exactly Eq. 3.22.

Proof. The orthogonal decomposition $L^2(p_i) = \operatorname{span}\{\mathbf{1}\} \oplus \mathcal{H}_i^0$ and its analogue at j induce four orthogonal tensor-product subspaces: constant, i-only, j-only, and centered bivariate. Applying P_i on the left and P_j^{T} on the right selects the last subspace. Existence and uniqueness follow from orthogonal projection. □

Terminology and the binary Ising case. Here *gauge* refers only to the additive non-identifiability of a redundant Potts pair-table parameterization. We call G_{ij} *reference-centered*, rather than "gauged," to avoid confusion with the spin-reversal gauge of spin glasses or with a $\mathbb{Z}_2$ lattice gauge theory; neither of those symmetries is used here. For two states with uniform reference, encode the categories by $s_i, s_j \in \{-1, +1\}$. Every table has the unique expansion

$$T(s_i, s_j) = c + h_i s_i + h_j s_j + J_{ij} s_i s_j, \qquad J_{ij} = \frac{T_{++} + T_{--} - T_{+-} - T_{-+}}{4}. \tag{3.26}$$

Double projection removes c, h_i, and h_j and leaves $G_{ij}(s_i, s_j) = J_{ij}s_is_j$. Thus the $q = 2$ categorical construction extracts the single Ising coupling coordinate from a redundant binary table. The "gauge" here remains the additive Potts parameter redundancy, distinct from an Ising gauge field.

The canonical decomposition is a functional-ANOVA statement relative to $p_ip_j^{\mathsf{T}}$ [53]. It identifies a bivariate component; direct contact, causal epistasis, and phylogenetic independence are separate empirical properties.

The underlying interaction is a gauge class rather than one selected representative. Define the additive subspace

$$\mathcal{A}_{ij} = \{T : T(a, b) = c + u(a) + v(b)\} \tag{3.27}$$

and the quotient $\mathcal{Q}_{ij} = \mathbb{R}^{q\times q}/\mathcal{A}_{ij}$. Every full-support pair of references selects one representative of the same $(q - 1)^2$-dimensional class.

Proposition 3.3 (Gauge quotient and reference transport)**.** *For fixed full-support references (p_i, p_j), two tables have the same projected block if and only if they define the same element of $\mathcal{Q}_{ij}$. For second full-support references (r_i, r_j), write $P_i^{[r]} = I - \mathbf{1}r_i^{\mathsf{T}}$ and similarly at j. The canonical representatives obey*

$$\boxed{G_{ij}^{[r]} = P_i^{[r]}G_{ij}^{[p]}P_j^{[r]\mathsf{T}}.} \tag{3.28}$$

Proof. Theorem 3.2 identifies the kernel of $T \mapsto P_iTP_j^{\mathsf{T}}$ with $\mathcal{A}_{ij}$, proving the quotient statement. The projectors for two references satisfy $P_i^{[r]}P_i^{[p]} = P_i^{[r]}$ and $P_j^{[p]\mathsf{T}}P_j^{[r]\mathsf{T}} = P_j^{[r]\mathsf{T}}$. Applying these identities to $G_{ij}^{[p]} = P_i^{[p]}T_{ij}P_j^{[p]\mathsf{T}}$ gives $P_i^{[r]}T_{ij}P_j^{[r]\mathsf{T}} = G_{ij}^{[r]}$. □

Reference-invariance of zero context dependence. Let a family of raw tables $T(C)$ be evaluated under one fixed background law, and let $G^{[p]}(C)$ and $G^{[r]}(C)$ be its representatives under two full-support reference pairs. If

$$V_{\mathrm{dyn}}^{[p]} = \mathbb{E}_C \left\| G^{[p]}(C) - \mathbb{E}_C G^{[p]}(C) \right\|_{p_i,p_j}^2, \tag{3.29}$$

and $V_{\mathrm{dyn}}^{[r]}$ is defined analogously, then

$$\boxed{V_{\mathrm{dyn}}^{[p]} = 0 \quad \Longleftrightarrow \quad V_{\mathrm{dyn}}^{[r]} = 0.} \tag{3.30}$$

Indeed, Eq. 3.28 maps a constant representative to a constant representative, and applying the same transport law in the reverse direction gives the converse. Thus absence of context dependence is reference independent, although every positive variance, norm, singular value, and dynamic fraction depends quantitatively on the chosen reference geometry. When the denominator in Eq. 4.22 is nonzero, the same equivalence holds with $V_{\mathrm{dyn}} = 0$ replaced by $\rho^{\mathrm{dyn}} = 0$.

Thus the gauge class is reference independent, while its centered representative, Hilbert norm, and singular geometry are reference dependent. Comparisons across families or contexts must either use one fixed reference or transport every representative to a declared anchor by Eq. 3.28.

3.5 From a table to an edge operator

A centered block becomes an edge only after its input and output are typed. The next equation feeds a centered substitution pattern at j into the block and returns a centered response pattern

at i. This is what the operator computes. Physical action and reciprocity are subsequent empirical and compatibility questions.

A gauged block induces the directed map

$$(\mathcal{K}_{i\leftarrow j} f)(a) = \sum_b p_j(b) G_{i\leftarrow j}(a,b) f(b) = \big(G_{i\leftarrow j} D_j f\big)(a). \tag{3.31}$$

source identity at $j \longrightarrow$ source contrast $f \in \mathcal{H}_j^0 \xrightarrow{\mathcal{K}_{i\leftarrow j}}$ target contrast in $\mathcal{H}_i^0 \longrightarrow$ change in target preference.

Figure 3.1: An edge is a typed map between local contrast spaces. Its scalar existence flag leaves the source-substitution and target-response directions unresolved.

Proposition 3.4 (Gauged tables are fiber maps)**.** *Equation 3.31 defines a linear map $\mathcal{K}_{i\leftarrow j} : \mathcal{H}_j^0 \to \mathcal{H}_i^0$ that annihilates the constant source mode. In whitened coordinates its matrix is*

$$\widetilde{G}_{i\leftarrow j} = D_i^{1/2} G_{i\leftarrow j} D_j^{1/2}, \tag{3.32}$$

and

$$\|\mathcal{K}_{i\leftarrow j}\|_{\mathrm{HS}} = \|\widetilde{G}_{i\leftarrow j}\|_F. \tag{3.33}$$

Proof. The left condition in Eq. 3.25 centers the output, while $G_{i\leftarrow j} p_j = 0$ gives $\mathcal{K}_{i\leftarrow j}\mathbf{1} = 0$. Conjugation by the isometries $f \mapsto D_i^{1/2} f$ and $f \mapsto D_j^{1/2} f$ gives the whitened matrix and preserves the Hilbert–Schmidt norm. □

For routing coefficients z_{ij}, the operator-valued graph is the block operator

$$(\mathbb{K}_z f)_i = \sum_{j\neq i} z_{ij} \mathcal{K}_{i\leftarrow j} f_j. \tag{3.34}$$

Its support, edge rank, and directionality are explicit. If $z_{ij} = z_{ji}$ and $G_{j\leftarrow i} = G_{i\leftarrow j}^{\mathsf{T}}$ under compatible references, $\mathbb{K}_z$ is self-adjoint. Predictive sequence models need not satisfy this reciprocity and may therefore yield genuinely directed block operators.

Coda: the graph after gauge

A declared reference selects one centered representative of a pair-table equivalence class and turns it into a map between categorical contrast spaces. In a compatible Potts Hamiltonian, adding $u(a) + v(b) + c$ merely reallocates terms among the pair, the one-body fields, and the energy zero. The quotient removes this freedom. Scalar contact scores are a later compression.

Gauge projection and endpoint invariance remain separate: centering a table does not control the information used to generate it. The next chapter therefore asks what happens when one residue is varied while another prediction is observed. A Potts model gives the same pair response in every background; a nonlinear sequence model can give a field of responses.

Chapter 4

Static and Dynamic Sequence Interactions

A static interaction graph describes a family by assigning one pair object to each edge. A deep sequence model does something more elusive. Its effective relation between two positions can change when a third residue changes, when the sequence enters another family-like background, or when information is rerouted through a different hidden path. The model parameters are fixed, but the response graph is not.

To see the distinction, hold all positions except i and j in a background C. Substitute the source residue at j, observe the model's categorical prediction at i, and project away target- and source-only components. The resulting operator

$$G_{i \leftarrow j}(C) \tag{4.1}$$

is a section of the model's behavior through that background. Repeating the audit over C does not estimate the same quantity with noise; it reveals whether the quantity itself changes.

This creates a clean comparison. An additive pairwise Potts conditional produces the same gauged response for every background, so its dynamic variance is zero. A nonlinear protein language model generally yields a distribution of edge operators. Their mean is the best static approximation in the chosen reference geometry, while their fluctuation measures the failure of a single static edge to describe the model.

The fluctuation has a precise higher-order interpretation. Under a product reference it can be tied exactly to higher-order source-coordinate components valued in the target contrast fiber. Under a correlated background law, orthogonality can fail and the same variance can contain cross terms. The reference distribution is therefore part of both the scientific statement and the estimator.

Two further questions run through the chapter. First, when does a directed response deserve the conditional meaning of a pair interaction? If the generator of the edge reads the endpoint being varied, the edge itself moves during the intervention. Second, when do the two directed responses $j \to i$ and $i \to j$ integrate into one undirected pair energy? That upgrade requires reciprocity after the relevant gauges have been aligned. Context dependence and directionality are therefore measurable coordinates, not inconveniences to be symmetrized away at the outset.

4.1 Potts models as static response fields

A pairwise Potts model has energy

$$E_{\mathrm{P}}(x) = -\sum_i h_i(x_i) - \frac{1}{2}\sum_{i\neq j} J_{ij}(x_i, x_j), \tag{4.2}$$

and single-site conditional

$$P_{\mathrm{P}}(X_i = a \mid x_{-i}) = \mathrm{softmax}_{a\in[q]}\left[h_i(a) + \sum_{j\neq i} J_{ij}(a, x_j)\right]. \tag{4.3}$$

The tables J_{ij} are fixed after fitting and are shared by all sequences in the family. Their canonical edge blocks are

$$G^{\mathrm{P}}_{ij} = P_i J_{ij} P_j^{\mathsf{T}}. \tag{4.4}$$

This is the sense in which a Potts interaction graph is static. The conditional probabilities still change with x_{-i}, but the rule mapping a source category to a target logit contrast does not. Static refers to the shared response law, not to constancy of the model output.

4.2 Categorical response of a sequence model

Let $c = x_{[L]\setminus\{i,j\}}$ be a sequence background excluding the endpoints. Evaluate a sequence model with target i withheld and source j set to category b, and denote its target logit by

$$T^c_{i\leftarrow j}(a, b) = \ell_i(a \mid X_j = b, C_{ij} = c). \tag{4.5}$$

The reference-centered categorical response is

$$G_{i\leftarrow j}(c) = P_i T^c_{i\leftarrow j} P_j^{\mathsf{T}}. \tag{4.6}$$

Equivalently, one may subtract any baseline column from T^c before projection. This gives the finite substitution table

$$R^c_{i\leftarrow j}(a, b) = \ell_i(a \mid b, c) - \ell_i(a \mid b_0, c), \tag{4.7}$$

with $P_i R^c P_j^{\mathsf{T}} = G_{i\leftarrow j}(c)$. The construction is sometimes called a categorical Jacobian. More precisely, it is a finite intervention, with an infinitesimal derivative as its local limit.

Proposition 4.1 (Exact recovery of an additive coupling)**.** *Suppose the target logits are additive,*

$$\ell_i(a \mid x_{-i}) = h_i(a) + \sum_{k\neq i} J_{i\leftarrow k}(a, x_k). \tag{4.8}$$

Then for every background c,

$$G_{i\leftarrow j}(c) = P_i J_{i\leftarrow j} P_j^{\mathsf{T}}. \tag{4.9}$$

Thus the gauged response is background independent and exactly recovers the canonical pair coupling.

Proof. In Eq. 4.5, the field $h_i(a)$ and all source terms with $k \neq j$ depend on a and c but not on b. They are annihilated by right projection. The remaining table is $J_{i\leftarrow j}(a, b)$, and left projection removes its target-only gauge component. □

For a nonlinear or higher-order conditional, the same response has the schematic expansion

$$G_{i\leftarrow j}(c) = G^{(2)}_{ij} + \sum_{k\notin\{i,j\}} G^{(3)}_{i,j,k}(c_k) + \sum_{\substack{k<l\\ k,l\notin\{i,j\}}} G^{(4)}_{i,j,k,l}(c_k,c_l) + \cdots . \tag{4.10}$$

It is therefore a well-defined *effective* edge at background c, assembled from all statistical and computational paths through which changing x_j changes the target preference at i.

4.3 The static–dynamic decomposition

Fix anchor references p_i, p_j and let C be distributed according to a declared family, model, or evaluation distribution μ_{ij}. The gauged blocks form one finite-dimensional Hilbert space with norm

$$\|G\|^2_{p_i,p_j} = \sum_{a,b} p_i(a)p_j(b)G(a,b)^2. \tag{4.11}$$

Assume $\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} < \infty$ and define

$$\overline{G}_{i\leftarrow j} = \mathbb{E}_C G_{i\leftarrow j}(C), \qquad \delta G_{i\leftarrow j}(C) = G_{i\leftarrow j}(C) - \overline{G}_{i\leftarrow j}. \tag{4.12}$$

Theorem 4.2 (Best static approximation and dynamic variance)**.** *The mean block $\overline{G}_{i\leftarrow j}$ is gauged and is the unique minimizer*

$$\overline{G}_{i\leftarrow j} = \arg\min_{A:\, p_i^{\mathsf{T}}A=0,\, Ap_j=0} \mathbb{E}_C\|G_{i\leftarrow j}(C) - A\|^2_{p_i,p_j}. \tag{4.13}$$

Moreover,

$$\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} = \|\overline{G}_{i\leftarrow j}\|^2_{p_i,p_j} + \mathbb{E}_C\|\delta G_{i\leftarrow j}(C)\|^2_{p_i,p_j}. \tag{4.14}$$

Proof. The gauged subspace is linear, so expectation preserves both zero-marginal constraints. Expanding $G(C)-A = \delta G(C)+(\overline{G}-A)$, the cross term vanishes because $\mathbb{E}_C\delta G(C) = 0$. The resulting Pythagorean identity is minimized uniquely at $A = \overline{G}$ and gives Eq. 4.14 when $A = 0$. □

The orthogonality in Theorem 4.2 is mean–residual orthogonality in $L^2(\mu_{ij})$. It holds for any declared background law μ_{ij}, including a correlated empirical MSA distribution, because the residual is centered under that same law. Independence enters only when the residual is further attributed to mutually orthogonal interaction orders, as in Corollary 5.2 below. Thus correlation changes the order-by-order interpretation of dynamic variance, while the static–dynamic Pythagorean identity remains exact.

Static component $\overline{G}_{i\leftarrow j}$	Dynamic component $\delta G_{i\leftarrow j}(C)$
Best single operator under the declared background law	Background-specific departure from that operator
Persists after averaging over C	Has mean zero over the same C
Comparable with a family-level static field	Measures the failure of one static edge to describe the response
Physical interpretation requires calibration	Origin may combine several mechanisms

Figure 4.1: The static–dynamic split separates a reusable mean relation from context dependence; biological importance is an independent validation coordinate.

A two-background calculation

The static–dynamic identity can be seen without fitting a large model. Continue with uniform three-state references and let the persistent pair component be

$$G_0 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{4.15}$$

Introduce a second centered direction

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{4.16}$$

Both matrices have zero row and column means, and they are orthogonal under the uniform weighted inner product. Let a third position k supply a binary background contrast $C \in \{-1, +1\}$ with equal probability, and define the effective edge

$$G_{i\leftarrow j}(C) = G_0 + \lambda C H. \tag{4.17}$$

The response of the pair (i, j) now changes when position k changes. Its mean and residual are

$$\overline{G}_{i\leftarrow j} = G_0, \qquad \delta G_{i\leftarrow j}(C) = \lambda C H. \tag{4.18}$$

Direct calculation gives

$$\|G_0\|^2_{p_i,p_j} = \frac{10}{9}, \qquad \|H\|^2_{p_i,p_j} = \frac{6}{9} = \frac{2}{3}. \tag{4.19}$$

Therefore

$$\mathbb{E}_C \|G(C)\|^2_{p_i,p_j} = \frac{10}{9} + \frac{2\lambda^2}{3}, \tag{4.20}$$

and the dynamic fraction is

$$\rho^{\mathrm{dyn}}_{i\leftarrow j} = \frac{6\lambda^2}{10 + 6\lambda^2}. \tag{4.21}$$

At $\lambda = 0$ the response is exactly static. At $\lambda = 1$, three eighths of its squared mass is dynamic. As $|\lambda|$ grows, the pair is increasingly described by its background modulation rather than by one persistent block.

The construction also displays interaction order. The term CH jointly involves the audited target i, source j, and background position k. It is stored observationally as a background-dependent edge, but functionally it is a third-order component. A scalar graph can record the variance of this edge across backgrounds; it cannot by itself say which categorical direction H carries or which position produced the modulation.

Finally, the calculation assumes that C is held fixed while the endpoint categories are varied. If the coefficient were emitted by a gate that also reads X_j, then enumerating the source state b would change both the message and the gate selecting the message. Double projection would still center the resulting table, but Eq. 4.17 would no longer describe one fixed conditional pair operator. This is the practical reason endpoint exclusion appears as a separate requirement rather than as a consequence of gauge fixing.

A machine-checked certificate accompanies this calculation (`lean/OperatorGraph/TwoBackground.lean`): the orthogonality of G_0 and H under the uniform weighted inner product, their squared norms,

and the dynamic-norm identity $\mathbb{E}_C\|G(C)\|^2 = \|G_0\|^2 + \lambda^2\|H\|^2$ are formalized in Lean 4, kernel-checked.

This theorem supplies an operational definition of static and dynamic interaction. Define

$$\rho^{\mathrm{dyn}}_{i\leftarrow j} = \frac{\mathbb{E}_C\|\delta G_{i\leftarrow j}(C)\|^2_{p_i,p_j}}{\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j}}, \qquad 0 \le \rho^{\mathrm{dyn}}_{i\leftarrow j} \le 1, \tag{4.22}$$

whenever the denominator is nonzero. In finite-precision estimation, the ratio should be reported only when

$$\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} > \tau_G, \tag{4.23}$$

for a preregistered absolute or null-calibrated threshold τ_G; otherwise it is undefined, not zero or one. This prevents a ratio of two numerical underflow terms on a null edge from appearing as strong dynamic response. An additive Potts conditional has $\rho^{\mathrm{dyn}}_{i\leftarrow j} = 0$. A positive value measures background dependence of the effective response. Higher-order epistasis, remote homologous patterns, and internal circuits are candidate sources to be separated by further interventions.

Two background laws expose a genuine interpretation–geometry tradeoff. With the same endpoint anchor references, define

$$\rho^{\mathrm{dyn,prod}}_{i\leftarrow j} = \rho^{\mathrm{dyn}}_{i\leftarrow j}\Big|_{\mu_C=\bigotimes_{k\notin\{i,j\}} p_k}, \qquad \rho^{\mathrm{dyn,emp}}_{i\leftarrow j} = \rho^{\mathrm{dyn}}_{i\leftarrow j}\Big|_{\mu_C=\mu_C^{\mathrm{emp}}}. \tag{4.24}$$

Only the product-background quantity has the exact order-by-order ANOVA interpretation in Corollary 5.2; its independently assembled backgrounds may, however, be far outside the natural sequence manifold. The empirical quantity evaluates natural or curated backgrounds and is therefore closer to the model's training support, although exhaustive endpoint substitutions can still leave that support. Correlations destroy order orthogonality, so the empirical dynamic variance remains a valid mean–residual quantity but not a sum of pure higher-order masses. Both should be reported, together with

$$\Delta\rho^{\mathrm{geom}}_{i\leftarrow j} = \rho^{\mathrm{dyn,emp}}_{i\leftarrow j} - \rho^{\mathrm{dyn,prod}}_{i\leftarrow j}, \tag{4.25}$$

as a diagnostic of sensitivity to the evaluation geometry rather than as a signed amount of biological interaction.

The distribution μ_{ij} is part of the claim. Averaging over one protein family, all natural proteins, or model-generated sequences produces different static components. A pLM can therefore have learned an amortized dynamic rule, while an MSA estimates a separate family-specific statistical model against which the response mean can be tested. Equality between an MSA-fitted Potts block and the pLM mean is an empirical transfer hypothesis in their common coordinates. The family distribution supplies the background law under which that hypothesis is evaluated.

4.4 A fixed random network with a dynamic response graph

The random recurrent network gives a solvable temporal analogue of the static–dynamic split [92]. Let

$$h(t+1) = \phi(Jh(t) + Bu(t))\,, \tag{4.26}$$

where J is fixed and ϕ acts coordinatewise. At the realized state define

$$D_t = \mathrm{diag}\big(\phi'(Jh(t) + Bu(t))\big)\,, \qquad K_t = D_tJ. \tag{4.27}$$

The support and entries of J belong to the parameterization. The effective one-step response K_t depends on the trajectory through D_t. A perturbation introduced at time s is transported to time $t > s$ by

$$\Phi_{t:s} = K_{t-1}K_{t-2}\cdots K_s. \tag{4.28}$$

Ordered multiplication matters: the same collection of local Jacobians in a different temporal order generally gives a different response.

This example separates three graphs that share the same neuron labels:

$$J \quad \text{(parameter graph)}, \qquad K_t \quad \text{(instantaneous response)}, \qquad \Phi_{t:s} \quad \text{(finite-time transport)}. \tag{4.29}$$

Sparsity of J constrains possible one-step paths. It does not determine the magnitude or long-time stability of the realized response. Saturated units can suppress a parameter edge, while products of individually modest Jacobians can amplify selected directions.

In the classical ensemble $J_{ij} \sim \mathcal{N}(0, g^2/n)$ with a sigmoidal nonlinearity, dynamical mean-field theory reduces the $n \to \infty$ network to a self-consistent stochastic process. For the zero-input hyperbolic-tangent model, the fixed state loses stability at the critical gain $g = 1$ and a chaotic regime appears beyond it [92]. This threshold belongs to the specified random ensemble and dynamics. A trained finite network with structured weights requires its own stability and finite-size analysis.

The correspondence with a deep sequence model is operational. For layer maps $h^{(\ell+1)} = F_\ell(h^{(\ell)})$, define

$$K_\ell(x) = \left.\frac{\partial F_\ell}{\partial h}\right|_{h=h^{(\ell)}(x)}, \qquad \Phi_{b:a}(x) = K_{b-1}(x)\cdots K_a(x). \tag{4.30}$$

This is a layerwise response audit. It becomes a Sompolinsky-style mean-field model only after randomness, width, stationarity, and scaling assumptions have been supplied. For a protein Transformer, the practical experiment is to perturb one residue representation, track the singular values and localization of $\Phi_{b:a}(x)$ across layers and backgrounds, and compare that internal transport with the finite categorical response $G_{i\leftarrow j}(C)$. Agreement is a mechanism hypothesis; the two operators live at different points in the encoder–readout chain.

The example gives the phrase "dynamic edge" a precise temporal reading. One may average K_t or a transported categorical block over a declared trajectory law, then apply the same mean–residual Pythagorean decomposition used for sequence backgrounds. Lyapunov exponents and finite-time singular values answer propagation and stability questions, while ρ^{dyn} answers variation across the declared contexts. They complement one another and should be reported in their respective state spaces.

4.5 Conditional meaning requires endpoint exclusion

Equation 4.6 is a strict conditional component only if the table-generating context remains fixed while the endpoint states are varied. Let $X \sim \nu$ under a declared joint sequence law, set

$$C_{ij} = X_{[L]\setminus\{i,j\}}, \qquad \mathcal{C}_{ij} = \sigma(C_{ij}), \qquad \mu_{ij} = \mathrm{Law}_\nu(C_{ij}), \tag{4.31}$$

and allow the reference, raw table, and any route decision to depend on this endpoint-excluding context.

Proposition 4.3 (Conditional gauge under an endpoint-excluding audit)**.** *Require the audit protocol to construct $T_{i\leftarrow j}(C_{ij})$, $p_i(C_{ij})$, $p_j(C_{ij})$, and every route decision from an endpoint-excluding*

representation measurable with respect to $\mathcal{C}_{ij}$; in particular, it may not reuse a hidden state that previously observed x_i or x_j. If the conditional references have full support, and the enumerated source category b enters only in the final endpoint evaluation used to assemble the table, then, for μ_{ij}-almost every context c,

$$G_{i\leftarrow j}(c) = P_i(c)T_{i\leftarrow j}(c)P_j(c)^{\mathsf{T}} \tag{4.32}$$

is the unique bivariate contrast relative to $p_i(\cdot \mid c)p_j(\cdot \mid c)$.

Proof. The protocol condition ensures that, after conditioning on $\mathcal{C}_{ij}$, the table, route, and reference measures are fixed while (X_i, X_j) vary, so Theorem 3.2 applies on each conditional fiber. □

If a hidden representation used to emit the table has already observed x_i or x_j, projection still gives a locally centered table, but not a global conditional decomposition. This caveat is especially important when interpreting ordinary attention heads: masking the target and explicitly enumerating source substitutions defines a cleaner response audit than double centering an attention-derived table computed from the unmodified endpoints.

When the conditional references in Proposition 4.3 depend on the random context C_{ij}, the blocks inhabit moving fibers. For a realization c, transport them into common anchor references $(\bar{p}_i, \bar{p}_j)$ before applying Eq. 4.12:

$$G^{\mathrm{anc}}_{i\leftarrow j}(c) = P_i^{[\bar{p}]}G^{[p(c)]}_{i\leftarrow j}(c)P_j^{[\bar{p}]\mathsf{T}}; \tag{4.33}$$

then apply the static–dynamic decomposition to G^{anc}. This convention separates a change of coordinates from genuine background dependence of the gauge class.

4.6 When directed responses define a joint pair energy

A masked or autoregressive predictor naturally produces directed blocks $G_{i\leftarrow j}$, whereas a pairwise Hamiltonian uses one reciprocal table for the unordered pair $\{i, j\}$. This distinction is another independent coordinate of the framework.

Proposition 4.4 (Reciprocity criterion for additive conditionals)**.** *Consider full-support additive categorical conditionals in one common reference-centered parameterization. Their pairwise interaction branch is compatible with a joint Potts energy if and only if*

$$G_{i\leftarrow j} = G^{\mathsf{T}}_{j\leftarrow i} \qquad \text{for every } i \neq j. \tag{4.34}$$

When this holds, the common block defines the gauged coupling of the joint energy; one-body fields are then fixed by the corresponding conditional biases up to a global additive constant.

Proof. Necessity follows because one joint pair table contributes $J_{ij}(a, b)$ to the conditional at i and $J_{ij}(a, b)$, with transposed endpoint order, to the conditional at j. Gauge projection therefore gives Eq. 4.34. Conversely, reciprocal gauged blocks can be inserted as the pair tables of Eq. 4.2; their single-site conditionals recover the specified pair branches, while endpoint-only differences are absorbed into the site fields. □

For arbitrary directed responses define the reciprocal and nonintegrable parts

$$G^{\mathrm{rec}}_{ij} = \frac{1}{2}\left(G_{i\leftarrow j} + G^{\mathsf{T}}_{j\leftarrow i}\right), \tag{4.35}$$

$$G^{\mathrm{irr}}_{ij} = \frac{1}{2}\left(G_{i\leftarrow j} - G^{\mathsf{T}}_{j\leftarrow i}\right). \tag{4.36}$$

Only the first is directly compatible with an undirected pair energy. The second measures failure of pairwise joint integrability in the chosen response coordinates. A thermodynamic irreversibility interpretation additionally requires a dynamical model.

Coda: a graph can be fixed in parameters and dynamic in response

The static–dynamic decomposition changes the comparison between family models and protein Transformers. Their difference is no longer summarized by saying that one is generative and the other contextual. Both can be interrogated through the same categorical response operator. The Potts model occupies the background-invariant limit; the nonlinear model supplies a mean operator and a measurable field of deviations.

This common coordinate does not erase the distinction between the models. A mean pLM response is an estimator built from interventions on a trained function, not a parameter fitted from an MSA. A small dynamic residual supports a static approximation under the audited background law, not a universal physical coupling. A large residual is evidence against that approximation and a clue to higher-order organization, but its decomposition depends on the reference geometry.

The next question is what kind of complexity the residual represents. Graph support, categorical rank, and interaction order are often compressed into one informal notion of complexity. They are independent. A single edge may carry a full-rank operator; a dense graph may use only a shared low-dimensional family of maps; and nonlinear composition can generate higher-order terms even when every local message was separately pair-centered. The next chapter separates these axes.

Chapter 5

Interaction Order, Rank, and Scalar Compression

Several different simplifications are routinely called "sparse interaction." A model may use few position pairs, compress the categorical action on each pair, truncate all dependence above order two, or reduce every edge to one scalar score. These operations save different resources and discard different information. None implies the others.

Graph support answers whether a position pair is represented. Categorical rank answers how many independent substitution-contrast directions its edge operator carries. Interaction order asks how many sequence coordinates participate irreducibly in one functional component. A sparse position graph can attach a full-rank $q \times q$ operator to each surviving edge. A complete graph can reuse a one-dimensional message family. A network built from pairwise messages can generate higher-order dependence after nonlinear aggregation.

These counterexamples are not pathological. They identify three distinct approximation budgets:

$$\text{number of edges}, \qquad \text{rank per edge}, \qquad \text{maximum interaction order}. \tag{5.1}$$

A claim of compression should say which budget is controlled and how the discarded component is measured.

The chapter then confronts the most common compression: replacing an operator G_{ij} by a scalar such as its norm. This step is often necessary for contact prediction, visualization, or ordinary graph spectra. It is also irreversible. Sign, singular directions, and the identity of favored substitutions disappear. Projection and scalarization do not commute: a large raw norm can come entirely from one-body structure that vanishes in the canonical pair gauge.

Scalar graphs are therefore not rejected here. They are located. A scalar shadow answers a declared question about the magnitude of an already identified operator. Trouble begins only when properties of the shadow are read back as though no categorical information had been discarded.

5.1 A three-variable ANOVA calculation

The relation between a dynamic edge and a higher-order interaction can be calculated with two binary source coordinates. Let $x_j, x_k \in \{-1, +1\}$ and let $A, B, C \in \mathcal{H}_i^0$ be three target-contrast vectors. Consider the target-centered score

$$\Phi(x_j, x_k) = \phi_0 + Ax_j + Bx_k + Cx_jx_k, \qquad \phi_0 \in \mathcal{H}_i^0. \tag{5.2}$$

Under the uniform product reference, the four scalar functions $1, x_j, x_k, x_j x_k$ are orthonormal. They respectively encode a target-only term, two pair terms, and one target–source–source term.

Audit the response to source j while holding x_k fixed. Since the centered binary source space is one-dimensional, identify the edge operator with its coefficient multiplying x_j. Equation 5.2 then gives

$$G_{i\leftarrow j}(x_k) = A + Cx_k. \tag{5.3}$$

The A direction persists across backgrounds. The C direction changes sign when the second source changes. With X_k uniform,

$$\overline{G}_{i\leftarrow j} = \mathbb{E}_{X_k} G_{i\leftarrow j}(X_k) = A, \qquad \delta G_{i\leftarrow j}(X_k) = CX_k, \tag{5.4}$$

and therefore

$$\mathbb{E}_{X_k} \|\delta G_{i\leftarrow j}(X_k)\|_{p_i}^2 = \|C\|_{p_i}^2. \tag{5.5}$$

The dynamic variance is exactly the squared mass of the component containing the audited source j and the additional source k. The general interaction-hierarchy theorem replaces these two binary coordinates by arbitrary categorical source sets and sums the corresponding orthogonal masses.

The word "exactly" uses the product reference. To see the role of that geometry, keep balanced binary marginals but let a correlated law μ satisfy

$$\mathbb{E}_\mu[X_j X_k] = \rho \neq 0. \tag{5.6}$$

Then

$$\langle x_j, x_k\rangle_{L^2(\mu)} = \rho, \qquad \langle 1, x_j x_k\rangle_{L^2(\mu)} = \rho, \tag{5.7}$$

and, for example,

$$\mathbb{E}_\mu \|AX_j + BX_k\|_{p_i}^2 = \|A\|_{p_i}^2 + \|B\|_{p_i}^2 + 2\rho\langle A, B\rangle_{p_i}. \tag{5.8}$$

The algebraic section $A+Cx_k$ still describes the finite response at a fixed background. Its attribution to a sum of mutually orthogonal interaction orders now requires a new conditional geometry or must retain the cross terms induced by μ.

A machine-checked certificate accompanies this primer: the orthonormality of the four characters, the coefficient-extraction identities, the response audit $G(x_k) = A + Cx_k$ with background mean A and dynamic variance exactly $\|C\|^2$ in an arbitrary target inner product space, and the correlated cross term above are formalized in Lean 4 (`lean/OperatorGraph/Anova.lean`), with every theorem kernel-checked against the standard axiomatic base.

5.2 From operator-valued graphs to hypergraphs

A graph retains target–source maps $\mathcal{K}_{i\leftarrow j} : \mathcal{H}_j^0 \to \mathcal{H}_i^0$. A general conditional model also contains hyperedge maps

$$\mathcal{K}_{i\leftarrow A} : \bigotimes_{j\in A} \mathcal{H}_j^0 \longrightarrow \mathcal{H}_i^0, \qquad |A| \geq 2. \tag{5.9}$$

For a fixed background outside $\{i\} \cup N$, let $\mathbb{E}_u$ average over x_u under a positive product reference and let $P_u = I - \mathbb{E}_u$. For $B \subseteq \{i\} \cup N$, define

$$\Pi_B = \left(\prod_{u\in B} P_u\right)\left(\prod_{v\notin B} \mathbb{E}_v\right). \tag{5.10}$$

Theorem 5.1 (Interaction hierarchy and pair-truncation error)**.** *For $T \in L^2(\bigotimes_{u\in\{i\}\cup N} p_u)$, the operators $\{\Pi_B\}$ are mutually orthogonal projections that sum to the identity, and $T = \sum_B \Pi_B T$ uniquely. Let*

$$\mathcal{I}_i = \sum_{\emptyset\neq A\subseteq N} T_{i,A}, \qquad T_{i,A} = \Pi_{\{i\}\cup A} T. \tag{5.11}$$

For the pairwise truncation $\mathcal{I}_i^{(1)} = \sum_{j\in N} T_{i,\{j\}}$,

$$\|\mathcal{I}_i - \mathcal{I}_i^{(1)}\|^2 = \sum_{A\subseteq N:\,|A|\geq 2} \|T_{i,A}\|^2. \tag{5.12}$$

Proof. The coordinate-wise averaging operators are commuting self-adjoint idempotents, and $P_u = I - \mathbb{E}_u$ is the orthogonal complement of $\mathbb{E}_u$. Expanding $\prod_u(\mathbb{E}_u + P_u) = I$ gives the decomposition. Distinct components contain a factor $\mathbb{E}_u P_u = 0$ and are orthogonal. Equation 5.12 is Pythagoras applied to the omitted terms. □

For a reference-centered block G, define the canonical currying map

$$\begin{aligned} \kappa_{ij} : \{G : p_i^{\mathsf{T}} G = 0,\ Gp_j = 0\} &\longrightarrow L_0^2(p_j; \mathcal{H}_i^0), \\ (\kappa_{ij} G)(b) &= G(\cdot, b). \end{aligned} \tag{5.13}$$

It is an isometric identification because

$$\|\kappa_{ij} G\|^2_{L^2(p_j;\mathcal{H}_i^0)} = \sum_b p_j(b) \|G(\cdot, b)\|^2_{p_i} = \|G\|^2_{p_i,p_j}. \tag{5.14}$$

Corollary 5.2 (Dynamic variance equals higher-order response mass)**.** *Let $N \subseteq [L] \setminus \{i\}$ contain j and define the centered, vector-valued target score*

$$\Phi(x_N) = P_i \ell_i(\cdot \mid x_N) \in \mathcal{H}_i^0, \qquad \Phi \in L^2\left(\bigotimes_{u\in N} p_u; \mathcal{H}_i^0\right). \tag{5.15}$$

For $B \subseteq N$, let $\Pi_B^N = (\prod_{u\in B} P_u)(\prod_{v\in N\setminus B} \mathbb{E}_v)$ act on the source coordinates of this Hilbert-valued function, and write $\Phi_B = \Pi_B^N \Phi$. Set $C = X_{N\setminus\{j\}}$ and take the mean and residual in Eq. 4.12 under the product law $\bigotimes_{u\in N\setminus\{j\}} p_u$. The audited response then satisfies, for every b and for almost every C under this product law,

$$(\kappa_{ij} G_{i\leftarrow j}(C))(b) = \sum_{B\subseteq N:\, j\in B} \Phi_B\big(b, C_{B\setminus\{j\}}\big), \tag{5.16}$$

$$(\kappa_{ij} \overline{G}_{i\leftarrow j})(b) = \Phi_{\{j\}}(b), \tag{5.17}$$

$$\mathbb{E}_C \|\delta G_{i\leftarrow j}(C)\|^2_{p_i,p_j} = \sum_{B\subseteq N:\, B\supsetneq\{j\}} \|\Phi_B\|^2. \tag{5.18}$$

Thus dynamic response variance is exactly the total functional-ANOVA mass of components containing the audited source and at least one additional source coordinate. The target category remains an output coordinate in $\mathcal{H}_i^0$; it is not assumed to be an observed model input. The norms on the right are the Bochner L^2 norms over the indicated source coordinates and the target fiber.

Proof. The commuting source-coordinate projections extend to the Bochner space in Eq. 5.15 and remain mutually orthogonal. Applying P_j retains exactly the components whose source set contains j; P_i has already centered the output fiber. For fixed C, the definition of the audited table gives, columnwise,

$$[\kappa_{ij} G_{i\leftarrow j}(C)]\,(b) = [P_j \Phi(\cdot, C)]\,(b), \tag{5.19}$$

where P_j acts on the jth source coordinate. Expanding the right-hand side by the source-coordinate ANOVA gives the first identity. Averaging over $C = X_{N\setminus\{j\}}$ retains only $\Phi_{\{j\}}$. Orthogonality of the remaining Hilbert-valued components, followed by the isometry in Eq. 5.14, gives the squared-norm identity. □

The theorem gives Eq. 4.10 a precise interpretation. A dynamic edge response is a section through the higher-order expansion with all other coordinates fixed. Its background variance is therefore an exact higher-order mass under the product reference. Under a correlated empirical background distribution the components can lose orthogonality and acquire cross terms; the product reference or a declared conditional geometry is part of the theorem. Even in the orthogonal case, resolving individual hyperedges requires a designed multi-substitution experiment beyond their total variance.

Remark 5.3 (Mean orthogonality versus order orthogonality). Theorem 4.2 and Corollary 5.2 use two different orthogonalities. The first centers one random operator around its mean and requires no coordinate independence. The second decomposes that operator by subsets of sequence coordinates and uses a product reference. If the same order-indexed components are evaluated under a correlated law μ, then in general

$$\begin{aligned}\mathbb{E}_\mu \left\| \sum_{B \supsetneq \{j\}} \Phi_B(C_B) \right\|^2 &= \sum_{B \supsetneq \{j\}} \mathbb{E}_\mu \|\Phi_B(C_B)\|^2 \\ &\quad + \sum_{\substack{B, D \supsetneq \{j\} \\ B \neq D}} \mathbb{E}_\mu \langle \Phi_B(C_B), \Phi_D(C_D) \rangle,\end{aligned} \tag{5.20}$$

and the final sum records cross terms induced by the correlated reference geometry. Dynamic variance remains measurable, but it is no longer identifiable with a simple sum of order-specific ANOVA masses.

5.3 Random energy: a joint law without a low-order graph

Derrida's random-energy model assigns an independent energy to every complete configuration [21]. For a q-state sequence of length L, take

$$E(x) \overset{\text{iid}}{\sim} \mathcal{N}(0, L\sigma^2), \qquad x \in [q]^L, \qquad P_\beta(x) = \frac{e^{-\beta E(x)}}{Z_L(\beta)}. \tag{5.21}$$

The model has one exact joint Gibbs law. Its low-temperature freezing transition follows from competition between the q^L configurations and the extreme tail of their energies. With the normalization in Eq. 5.21, the critical inverse temperature is $\beta_c = \sqrt{2 \log q}/\sigma$. Positional locality plays no role in this result.

The operator framework reveals where the random energy resides in interaction order. Use the uniform product reference and let Π_A be the functional-ANOVA projector for a subset $A \subseteq [L]$.

Proposition 5.4 (ANOVA order profile of a random energy table)**.** *For the ensemble in Eq. 5.21,*

$$\mathbb{E}_E\|\Pi_A E\|_{L^2}^2 = \frac{L\sigma^2}{q^L}(q-1)^{|A|}. \tag{5.22}$$

Consequently the expected mass at interaction order k is

$$W_k = \frac{L\sigma^2}{q^L}\binom{L}{k}(q-1)^k, \qquad \sum_{k=0}^{L} W_k = L\sigma^2. \tag{5.23}$$

After normalization by total mass, the order profile is the binomial law Binomial$(L,(q-1)/q)$*.*

Proof. The product ANOVA subspace indexed by A has dimension $(q-1)^{|A|}$. The random vector of q^L energy values is isotropic with coordinate variance $L\sigma^2$. The expected Euclidean squared norm of its projection onto a rank-r subspace is $L\sigma^2 r$. The uniform L^2 norm divides by q^L, giving Eq. 5.22. Summing over the $\binom{L}{k}$ subsets of size k gives Eq. 5.23; the binomial theorem gives the final identity. □

For protein alphabets, $(q-1)/q$ is close to one, so the ensemble mass is concentrated at very high order. This is an exact statement about a random function on sequence space, independent of whether its Gibbs measure is in the frozen phase.

The same calculation sharpens the static–dynamic distinction. Fix a pair (i,j) and double center the $q \times q$ energy table obtained at each background $C = x_{[L]\setminus\{i,j\}}$. The expected total conditional pair-response mass is

$$\mathbb{E}_E\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2 = \beta^2 L\sigma^2\frac{(q-1)^2}{q^2}, \tag{5.24}$$

whereas the expected squared norm of the background mean is

$$\mathbb{E}_E\|\overline{G}_{i\leftarrow j}\|^2 = \beta^2 L\sigma^2\frac{(q-1)^2}{q^L}. \tag{5.25}$$

Thus the ratio of expected static mass to expected total response mass is $q^{-(L-2)}$. A typical background displays a substantial finite pair response, while averaging across backgrounds removes almost all of it. The apparent edge is a section through a high-order landscape.

The random-energy model is therefore a useful null for protein sequence models. It preserves a globally normalized energy landscape and destroys reusable positional structure. A learned model has gone beyond this null when low-order operator mass, cross-background singular directions, or family-transferable static blocks exceed matched random-function and sequence-shuffling controls. Freezing, low-order interaction, and biological constraint remain separate empirical coordinates.

The combinatorial spine of Eq. 5.23—the identity $\sum_{k=0}^{L}\binom{L}{k}(q-1)^k = q^L$ that normalizes the order profile—is machine-checked in Lean 4 (`lean/OperatorGraph/Rem.lean`). The full ensemble statement of Eq. 5.22 (functional-ANOVA projectors and isotropic Gaussian vectors) is outside the certificate companion.

Proposition 5.5 (Nonlinear composition can create higher order)**.** *Separately centered pair messages can generate higher-order components after nonlinear aggregation.*

Proof. For independent Rademacher sources $x_j, x_k \in \{-1, 1\}$ under the uniform product reference and a nonzero target contrast $P_i u$,

$$(P_i u)(a)(x_j + x_k)^2 = 2(P_i u)(a) + 2(P_i u)(a)x_j x_k. \tag{5.26}$$

After assigning the first term to the target-only component, the second is a nonzero target–source–source interaction. □

Consequently, pairwise first-layer messages can produce a globally higher-order deep network. Its final response must be decomposed at the output, or its higher-order residual measured by multi-background and multi-substitution audits.

5.4 Support, categorical rank, and order are independent

For a nonzero whitened block, write

$$\widetilde{G}_{ij} = \sum_{r=1}^{q-1} \sigma_{ijr} u_{ijr} v_{ijr}^{\mathsf{T}}. \tag{5.27}$$

Graph support counts which position pairs are represented. Categorical rank describes the number of substitution-contrast directions retained on one edge. Interaction order distinguishes edges from hyperedges. These are independent coordinates: a sparse graph may have full-rank categorical blocks, while a dense graph may share only a low-dimensional message family.

For a relative tolerance ϵ, an error-controlled edge rank is

$$r_{ij}(\epsilon) = \min \left\{ r : \frac{\sum_{t>r} \sigma_{ijt}^2}{\sum_t \sigma_{ijt}^2} \le \epsilon \right\}. \tag{5.28}$$

Low rank is therefore a measurable compression hypothesis independent of graph sparsity.

5.5 The scalar position graph is a shadow

The common scalar compression is

$$M_{ij} = \|G_{ij}\|_{p_i,p_j} = \|D_i^{1/2} G_{ij} D_j^{1/2}\|_F. \tag{5.29}$$

Proposition 5.6 (Loss and order of scalar compression)**.** *The map $G_{ij} \mapsto M_{ij}$ discards sign, singular directions, and categorical action. In addition, projection must precede the norm: for any nonzero $v \in \mathbb{R}^q$, the one-body table $T = \mathbf{1}v^{\mathsf{T}}$ satisfies*

$$P_i T P_j^{\mathsf{T}} = 0, \qquad \|T\|_{p_i,p_j} = \|v\|_{p_j} > 0. \tag{5.30}$$

Proof. Norms are invariant under sign and orthogonal changes of singular vectors, so the compression is many-to-one. The final statement follows from $P_i\mathbf{1} = 0$ and direct evaluation of the weighted norm: $\sum_{a,b} p_i(a)p_j(b)v(b)^2 = \sum_b p_j(b)v(b)^2$. □

Two operators with the same scalar shadow

Return to uniform three-state references and consider two gauged blocks

$$G_A = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad G_B = \sqrt{\frac{5}{3}} \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{5.31}$$

Both have zero row and column means. Under the uniform references used here, their ordinary squared Frobenius norms are both 10, and hence their weighted squared norms are both 10/9. Their weighted scalar shadows are therefore identical:

$$M_A = M_B = \frac{\sqrt{10}}{3}. \tag{5.32}$$

The operators are nevertheless different. The two nonzero singular values of the whitened G_A are 1 and 1/3, whereas those of the whitened G_B are both $\sqrt{5}/3$. One concentrates more strength in a leading substitution direction; the other distributes its strength equally across the two-dimensional contrast fiber.

Their action on the same source contrast $f = (1, -1, 0)^{\mathsf{T}}$ makes the difference explicit:

$$\mathcal{K}_A f = \begin{pmatrix} 1 \\ -2/3 \\ -1/3 \end{pmatrix}, \qquad \mathcal{K}_B f = \frac{1}{3}\sqrt{\frac{5}{3}} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}. \tag{5.33}$$

The first operator favors the first target contrast; the second suppresses it and favors the second. A contact map built only from M_{ij} assigns the two edges exactly the same value.

This is not an exotic collision. Sign reversal, rotation of singular vectors, and redistribution among singular values all create further operators with the same norm. Scalarization is suitable for asking where strong pair structure is located. It cannot answer which substitutions are compatible, compensatory, or antagonistic. Those are operator-level questions.

A machine-checked certificate accompanies this example (`lean/OperatorGraph/Shadow.lean`): both squared Frobenius norms, the shadow identity, the zero row and column means, the spectral witnesses with independence of the exhibited eigenvectors, and the differing actions on the same source contrast are formalized in Lean 4, kernel-checked.

This proposition explains why unprojected coupling norms can contain a large conservation- or entropy-correlated mode [100]. Removing such a mode may improve contact ranking; an independent null model determines whether the removed component merits a nuisance interpretation.

For dynamic responses, Eq. 4.14 first yields two directed scalar graphs,

$$M_{i\leftarrow j}^{\text{static}} = \|\overline{G}_{i\leftarrow j}\|_{p_i,p_j}, \qquad M_{i\leftarrow j}^{\text{dynamic}} = \Big(\mathbb{E}_C \|\delta G_{i\leftarrow j}(C)\|_{p_i,p_j}^2\Big)^{1/2}. \tag{5.34}$$

The first ranks persistent family-level response; the second ranks background sensitivity. Their different biological predictions call for separate reporting. Any undirected contact shadow additionally requires a declared symmetrization of the two directions; reciprocity should be tested before that symmetrization is interpreted as an energy coupling. Chapter 11 shows that scalar, signed, and connection Laplacians built from these shadows or blocks retain different amounts of the original categorical geometry.

Coda: compression is a scientific choice

Support thresholding selects edges; low-rank approximation selects categorical directions; pair truncation selects interaction orders; scalarization selects one summary per edge. Agreement after one reduction can conceal disagreement before it, as the same-shadow example demonstrates.

An ANOVA component is irreducible to lower orders under its declared variables and reference. Its origin may be a many-body energy, a marginalized latent variable, or nonlinear computation. The decomposition alone does not choose among them.

These constructions have used linear projection. A finite pair probability law must also remain positive and obey its marginal constraints. The next chapter connects the linear gauge to this nonlinear probability geometry.

Chapter 6

Probability-Exact Edge Geometry

A centered interaction table lives in a linear space. A joint probability table occupies a constrained subset: its entries remain nonnegative, sum to one, and often preserve two prescribed marginals. These constraints form a transport polytope. The two geometries describe an infinitesimal interaction coordinate and a finite probabilistic model.

Begin with independent marginals $p_i(a)p_j(b)$ and tilt them by a pair score. At weak coupling, double centering removes directions that only change one marginal, leaving the tangent interaction inside the zero-marginal subspace. At finite coupling, however, linear projection is insufficient: exponentiation bends the geometry and generally changes both marginals.

Exact marginal preservation requires multiplicative row and column factors. Matrix scaling finds those factors and returns the unique information projection onto the transport polytope under the usual positivity assumptions. The nonlinear construction and the linear gauge are complementary resolutions: the gauge-fixed operator is the local tangent coordinate through which the probability-exact family passes.

This relation matters whenever a pair table is given probabilistic language. Tangent-space norms and finite-strength mutual information are different functionals. A centered logit table reaches a joint law with desired marginals through matrix scaling, and separately fitted conditionals reach one global distribution only when their compatibility equations hold. Probability geometry supplies these finite constraints around the local operator coordinate.

The chapter develops that distinction without abandoning the operator view. Exact pair laws, information projections, and tangent operators describe different resolutions of the same local departure from independence. Keeping the resolution explicit prevents a convenient linear score from being mistaken for a fully normalized physical model.

A two-state probability table

For two binary variables, a joint law is a nonnegative 2×2 table Q whose four entries sum to one. Row sums give the marginal p_i, column sums give p_j, and for $p_j(b) > 0$ the conditional law is

$$q_{i|j}(a \mid b) = \frac{Q(a,b)}{p_j(b)}. \tag{6.1}$$

Independence means $Q(a,b) = p_i(a)p_j(b)$. Departure from independence can be measured by

$$I(X_i; X_j) = \mathrm{KL}(Q \| p_i p_j^{\mathsf{T}}) = \sum_{a,b} Q(a,b) \log \frac{Q(a,b)}{p_i(a)p_j(b)}, \tag{6.2}$$

with the convention $0 \log 0 = 0$. This quantity is nonnegative and vanishes exactly at independence when the reference has full support.

The difference between tangent geometry and an exact pair law is already visible for two uniform binary variables. Let

$$p_i = p_j = (1/2, 1/2)^{\mathsf{T}} \tag{6.3}$$

and choose the centered score

$$S_\theta = \theta \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{6.4}$$

Because symmetry gives equal row and column sums after exponentiation, exact matrix scaling can be written in closed form:

$$Q_\theta = \frac{1}{4\cosh\theta} \begin{pmatrix} e^{\theta} & e^{-\theta} \\ e^{-\theta} & e^{\theta} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1+t & 1-t \\ 1-t & 1+t \end{pmatrix}, \qquad t = \tanh\theta. \tag{6.5}$$

Every row and column sums exactly to $1/2$ for every finite θ.

Near independence, $t = \theta + O(\theta^3)$ and

$$Q_\theta - p_i p_j^{\mathsf{T}} = \frac{\theta}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + O(\theta^3), \tag{6.6}$$

which is exactly the tangent direction predicted by Eq. 6.12, derived later in this chapter. At finite strength the linear coordinate θ is replaced in the probability table by the bounded coordinate $\tanh\theta$. The transport polytope bends and eventually saturates as probability concentrates on the two matching states.

Substituting Eq. 6.5 into Eq. 6.2 gives the exact mutual information

$$I(\theta) = \frac{1+t}{2}\log(1+t) + \frac{1-t}{2}\log(1-t). \tag{6.7}$$

For weak coupling, $I(\theta) = \theta^2/2 + O(\theta^4)$, agreeing with the quadratic tangent norm. At strong coupling, the nonlinear probability geometry matters and the quadratic approximation no longer describes the finite law.

Finally, add arbitrary row and column scores to S_θ. They change the unscaled exponential matrix, but exact marginal scaling absorbs them and returns the same Q_θ. The nonlinear probability model therefore respects precisely the same gauge quotient as the linear operator, while encoding finite displacement on that quotient rather than only its tangent direction.

6.1 From tangent coordinates to finite pair laws

The linear gauge identifies a function-space component. Exact preservation of prescribed one-site marginals after exponentiation and normalization additionally requires the transportation polytope

$$\mathcal{U}(p_i, p_j) = \{Q \geq 0 : Q\mathbf{1} = p_i,\ Q^{\mathsf{T}}\mathbf{1} = p_j\}. \tag{6.8}$$

It has dimension $(q-1)^2$, matching the dimension of the categorical tangent-product space. Independence is $Q^0_{ij} = p_i p_j^{\mathsf{T}}$. The standardized residual

$$R_{ij} = D_i^{-1/2}(Q_{ij} - p_i p_j^{\mathsf{T}})D_j^{-1/2} \tag{6.9}$$

has zero constant modes and rank at most $q-1$. It is the correspondence-analysis coordinate of pair association.

6.2 Matrix scaling on the gauge quotient

Given a finite score table S_{ij}, let $K_{ij} = \exp S_{ij}$ elementwise. For strictly positive K_{ij}, p_i, p_j, matrix scaling supplies positive vectors u_{ij}, v_{ij} such that

$$Q_{ij} = \operatorname{diag}(u_{ij}) K_{ij} \operatorname{diag}(v_{ij}) \tag{6.10}$$

has the prescribed marginals [91, 16, 13, 14]. The coupling is unique, while (u, v) are unique up to reciprocal scaling.

Proposition 6.1 (Exact gauge invariance of matrix scaling)**.** *Let $\mathscr{S}_{p_i,p_j}(S)$ denote the coupling obtained by scaling* $\exp S$ *to marginals (p_i, p_j). For arbitrary vectors a, b and scalar c,*

$$\mathscr{S}_{p_i,p_j}\left(S + a\mathbf{1}^\mathsf{T} + \mathbf{1}b^\mathsf{T} + c\mathbf{1}\mathbf{1}^\mathsf{T}\right) = \mathscr{S}_{p_i,p_j}(S). \tag{6.11}$$

Consequently, the exact nonlinear map $S \mapsto Q$ factors through the same quotient $\mathcal{Q}_{ij}$ as the linear Potts parameter-gauge quotient. The induced map is a bijection from $\mathcal{Q}_{ij}$ onto the relative interior $\operatorname{relint} \mathcal{U}(p_i, p_j)$*, consisting of couplings with strictly positive entries. Finite score tables do not reach the boundary of the transport polytope.*

Proof. Exponentiating the added row, column, and constant terms only left- and right-scales K by positive diagonal matrices and a positive scalar. These factors are absorbed into the Sinkhorn vectors. Uniqueness of the coupling with the prescribed marginals gives Eq. 6.11. Conversely, for any strictly positive $Q \in \mathcal{U}(p_i, p_j)$, choosing $S = \log Q$ returns Q without nontrivial scaling. If two finite score tables return the same positive coupling, their exponentials differ only by the positive diagonal scaling absorbed by Sinkhorn, so their logarithms differ by row, column, and constant terms. This proves bijectivity on the quotient. □

Along $K_\epsilon = \exp(\epsilon S_{ij})$ around independence,

$$\left.\frac{dQ_{ij}(\epsilon)}{d\epsilon}\right|_{\epsilon=0} = D_i(P_i S_{ij} P_j^\mathsf{T}) D_j. \tag{6.12}$$

Thus the reference-centered representative is both the tangent-space linearization of exact marginal-preserving pair geometry and a linear coordinate on the same gauge quotient through which the exact Sinkhorn map factors.

6.3 Local pair laws and global compatibility

The exact conditional $q_{i|j}(a \mid b) = Q_{ij}(a, b)/p_j(b)$ is normalized for each b, and averaging over $b \sim p_j$ returns p_i. Its mutual information

$$I_{ij} = \mathrm{KL}(Q_{ij} \| p_i p_j^\mathsf{T}) \tag{6.13}$$

is an exact nonnegative scalar association. Near independence,

$$I_{ij} = \frac{1}{2}\|R_{ij}\|_F^2 + O(\|R_{ij}\|_F^3), \tag{6.14}$$

with a remainder controlled only when the smallest marginal probability stays away from zero. Rare and unsupported categories therefore require an explicit smoothing convention.

This construction separates three levels:

1. $P_i S_{ij} P_j^{\mathsf{T}}$ is a marginal-orthogonal logit contrast;
2. $Q_{ij} \in \mathcal{U}(p_i, p_j)$ is one exact pair law with prescribed marginals;
3. a collection of pair laws is globally probabilistic only if it lies in the marginal polytope of one compatible sequence distribution.

The third property is a global compatibility constraint beyond a sparse pair layer or pairwise Sinkhorn scaling. Marginal log-linear, discrete dependence, and minimum-information models provide global alternatives but introduce a separate estimation problem [5, 45, 87].

Three valid pair laws with no joint law

Pairwise normalization does not guarantee global compatibility. Let $X_1, X_2, X_3 \in \{0, 1\}$ have uniform one-site marginals. Consider three individually valid pair laws:

$$X_1 = X_2 \quad \text{almost surely}, \qquad X_2 = X_3 \quad \text{almost surely}, \qquad X_1 \neq X_3 \quad \text{almost surely}. \tag{6.15}$$

Each statement is realized by a normalized 2×2 table with the required marginals. The first two statements imply $X_1 = X_3$ for every triple in the support, contradicting the third. No joint law on (X_1, X_2, X_3) can therefore have all three pair marginals.

The example separates local and global probability claims. Matrix scaling can turn each positive pair score into a valid coupling. A collection of those couplings belongs to one sequence law only when their shared marginals and higher-order compatibility constraints can be satisfied simultaneously. A sparse layer of pair tables does not supply that global certificate by itself. This does not prevent finite pair *potentials* from being added into a globally normalized Gibbs Hamiltonian. It says that separately normalized pair *marginals* need not be the marginals of that Gibbs law, or of any common joint law.

A machine-checked certificate accompanies this example. The normalized 2×2 tables and the incompatibility statement are formalized in Lean 4 (`lean/OperatorGraph/PairLaw.lean` in the repository): the three almost-sure constraints are contradictory for normalized nonnegative tables on the eight-point space, in pair-marginal form, and for measurable coordinates on an arbitrary probability space. The contradiction is certified by the proof checker, not by the author.

6.4 Cavity inference on a declared factor graph

Matrix scaling solved a local problem: given one pair score and two marginals, construct a probability-exact pair law. The preceding compatibility example showed that solving this problem edge by edge does not guarantee one joint law for all vertices. Cavity inference starts after that additional joint factorization has been declared. It propagates the effect of the rest of the factor graph into a local marginal, thereby separating the pair factor, the inference message, and the resulting pair law.

The cavity method begins from a joint factorization. For a pairwise categorical model on a graph $G = (V, E)$, write

$$P(x) = \frac{1}{Z} \prod_{i \in V} \psi_i(x_i) \prod_{(i,j) \in E} \psi_{ij}(x_i, x_j). \tag{6.16}$$

On a tree, deleting the factor on (i, j) separates the two sides. The cavity message carried by the directed edge $i \to j$ obeys

$$m_{i\to j}(a) \propto \psi_i(a) \prod_{k\in\partial i\setminus\{j\}} \sum_b \psi_{ik}(a, b)m_{k\to i}(b). \tag{6.17}$$

The exact pair marginal is then

$$P_{ij}(a, b) \propto \psi_{ij}(a, b)m_{i\to j}(a)m_{j\to i}(b). \tag{6.18}$$

These equations are exact on trees and form the Bethe approximation on loopy graphs [69, 68].

One edge now carries several related objects:

Object	Definition	Meaning
$\log \psi_{ij}(a, b)$	factor parameter, projected into a gauge when needed	direct pair term in the declared joint law
$m_{i\to j}(a)$	marginal state of i with the (i, j) factor removed	directed inference message
$P_{ij}(a, b)$	compatible marginal after messages are recombined	exact or approximate pair law
$\chi_{i\leftarrow j}$	derivative of a marginal with respect to a field	propagated statistical susceptibility
$G_{i\leftarrow j}(C)$	finite endpoint substitution at fixed background	background-conditioned model response

The distinction matters even when all five objects are computed from one Potts model. The factor is static. Messages vary with boundary conditions and the rest of the graph. Susceptibility sums all paths permitted by the joint law. The finite-background audit conditions on one realized context rather than marginalizing it. Agreement among these objects is a theorem only under the corresponding factorization, topology, and conditioning operation.

There is also a terminology boundary. Graphical-model cavity inference deletes an edge or factor while retaining an endpoint variable as the argument of a message. The endpoint-excluding audit in this book withholds the realized identities of both X_i and X_j from support selection and reference construction. Both use deletion to control information flow, but they delete different objects and license different conditional statements. We reserve *cavity message* for Eq. 6.17 and use *endpoint exclusion* for the architectural measurability condition.

This supplies a controlled protein benchmark. Generate categorical sequences on a tree from a known Potts law, then report the projected factor G^{P}_{ij}, the BP marginal, the susceptibility, and the finite substitution response in common anchor coordinates. Exact tree identities test the implementation. Adding loops exposes the Bethe residual, which can be measured by marginal error, calibration, or discrepancy from exact enumeration on small systems. For dense protein-family models, tree exactness is unavailable unless a sparse factorization has been declared and validated.

Replica-symmetry breaking moves the object one level higher: a single cavity message is replaced by a distribution over messages or pure states. Survey propagation uses this idea to derive algorithms for random constraint problems [70]. In the language of this book it suggests an ensemble of operator graphs or message fields, with uncertainty attached to the edge object itself. That extension belongs to the open problem of graph ensembles rather than to a single deterministic interaction graph.

Coda: local coordinates and finite laws

Linear double projection preserves marginals to first order. Multiplicative matrix scaling factors through the same quotient and enforces those constraints at finite displacement. It also has the exact variational form

$$Q^* = \operatorname*{argmin}_{Q \in \mathcal{U}(p_i, p_j)} \mathrm{KL}(Q \| \overline{K}), \qquad \overline{K} = \frac{\exp(S)}{\sum_{a,b} \exp S(a,b)}. \tag{6.19}$$

For $S = -E/\tau$, the objective is an energy–entropy competition. A physical equilibrium reading requires the corresponding energy, temperature, and ensemble; the information projection is exact without that reading. Even exact pair laws need not assemble into one joint law, as the three-pair counterexample shows.

The next part applies these coordinates to familiar models, distinguishing their native probability laws, estimators, and update rules before comparing their edge operators.

Part III

Models, Estimation, and Spectra

Chapter 7

Model Correspondences in Operator Coordinates

This chapter compares the native states, edge objects, and laws of familiar models before extracting their common operator coordinates. A static Potts block, a Gaussian precision block, a marginalized RBM component, and a Transformer route enter through different constructions. The question is which correspondence survives a declared extraction.

The chapter has four blocks: probabilistic models; neural routes, state transport, and categorical response; the 2019 reconstruction framework; and the graph-network lineage. The two historical cases show what exact correspondences made computationally transferable and which differences remained afterward. Throughout, the audit names the state space, normalized variable, edge carrier, estimator, and first claim that needs additional evidence.

The framework compares sequence models coordinate by coordinate. A model M enters through the typed extraction

$$\mathrm{Audit}(M) = (h_M, \mathcal{K}_M, E_M, (\mathsf{C}_M, \Sigma_{\mathsf{C}_M}), \mathcal{O}_M, \mathcal{D}_M, \mathcal{N}_M, \mathcal{E}_{\mathrm{st},M}, \mathcal{V}_{\mathrm{id},M}), \tag{7.1}$$

This full record replaces comparison by one displayed matrix.

The comparison below locates each model's native object before the detailed correspondences.

7.1 Model family portraits: one residue pair through six models

For the running pair (i, j), each model supplies a different native object. Table 7.1 fixes those objects before the derivations below compare them.

For neural models, the distinction needed throughout the chapter is

$$\begin{aligned} \alpha_{ij}^{(h)}(x): &\quad \text{where one layer reads,} \\ \Phi_{t:s}(x): &\quad \text{how stored information crosses later transitions,} \\ G_{i\leftarrow j}(C): &\quad \text{how the complete categorical prediction changes.} \end{aligned} \tag{7.2}$$

The first two are internal computational objects. The last is the externally comparable response. The following sections derive the correspondences and specify the assumptions under which their native probability laws or update rules can also be matched.

Model	Native object at (i, j)	Route into operator coordinates
PSSM	one-site fields $h_i(a)$; no represented pair edge	$\mathcal{K}_{i\leftarrow j} = 0$, even if the observed family is correlated
Potts	one static table $J_{ij}(a, b)$ in a joint sequence law	$P_i J_{ij} P_j^{\mathsf{T}}$ gives the reference-centered pair block
Gaussian	a precision block between continuous variables	conditional linear coupling; one-hot categories require a declared Gaussian approximation
Hopfield–Potts	collective patterns factorizing static pair tables	center the induced blocks; dense support can coexist with low global rank
RBM	hidden motifs whose marginalization generates several interaction orders	extract the visible pair component and retain the higher-order remainder
Transformer	instantaneous routes and, in recurrent variants, state propagators	audit categorical output under source substitutions to obtain $G_{i\leftarrow j}(C)$

Table 7.1: Native edge objects and their extraction into comparable operator coordinates.

7.2 Static fields, Gaussian conditionals, and low-rank patterns

A PSSM retains only the one-body branch $h_i(a)$ and therefore occupies the field coordinate with zero off-diagonal edge operator. A Potts model adds static full categorical edge blocks. We first retain the model-specific coordinates before asking which of them can be collapsed into a shared reconstruction diagram.

For a nondegenerate Gaussian vector with covariance Σ and precision $\Omega = \Sigma^{-1}$,

$$\Omega_{ij} = 0 \quad \Longleftrightarrow \quad X_i \perp\!\!\!\perp X_j \mid X_{[L]\setminus\{i,j\}}. \tag{7.3}$$

This is a distributional theorem under a Gaussian law [20]. Leading-mode deletion is a separate spectral operation, while a Gaussian approximation to one-hot protein variables still requires reference-centered Potts coordinates for categorical comparison.

A classical Hopfield–Potts restriction factorizes the static coupling as

$$J_{ij}(a, b) = \sum_{\mu=1}^{p} \lambda_\mu \xi_{i\mu}(a) \xi_{j\mu}(b). \tag{7.4}$$

It is a low-rank parameterization of a pairwise Potts energy [54, 10]. Attention belongs to a separate state-dependent routing construction. Positive and negative pattern weights select different operator directions, and gauge projection is still required. At the pairwise level one may test

$$\mathbb{K}^{(2)} = \mathbb{K}_{\mathrm{LR}} + \mathbb{K}_{\mathrm{sp}}, \tag{7.5}$$

where the two terms represent distributed collective modes and localized typed constraints. This decomposition is a model hypothesis: low-rank and sparse classes overlap, so recovery needs declared penalties, incoherence assumptions, synthetic ground truth, or independent interventions.

The next block moves from fitted probabilistic fields to neural computation. Its order is instantaneous route, end-to-end latent transport, positional observability, and current-family conditioning. The output response remains the externally comparable operator throughout.

7.3 Transformer computation and interaction graphs

A self-attention layer is a message-passing operation whose nodes are tokens. Its candidate support is usually the complete directed graph (or the graph allowed by a mask), whereas its effective weighted route is generated anew from the current representations. For head h,

$$\alpha_{ij}^{(h)}(x) = \mathrm{softmax}_j\left(\frac{Q_i^{(h)}(x)^{\mathsf{T}} K_j^{(h)}(x)}{\sqrt{d_h}}\right), \qquad y_i = \sum_{j,h} \alpha_{ij}^{(h)} B_h x_j. \tag{7.6}$$

In this computational sense a Transformer instantiates a special dynamic GN block: the same query–key and value rules are shared across token pairs, the positional aggregation is permutation-respecting before positional encodings or masks are supplied, and the effective edge weights depend on the current sequence and layer [39, 47, 38, 24]. Training therefore learns a shared graph-generating rule $X \mapsto P_h(X)$. The resulting route is input-specific, while the statistical operator in Eq. 3.31 is obtained through a separate output audit.

There are at least four non-equivalent symmetry questions inside Eq. 7.6. For a head with shared linear query–key maps and no additional position-dependent score transform, let X collect the current token states and set

$$C_h = W_Q^{(h)} W_K^{(h)\mathsf{T}}, \qquad S_h(X) = \frac{X C_h X^{\mathsf{T}}}{\sqrt{d_h}}, \qquad P_h(X) = \mathrm{softmax}_{\mathrm{row}}\, S_h(X), \tag{7.7}$$

one may ask whether (i) C_h is symmetric, (ii) the token logit matrix $S_h(X)$ is symmetric, (iii) the normalized route $P_h(X)$ is reversible, or (iv) the audited categorical response obeys Eq. 4.34. Symmetry of C_h implies symmetry of $S_h(X)$ before an asymmetric mask, relative-position bias, or position-dependent query–key transform is applied, but row normalization generally gives $P_{ij} \neq P_{ji}$. Stationary flux, rather than entrywise symmetry, determines whether this asymmetry carries nonreciprocal circulation. Indeed, if $K_{ij} = \exp S_{ij} = K_{ji}$, $z_i = \sum_j K_{ij}$, and

$$P_{ij} = \frac{K_{ij}}{z_i}, \qquad \pi_i = \frac{z_i}{\sum_k z_k}, \tag{7.8}$$

then

$$\pi_i P_{ij} = \pi_j P_{ji}. \tag{7.9}$$

Thus a symmetric score followed by row-softmax is generally asymmetric in Euclidean coordinates but reversible in its stationary geometry. An asymmetric query–key kernel or mask can break this property, although either can also produce accidental reversibility on a particular input.

This distinction sharpens recent empirical results. Saponati et al. analyze symmetry and column dominance primarily in C_h, reporting that bidirectional objectives favor symmetric structure whereas autoregressive objectives favor directionality; symmetric initialization improves several encoder-language training runs, while the reported vision gains are not significant and large-scale behavior remains open [83]. Courtois et al. similarly improve BERT-style efficiency with a symmetric dot-product parameterization [12]. Neither result addresses the normalized token route or the end-to-end substitution response directly. Doubly stochastic attention, in turn, balances incoming and outgoing mass while allowing asymmetric directed cycles. Sinkformers implement this constraint by iterative matrix scaling, while ESPFormer uses parallel expected-sliced transport and reports improvements across several modalities [82, 88]. Marginal balance and detailed balance remain two independently testable properties.

At fixed attention coefficients, the linear value-path map is

$$\mathcal{A}_{i\leftarrow j} = \sum_{h=1}^{H} \alpha_{ij}^{(h)} B_h. \tag{7.10}$$

Proposition 7.1 (Shared span of a frozen attention layer)**.** *Every map in Eq. 7.10 lies in the shared subspace* $\mathrm{span}\{B_1, \dots, B_H\}$*. Dense connectivity therefore distributes a shared* H*-dimensional operator family across its edges.*

The complete input–output response also contains derivatives or finite differences through queries, keys, earlier layers, residual paths, recurrent state, and nonlinear blocks. This is why the categorical output audit in Eq. 4.6 is the proper model-level object for comparison with a Potts block. The attention map supplies an instantaneous route graph, while a state transition describes how stored information is transported between steps. Route, state transport, and response should therefore be reported separately.

7.3.1 Encoder–transport–readout closes the type system

The categorical edge operator and the internal neural operators live in different fibers. A complete architectural realization therefore includes an augmented latent state space

$$\mathscr{Z}^{(n)} = \left(\bigoplus_{i=1}^{L} \mathscr{Z}_i^{(n)} \right) \oplus \mathscr{S}^{(n)}, \tag{7.11}$$

where $\mathscr{Z}_i^{(n)}$ is a token or residue fiber and $\mathscr{S}^{(n)}$ contains any recurrent, pair, or global state. Fix an ordered audit pair $i \leftarrow j$, an audit context $C = C_{ij}$ constructed without the two endpoint identities, and the complete target protocol, and write $q_k = |\Omega_k|$. Let

$$\iota_{i\leftarrow j,C} : \Omega_j \longrightarrow \mathscr{Z}^{(0)}, \qquad \Phi_{i,C} : \mathscr{Z}^{(0)} \longrightarrow \mathscr{Z}^{(N)}, \qquad \varrho_i : \mathscr{Z}^{(N)} \longrightarrow \mathbb{R}^{q_i} \tag{7.12}$$

denote source-category injection, the complete latent computation, and the target-logit readout. The pair and target indices record the audit intervention: for masked prediction, the input at i is withheld before any latent state or route is formed. They do not introduce learned parameters specific to the pair. The exact nonlinear table already used in Eq. 4.5 is written columnwise as

$$T_{i\leftarrow j}^{C}(:, b) = \varrho_i(\Phi_{i,C}(\iota_{i\leftarrow j,C}(b))), \qquad b \in \Omega_j, \tag{7.13}$$

and its external categorical response remains

$$G_{i\leftarrow j}(C) = P_i T_{i\leftarrow j}^{C} P_j^{\mathsf{T}}. \tag{7.14}$$

This factorization is exact for finite categorical substitutions and requires no differentiability. It also makes endpoint exclusion operational: the injection, latent computation, route decisions, stored state, and readout used for this table must all obey the same targetwise context protocol.

The local linear bridge requires one additional declaration. Choose a differentiable extension

$$\bar{\iota}_{i\leftarrow j,C} : \Delta_{q_j-1} \longrightarrow \mathscr{Z}^{(0)} \tag{7.15}$$

that agrees with $\iota_{i\leftarrow j,C}$ at the simplex vertices, and use p_j as the local anchor. Assume that this extension, $\Phi_{i,C}$, and ϱ_i are differentiable along the resulting forward orbit. The extension is part

of the local audit and is not determined by the vertex table. Hard routing boundaries require a regionwise derivative or the finite audit above. A source contrast $f \in \mathcal{H}_j^0$ determines the zero-mass simplex tangent $D_j f$ because $\mathbf{1}^\mathsf{T} D_j f = p_j^\mathsf{T} f = 0$. Set

$$z_{i\leftarrow j,C}^{(0)} = \bar{\iota}_{i\leftarrow j,C}(p_j), \qquad z_{i\leftarrow j,C}^{(N)} = \Phi_{i,C}(z_{i\leftarrow j,C}^{(0)}). \tag{7.16}$$

The source injection, latent Jacobian, and centered target readout have the explicit types

$$\mathsf{E}_{i\leftarrow j,C} : \mathcal{H}_j^0 \longrightarrow T_{z_{i\leftarrow j,C}^{(0)}} \mathscr{Z}^{(0)}, \qquad \mathsf{E}_{i\leftarrow j,C} f = D\bar{\iota}_{i\leftarrow j,C}(p_j)[D_j f], \tag{7.17}$$

$$\mathbb{J}_{i\leftarrow j,C}^{\mathrm{lat}} : T_{z_{i\leftarrow j,C}^{(0)}} \mathscr{Z}^{(0)} \longrightarrow T_{z_{i\leftarrow j,C}^{(N)}} \mathscr{Z}^{(N)}, \qquad \mathbb{J}_{i\leftarrow j,C}^{\mathrm{lat}} = D\Phi_{i,C}(z_{i\leftarrow j,C}^{(0)}), \tag{7.18}$$

$$\mathsf{R}_{i\leftarrow j,C} : T_{z_{i\leftarrow j,C}^{(N)}} \mathscr{Z}^{(N)} \longrightarrow \mathcal{H}_i^0, \qquad \mathsf{R}_{i\leftarrow j,C} = P_i D\varrho_i(z_{i\leftarrow j,C}^{(N)}). \tag{7.19}$$

Define the anchor-dependent local categorical response by

$$\mathcal{K}_{i\leftarrow j}^{\mathrm{lin}}(C) f = P_i D[\varrho_i \circ \Phi_{i,C} \circ \bar{\iota}_{i\leftarrow j,C}]\,(p_j)[D_j f]. \tag{7.20}$$

Proposition 7.2 (Encoder–transport–readout factorization)**.** *The local categorical edge operator factors as*

$$\boxed{\mathcal{K}_{i\leftarrow j}^{\mathrm{lin}}(C) = \mathsf{R}_{i\leftarrow j,C}\, \mathbb{J}_{i\leftarrow j,C}^{\mathrm{lat}}\, \mathsf{E}_{i\leftarrow j,C} : \mathcal{H}_j^0 \longrightarrow \mathcal{H}_i^0.} \tag{7.21}$$

Consequently,

$$\operatorname{rank} \mathcal{K}_{i\leftarrow j}^{\mathrm{lin}}(C) \le \min\left\{ \operatorname{rank} \mathsf{E}_{i\leftarrow j,C}, \operatorname{rank} \mathbb{J}_{i\leftarrow j,C}^{\mathrm{lat}}, \operatorname{rank} \mathsf{R}_{i\leftarrow j,C}, q_j - 1, q_i - 1 \right\}. \tag{7.22}$$

If the composite map $u \mapsto \varrho_i(\Phi_{i,C}(\bar{\iota}_{i\leftarrow j,C}(u)))$ is affine on Δ_{q_j-1}, then $\mathcal{K}_{i\leftarrow j}^{\mathrm{lin}}(C)$ equals the finite categorical operator induced by $G_{i\leftarrow j}(C)$ through Eq. 3.31.

Proof. The chain rule gives Eq. 7.21; P_i centers its range and the source map is defined only on $\mathcal{H}_j^0$. The rank of a composition cannot exceed the rank of any factor. Let g denote the composite map in the affine case. Barycentric coordinates on the simplex give $g(u) = \sum_b u_b g(e_b) = T_{i\leftarrow j}^C u$, so $Dg(u)[\delta u] = T_{i\leftarrow j}^C \delta u$ for every zero-mass tangent δu. For $f \in \mathcal{H}_j^0$, $P_j^\mathsf{T} D_j f = D_j P_j f = D_j f$, so the resulting action is $P_i T_{i\leftarrow j}^C P_j^\mathsf{T} D_j f = G_{i\leftarrow j}(C) D_j f$. □

When all sources and targets are audited at one base point of a joint differentiable encoder and a common forward map, the factors assemble over $\mathcal{H}^0 = \bigoplus_i \mathcal{H}_i^0$. More precisely, let

$$\bar{\iota}_C : \prod_j \Delta_{q_j-1} \longrightarrow \mathscr{Z}^{(0)}, \qquad p = (p_1, \dots, p_L), \qquad z_C^{(0)} = \bar{\iota}_C(p), \qquad z_C^{(N)} = \Phi_C(z_C^{(0)}) \tag{7.23}$$

be the joint encoder and common orbit, let $\varrho = (\varrho_1, \dots, \varrho_L)$, and define $\mathbb{D}_p(f_i)_i = (D_i f_i)_i$ and $\mathbb{P} = \bigoplus_i P_i$. Then

$$\mathfrak{E}_C = D\bar{\iota}_C(p)\mathbb{D}_p : \mathcal{H}^0 \longrightarrow T_{z_C^{(0)}} \mathscr{Z}^{(0)}, \tag{7.24}$$

$$\mathbb{J}_C^{\mathrm{lat}} = D\Phi_C(z_C^{(0)}) : T_{z_C^{(0)}} \mathscr{Z}^{(0)} \longrightarrow T_{z_C^{(N)}} \mathscr{Z}^{(N)}, \tag{7.25}$$

$$\mathfrak{R}_C = \mathbb{P} D\varrho(z_C^{(N)}) : T_{z_C^{(N)}} \mathscr{Z}^{(N)} \longrightarrow \mathcal{H}^0. \tag{7.26}$$

The complete local response Jacobian has the compact realization

$$\boxed{\mathbb{K}_C^{\mathrm{lin}} = \mathfrak{R}_C \, \mathbb{J}_C^{\mathrm{lat}} \, \mathfrak{E}_C.} \tag{7.27}$$

Its declared off-diagonal blocks form the operator graph. Target-specific masking or pair-specific endpoint removal gives the factors in Eq. 7.21; their block assembly is useful bookkeeping but represents a collection of audit runs, not one physical forward pass.

The latent Jacobian itself is an ordered network object. Suppress the pair indices for one audit run. If $\Phi_C = F_{N-1,C} \circ \cdots \circ F_{0,C}$ and $z_C^{(n+1)} = F_{n,C}(z_C^{(n)})$, then

$$\mathbb{J}_C^{\mathrm{lat}} = DF_{N-1,C}(z_C^{(N-1)}) \cdots DF_{0,C}(z_C^{(0)}). \tag{7.28}$$

For an additive residual block, its derivative can be grouped into identity, frozen value or dictionary transport, route derivatives, state derivatives, and nonlinear channel derivatives. For ordinary attention, the frozen value contribution on the token-state subspace is exactly

$$\mathbb{J}_{n,C}^{\mathrm{value}} = \sum_h B_h \otimes P_h(C) \tag{7.29}$$

under column-stacking vectorization. Expanding Eq. 7.28 gives the end-to-end computational path sum. Equation 7.29 is the token-to-token value-path block and is embedded into the larger augmented-state Jacobian when $\mathscr{S}^{(n)}$ is present. Applying $\mathfrak{R}_C$ and $\mathfrak{E}_C$ converts internal transport into the external categorical operator.

This closes the operator hierarchy without identifying its levels. A Potts block is a static categorical realization with no latent routing requirement. Attention, operator dictionaries, and state propagators are internal factors of $\mathbb{J}_C^{\mathrm{lat}}$. Scalar, signed, and connection Laplacians are downstream contractions or transports built from a declared categorical or route operator. Phylogeny enters the context and estimator coordinates, or enters $\mathbb{J}_C^{\mathrm{lat}}$ only when tree messages are actually executed. Diffusion belongs to the state-update coordinate and reaches a sequence interaction claim through the same readout and validation ladder. The finite table in Eq. 7.13 is exact at the categorical vertices; the Jacobian factorization is local, anchor dependent, and contingent on the chosen simplex extension.

7.4 Where position lives: coordinates, support, and dynamics

An integer token index is only one realization of sequence order. A model may carry positional information in a token coordinate, an edge score, the allowed support, a local ordered filter, a persistent state, or the order in which updates are composed. These mechanisms break different symmetries and impose different memory geometries. It is useful to separate three questions: whether a computation can distinguish an ordering, where that distinction is represented, and whether the training task forces the representation to use it.

7.4.1 Support symmetry before positional coordinates

Let $A \in \{0,1\}^{L \times L}$ declare the directed support of a shared message-passing rule and let Π be a permutation matrix. Write

$$\mathrm{Aut}(A) = \{\Pi : \Pi A \Pi^{\mathsf{T}} = A\}. \tag{7.30}$$

Proposition 7.3 (Support automorphisms bound positional identifiability)**.** *Suppose an architecture has no coordinate-labelled input, uses shared node maps, and aggregates each allowed neighborhood by a permutation-symmetric rule. Then, for every* $\Pi \in \mathrm{Aut}(A)$*,*

$$F_A(\Pi H) = \Pi F_A(H). \tag{7.31}$$

For complete bidirectional support, $\mathrm{Aut}(A)$ *is the full symmetric group* S_L*, so a token permutation only relabels the outputs; the architecture cannot attach meaning to the order itself without another symmetry-breaking input. For the finite causal support* $A_{ts} = \mathbf{1}\{s \le t\}$*, the automorphism group is trivial.*

Proof. Relabelling nodes by an automorphism preserves every allowed neighborhood. Shared node maps commute with the relabelling, and symmetric aggregation removes the ordering of messages within each neighborhood, which gives Eq. 7.31. Every permutation preserves complete support. Under causal support, node t has in-degree t; these degrees are all distinct, so every support automorphism fixes every node. □

Trivial support symmetry makes position structurally available, but it does not guarantee that a particular computation reads it. A single content-only causal attention layer has

$$\mathrm{Attn}_t(H) = \frac{\sum_{s \le t} \exp(q(h_t)^\mathsf{T} k(h_s)) v(h_s)}{\sum_{s \le t} \exp(q(h_t)^\mathsf{T} k(h_s))}. \tag{7.32}$$

At fixed target state, this expression depends on the prefix source states only through their multiset. Its target output therefore agrees on (a, b, c) and (b, a, c) when the target c and the source embeddings are otherwise unchanged. Later layers can distinguish the two sequences once the source states themselves contain different prefix histories. Boundary tokens, causal local convolutions, and recurrent updates provide additional carriers. Conversely, a composition of fully S_L-equivariant layers remains S_L-equivariant; a noncommuting product across depth transports order only after some coordinate, support, boundary, or state update has broken that symmetry.

Explicit relative coordinates provide a different construction. Rotary position embedding writes the displacement directly into the query–key bilinear form,

$$s_{ts}^{\mathrm{RoPE}} = q_t^\mathsf{T} R_{t-s} k_s, \tag{7.33}$$

where R_{t-s} is a prescribed orthogonal transport depending on relative index [96]. The causal mask supplies precedence, while Eq. 7.33 supplies an explicit relative coordinate.

7.4.2 Time-ordered affine state transport

Recurrent linear attention places order in a state transition. Kimi Delta Attention provides a concrete example. For one head, let $q_t, k_t \in \mathbb{R}^{d_k}$, $v_t \in \mathbb{R}^{d_v}$, and $S_t \in \mathbb{R}^{d_k \times d_v}$. KDA applies ℓ_2 normalization to queries and keys; write $\|q_t\|_2 = \|k_t\|_2 = 1$. Its recurrent core is

$$S_t = M_t S_{t-1} + B_t, \qquad M_t = (I - \beta_t k_t k_t^\mathsf{T}) \operatorname{diag}(\alpha_t), \qquad B_t = \beta_t k_t v_t^\mathsf{T}, \qquad \widetilde{o}_t = S_t^\mathsf{T} q_t. \tag{7.34}$$

Here $\alpha_t \in (0,1)^{d_k}$ controls channelwise retention and $\beta_t \in (0,1)$ controls write strength [34, 35]. Key normalization gives

$$\left\| I - \beta_t k_t k_t^\mathsf{T} \right\|_2 \le 1, \qquad \|M_t\|_2 \le \|\alpha_t\|_\infty < 1, \tag{7.35}$$

which supplies the contraction assumption used below. Without a norm bound on k_t, the interval $0 < \beta_t < 1$ alone does not imply stability. With $\overline{S}_{t-1} = \mathrm{diag}(\alpha_t)S_{t-1}$, the same update is

$$S_t = \overline{S}_{t-1} + \beta_t k_t \big(v_t - \overline{S}_{t-1}^{\mathsf{T}} k_t\big)^{\mathsf{T}}. \tag{7.36}$$

Thus retention first attenuates the previous state, after which the rank-one delta rule corrects the value associated with the current key. This is an online least-squares correction after retention. The state compresses key–value associations, with interference between nonorthogonal keys.

Define the path-ordered propagator

$$\Phi_{t:s+1} = M_t M_{t-1} \cdots M_{s+1}, \qquad \Phi_{t:t+1} = I. \tag{7.37}$$

Unrolling Eq. 7.34 gives

$$S_t = \Phi_{t:1} S_0 + \sum_{s=1}^{t} \Phi_{t:s+1} B_s, \tag{7.38}$$

and therefore

$$\widetilde{o}_t = S_0^{\mathsf{T}} \Phi_{t:1}^{\mathsf{T}} q_t + \sum_{s=1}^{t} w_{t\leftarrow s}(x) v_s, \qquad w_{t\leftarrow s}(x) = \beta_s q_t^{\mathsf{T}} \Phi_{t:s+1} k_s. \tag{7.39}$$

The matrix $W(x) = (w_{t\leftarrow s}(x))$ is a lower-triangular effective transport kernel. Its entries may be signed, need not sum to one, and depend on every intervening transition between s and t. It is an implicit finite-state Green kernel rather than a softmax route. At frozen gates Eq. 7.39 is an exact algebraic contribution of each write. A substitution of x_s also changes local convolutions, gates, keys, queries, and later transitions; that complete effect belongs to the audited response $G_{t\leftarrow s}(C)$.

The fixed state dimension imposes a precise capacity constraint even though $W(x)$ can be dense.

Proposition 7.4 (Cross-cut rank of recurrent transport)**.** *Fix the gates, keys, and queries in Eq. 7.39. For every cut $1 \le u < T$, the strictly past-to-future block factors as*

$$W_{u+1:T,\,1:u} = \mathcal{O}_u \mathcal{C}_u, \qquad [\mathcal{O}_u]_{t,:} = q_t^{\mathsf{T}} \Phi_{t:u+1}, \qquad [\mathcal{C}_u]_{:,s} = \beta_s \Phi_{u:s+1} k_s, \tag{7.40}$$

where rows of $\mathcal{O}_u$ are indexed by $t > u$ and columns of $\mathcal{C}_u$ by $s \le u$. Consequently,

$$\operatorname{rank} W_{u+1:T,\,1:u} \le d_k. \tag{7.41}$$

The corresponding linear map from past value arrays to future outputs has rank at most $d_k d_v$.

Proof. For $t > u \ge s$, path ordering gives $\Phi_{t:s+1} = \Phi_{t:u+1}\Phi_{u:s+1}$. Substitution into Eq. 7.39 yields Eq. 7.40. The scalar kernel block has inner dimension d_k; the value-array map is its Kronecker product with I_{d_v}. □

Thus constant recurrent state and dense effective support coexist with a bounded communication rank across every time cut. Direct token memory, additional heads, or periodic global attention can add channels beyond this bottleneck.

The complete local response follows from the differential of the recurrence, rather than from the frozen kernel alone. For an arbitrary input perturbation, define $\Xi_r = (\delta M_r) S_{r-1} + \delta B_r$. Then

$$\delta S_t = M_t \delta S_{t-1} + \Xi_t = \Phi_{t:1} \delta S_0 + \sum_{r=1}^{t} \Phi_{t:r+1} \Xi_r, \tag{7.42}$$

$$\delta \widetilde{o}_t = (\delta S_t)^{\mathsf{T}} q_t + S_t^{\mathsf{T}} \delta q_t. \tag{7.43}$$

The δB_r terms include value and write changes, the δM_r terms include retention and key changes, and the last term includes query changes. Local convolutions and upstream token maps enter through these variations. Equations 7.42–7.43 are the KDA block inside $\mathbb{J}_C^{\mathrm{lat}}$; composing them with $\mathsf{E}_{i\leftarrow j,C}$ and $\mathsf{R}_{i\leftarrow j,C}$ gives the local categorical response in Eq. 7.21.

The affine form also identifies the precise source of order sensitivity. If token x induces $F_x(S) = M_xS + B_x$, then

$$(M_2, B_2) \circ (M_1, B_1) = (M_2M_1,\ M_2B_1 + B_2)\,. \tag{7.44}$$

This affine composition is associative. Reversing two input-conditioned transitions gives

$$F_b(F_a(S)) - F_a(F_b(S)) = [M_b, M_a]S + (M_b - I)B_a - (M_a - I)B_b. \tag{7.45}$$

The first term is a linear commutator. The remaining terms record the order in which writing and forgetting occur. Affine transitions can consequently distinguish order even when a particular pair of linear factors commutes. The native object is the time-ordered affine semigroup in Eq. 7.44.

Equations 7.33 and 7.39 expose two realizations of relative transport:

$$q_t^{\mathsf{T}} R_{t-s} k_s \quad \longleftrightarrow \quad q_t^{\mathsf{T}} \mathcal{P} \prod_{r=s+1}^{t} M_r k_s. \tag{7.46}$$

The left side is a fixed, stationary, orthogonal coordinate transport. The right side is content-dependent, dissipative, and path ordered. In the special case $M_r = \alpha I$,

$$w_{t\leftarrow s} = \beta_s \alpha^{t-s} q_t^{\mathsf{T}} k_s, \tag{7.47}$$

so the recurrent construction reduces to an exponential relative-distance kernel. For variable M_r, finite-horizon memory depends on singular values of $\Phi_{t:s}$ and nonnormal transients. These channel-state spectra differ from positional Laplacian spectra and categorical response spectra.

7.4.3 What NoPE establishes

Kimi K3 combines KDA recurrence and causal short convolutions with periodic global Gated MLA layers whose queries and keys use no explicit positional embedding [35]. The global layers still receive states shaped by ordered computation. They also retain addressable token memory, supplying paths outside the fixed-state cross-cut bound in Proposition 7.4.

Position use should be audited at three levels:

$$\begin{aligned} \text{structural identifiability}: &\quad \mathrm{Aut}(A) \text{ and other exact symmetries,} \\ \text{representational observability}: &\quad \text{whether order or displacement is decodable from hidden state,} \\ \text{task identifiability}: &\quad \text{whether data and loss distinguish order-sensitive alternatives.} \end{aligned} \tag{7.48}$$

Earlier controlled studies showed that causal NoPE Transformers can recover position and support length generalization on selected tasks [51, 58]. K3 supplies a large-scale hybrid example, but its joint architectural and training changes do not isolate a universal NoPE mechanism. The relevant claim is that explicit position embeddings are one carrier of order among several.

7.4.4 Protein sequences and MSA axes

The symmetry audit transfers directly to protein models. A bidirectional, fully connected NoPE model satisfying the hypotheses of Proposition 7.3 remains S_L-equivariant and therefore lacks

a carrier of residue order. Protein architectures may supply that carrier through relative chain coordinates, local sequence neighborhoods, bidirectional state propagation, backbone adjacency, or three-dimensional geometry. These choices encode different invariances and should be compared by matched training and intervention. The recurrence index is an aligned token coordinate, not evolutionary time or physical distance; gaps and indels can change the coordinate path length across homologs and therefore belong to the MSA alignment and phylogeny audit.

An MSA has two base geometries. The residue-column axis is ordered along the chain. The homolog-row axis has no meaningful file order: an exchangeable family model should be equivariant to arbitrary row permutations, while a phylogeny-aware model may replace that symmetry by tree-constrained transport. Removing arbitrary row coordinates can therefore be a correct inductive bias even when the column axis still requires chain geometry. The unified architectural record becomes

$$\boxed{\text{support geometry} + \text{edge transport} + \text{state propagator} + \text{interventional response.}} \tag{7.49}$$

Instantaneous attention, recurrent memory, and categorical substitution effects occupy separate coordinates of this record.

7.5 MSA Transformer as amortized family inference

MSA Transformer makes the family sample an explicit neural state rather than compressing it into fixed counts before the network [79]. For an alignment with N homologous sequences and L positions, a layer acts on

$$H_f \in \mathbb{R}^{N\times L\times d}. \tag{7.50}$$

Column attention compares alignment rows at one fixed position,

$$A^{\mathrm{col}}_{i,rn} = \mathrm{softmax}_n\left(\frac{q_{ri}^{\mathsf{T}} k_{ni}}{\sqrt{d}}\right), \qquad y^{\mathrm{col}}_{ri} = \sum_n A^{\mathrm{col}}_{i,rn} v_{ni}, \tag{7.51}$$

whereas tied row attention aggregates evidence across homologues to produce one position route shared by all rows, schematically

$$S^{\mathrm{row}}_{ij} = \frac{1}{N}\sum_{r=1}^{N} \frac{q_{ri}^{\mathsf{T}} k_{rj}}{\sqrt{d}}, \qquad A^{\mathrm{row}}_{ij} = \mathrm{softmax}_j\, S^{\mathrm{row}}_{ij}, \qquad y^{\mathrm{row}}_{ri} = \sum_j A^{\mathrm{row}}_{ij} v_{rj}. \tag{7.52}$$

Thus the native computation contains two graph families: an N-node sequence graph for each position and an L-node position graph conditioned on the complete MSA. Axial factorization reduces full attention over NL cells from $O(N^2L^2)$ to $O(LN^2 + NL^2)$ per layer. Alternating the axes still permits information to travel from (r, i) to (n, j) through two-hop paths, but it is an architectural factorization rather than an exact equality with unrestricted joint attention.

The cross-family commonality is most cleanly expressed hierarchically. Let Θ_f denote the family-specific constraint system—fields, typed couplings, structural organization, and functional restrictions—and let $\mathcal{T}_f$ denote phylogeny and sampling. A schematic population model is

$$\Theta_f \sim P_\star(\Theta), \qquad \mathcal{M}_f \sim p(\mathcal{M} \mid \Theta_f, \mathcal{T}_f). \tag{7.53}$$

Families do not share one coupling graph. They share recurring amino-acid response motifs, structural graph organization, conservation and compensation patterns, and the statistical task of separating constraint from finite depth, gaps, subfamilies, and common ancestry. Training across many

alignments learns an amortized inference rule

$$\boxed{\mathcal{A}_\theta : \mathcal{M}_f \longmapsto \widehat{\Theta}_f \text{ or directly to family-conditioned predictions.}} \tag{7.54}$$

Here $\widehat{\Theta}_f$ is a conceptual audit coordinate, not necessarily an explicit stored model variable. The current MSA supplies family evidence; the shared parameters supply a cross-family prior over how such evidence should be interpreted.

This gives a precise intermediate position between DCA and a single-sequence pLM:

$$\begin{aligned} \text{Potts / DCA:} \quad & \mathcal{M}_f \xrightarrow{\text{family optimization}} \widehat{J}_f, \\ \text{MSA Transformer:} \quad & (\theta_\star, \mathcal{M}_f) \xrightarrow{\text{forward inference}} G_f(C; \mathcal{M}_f), \\ \text{single-sequence pLM:} \quad & (\theta_\star, x) \xrightarrow{\text{amortized inference}} G(C; x). \end{aligned} \tag{7.55}$$

The first estimates one static family field, the second combines an explicit family sample with a cross-family learned estimator, and the third relies on information already amortized into model parameters. Tied row attention can contain contact-predictive statistics, but its scalar route is not a $q \times q$ Potts block. Column attention is likewise a normalized route over homologues, not a phylogenetic tree. Output substitution audits, clade-aware nulls, and row/column interventions are required to distinguish reusable family constraints from ancestry and sampling shortcuts.

7.6 Softmax over different sample spaces

The Potts conditional in Eq. 4.3 normalizes over residue identities at one position. Attention in Eq. 7.6 normalizes over source positions. Modern Hopfield retrieval normalizes over stored memories,

$$q^{\text{new}} = \Xi \,\text{softmax}_\mu(\beta \Xi^\mathsf{T} q), \tag{7.56}$$

and can be derived from the Lyapunov function

$$E_{\text{MH}}(q) = \frac{1}{2}\|q\|_2^2 - \frac{1}{\beta}\log \sum_\mu e^{\beta \xi_\mu^\mathsf{T} q} + C \tag{7.57}$$

under the assumptions of the modern Hopfield construction [29]. One attention readout matches this update when keys and returned values are tied or compatibly mapped. Untied values, changing memories, residual connections, and feed-forward layers generally define a different dynamics whose energy descent must be established separately.

The bridge between latent-variable models and modern Hopfield retrieval becomes exact after the hidden sample space is specified. Replace the independent Bernoulli hidden vector of an RBM by one categorical memory variable $Z \in [M]$ and define

$$E(q, Z = \mu) = \frac{1}{2}\|q\|_2^2 - \xi_\mu^\mathsf{T} q. \tag{7.58}$$

Its posterior is

$$P(Z = \mu \mid q) = \text{softmax}_\mu(\beta \Xi^\mathsf{T} q), \tag{7.59}$$

and marginalizing Z gives Eq. 7.57 up to a constant. The essential distinction is therefore

$$\boxed{\begin{aligned} \text{Bernoulli RBM:} \quad & \log \sum_{r \in \{0,1\}^M} e^{r^\mathsf{T} s} = \sum_\mu \log(1 + e^{s_\mu}), \\ \text{modern Hopfield:} \quad & \log \sum_{\mu=1}^{M} e^{\beta \xi_\mu^\mathsf{T} q}. \end{aligned}} \tag{7.60}$$

The first permits multiple motifs to be active independently; the second makes memories compete on one simplex. Transformer attention applies the same categorical-memory posterior separately at each target, but its memories are generated by the current sequence rather than stored as fixed parameters. Multi-head attention repeats retrieval in several learned memory spaces.

The three energy-like quantities must therefore remain distinct:

$$\boxed{E_{\text{Potts}}(x) \neq E_{\text{Hopfield}}(q) \neq -\log \alpha_{ij}.} \tag{7.61}$$

They score a sequence configuration, a retrieval state, and one normalized routing coordinate, respectively.

7.7 Latent factors and interaction order

For binary hidden variables r_μ, an RBM energy can be written

$$E_{\text{RBM}}(x, r) = -\sum_i g_i(x_i) - \sum_\mu c_\mu r_\mu - \sum_{i,\mu} W_{i\mu}(x_i) r_\mu. \tag{7.62}$$

Marginalizing the hidden layer gives

$$\log \widetilde{P}_{\text{RBM}}(x) = \sum_i g_i(x_i) + \sum_\mu \log \left[1 + \exp\left(c_\mu + \sum_i W_{i\mu}(x_i) \right) \right]. \tag{7.63}$$

The nonlinear softplus generates one-body, pairwise, and higher-order terms. Its weak-coupling quadratic component is low rank across positions; a Potts approximation additionally requires a negligible higher-order remainder. Protein RBMs can therefore represent extended motifs without identifying every pair touched by a hidden unit as a direct contact [52, 98].

The useful unification is consequently typed: Potts couplings, low-rank Hopfield–Potts patterns, RBM-induced pair components, and pLM substitution responses can all be projected into the same edge-operator coordinates. Equality in those coordinates is then an empirical hypothesis tested after the shared matrix multiplication or softmax has been typed.

7.8 Longitudinal case study: the 2019 unified framework

Dauparas et al.'s 2019 *Unified Framework for Modeling Multivariate Distributions in Biological Sequences* placed PSSMs, Potts models, multivariate Gaussian models, and autoencoders inside one reconstruction diagram [17]. The resulting objective equivalences allowed regularization choices to be transferred between models. They also leave a concrete question for the operator audit: which probability and response objects survive the common implementation?

7.8.1 The common reconstruction diagram

Let an MSA contain N sequences of length L over an alphabet of size q. Flatten the one-hot tensor to $X \in \{0, 1\}^{N \times d}$ with $d = Lq$. A linear reconstruction system has preactivation

$$H = B + XW, \tag{7.64}$$

where B supplies one-site fields and W carries cross-coordinate dependence. The models in the 2019 framework are obtained by changing four choices: the admissible blocks of W, the output loss,

the treatment of self-connections, and the regularizer. Adding hidden layers replaces the single linear map by a nonlinear encoder–decoder while preserving the reconstruction view.

For a PSSM, $W = 0$ and a sitewise categorical softmax learns only $B_i(a)$. For a Potts pseudolikelihood model, the logits at target position i are

$$H_i(a; x_{-i}) = B_i(a) + \sum_{j \neq i} \sum_b x_j(b) W_{j \to i}(b, a), \tag{7.65}$$

and categorical cross-entropy is applied over the residue alphabet at i. Because one-hot targets satisfy $\sum_a x_i(a) = 1$, the loss is exactly the negative sitewise Potts log pseudolikelihood:

$$\mathcal{L}_{\mathrm{CCE}} = -\frac{1}{N} \sum_{n,i} \log \frac{\exp H_i(x_i^{(n)}; x_{-i}^{(n)})}{\sum_a \exp H_i(a; x_{-i}^{(n)})} = \mathcal{L}_{\mathrm{PL}}. \tag{7.66}$$

The equality is an exact identity of objectives after the categorical sample space and the targetwise normalization have been matched.

The Gaussian branch begins from a different law. Center the flattened observations to $\widetilde{X}$ and consider a regularized self-reconstruction objective of the form

$$\mathcal{L}_{\mathrm{MSE}}(W) = \frac{1}{N} \|\widetilde{X} - \widetilde{X} W\|_F^2 + \lambda \|W\|_F^2 - 2\gamma \operatorname{tr}(W). \tag{7.67}$$

Writing $C = N^{-1} \widetilde{X}^{\mathsf{T}} \widetilde{X}$, its stationary solution is

$$W = (\gamma - \lambda)(C + \lambda I)^{-1} + I. \tag{7.68}$$

After the identity map is separated, the learned cross-coordinate map is therefore an affine transform of a regularized precision matrix. With the corresponding choice of γ, the MSE system and Gaussian likelihood share the same global optimum, even though one is written as a neural self-map and the other as covariance inversion.

The resulting family resemblance can be summarized without erasing the differences:

Model	Reconstruction map	Objective	Native probability object	First semantic remainder
PSSM	bias field only	sitewise CCE	independent categorical sites	no represented pair edge
Potts/MRF	linear cross-position blocks	categorical CCE = pseudolikelihood	compatible categorical conditionals, when parameters are tied into one law	raw block retains gauge freedom; contact is a separate endpoint
Gaussian/MG	centered linear self-map	regularized MSE	continuous Gaussian law through precision	one-hot simplex and categorical contrasts are approximated
Autoencoder	factorized or nonlinear reconstruction	chosen output loss	generally a conditional predictor unless a joint law is supplied	hidden paths can induce higher-order final responses

7.8.2 Regularization transfer and the remaining distinctions

The contact-prediction experiment tested the equivalence by transferring regularization. The MRF literature commonly used L_2 penalties, while Gaussian DCA often used covariance pseudocounts. Transferring MRF-style shrinkage to the MSE formulation improved contact prediction over GaussDCA; removing the diagonal self-map gave a further small improvement and connected the score to partial correlation. Categorical cross-entropy still performed better, but the gap narrowed under more closely matched regularization. The fitted matrices were scalarized, average-product corrected, and ranked against structural contacts, so the experiment compared their contact-prediction utility after those operations.

Three distinctions remain after the objectives have been matched. Gaussian reconstruction uses continuous coordinates, whereas Potts pseudolikelihood normalizes categorical identities. A generic CCE predictor defines one joint pair energy only when its directed blocks satisfy the reciprocity condition in Eq. 4.34. Finally, centering removes one-body gauge directions from the visible blocks, while hidden nonlinear layers require an audit of the complete output response because they can introduce higher-order dependence.

The reconstruction diagram thus supplies transferable estimation procedures. The categorical operator audit supplies the edge object to compare after fitting. The next case follows the corresponding changes in graph-network computation.

7.9 A second longitudinal case: from spectral filters to geometric messages

Graph-network models range from fixed spectral filters and local propagation to adaptive routes, typed messages, and geometric maps. The following sequence keeps the node and channel axes explicit so that these computations can later be compared with AlphaFold's pair state.

7.9.1 Spectral convolution is functional calculus on a declared graph

Let A be a symmetric nonnegative adjacency matrix, $D_{ii} = \sum_j A_{ij}$, and

$$L_{\mathrm{sym}} = I - D^{-1/2} A D^{-1/2} = U \Lambda U^{\mathsf{T}} \tag{7.69}$$

be the normalized graph Laplacian. The columns of U replace Fourier modes: $U^{\mathsf{T}} x$ are the graph-frequency coefficients of a node signal x. A spectral convolution is defined by

$$g_\theta *_G x = U g_\theta(\Lambda) U^{\mathsf{T}} x = g_\theta(L_{\mathrm{sym}}) x. \tag{7.70}$$

Low values of λ describe signals that vary slowly across heavily weighted edges, so a low-frequency filter smooths the node signal relative to the declared graph [6, 18]. A covalent, contact, or learned-similarity graph gives the same calculus a different meaning. An unrestricted spectral multiplier is generally dense in node coordinates; locality requires a restricted multiplier.

7.9.2 Chebyshev filters convert frequency response into finite-hop propagation

ChebNet replaces the unrestricted spectral multiplier by a polynomial. Rescale the spectrum to $[-1, 1]$ with

$$\widetilde{L}_{\mathrm{sym}} = \frac{2}{\lambda_{\max}} L_{\mathrm{sym}} - I \tag{7.71}$$

and use Chebyshev polynomials T_k:

$$g_\theta(L_{\text{sym}})x \approx \sum_{k=0}^{K} \theta_k T_k(\widetilde{L}_{\text{sym}})x. \tag{7.72}$$

Because a degree-k polynomial in L_{sym} connects only nodes reachable within k graph steps, a degree-K filter is K-hop localized. Spectral approximation has become a spatial propagation rule without ceasing to be a function of the Laplacian.

The graph supplies the support, while shared coefficients combine path lengths and spectral scales.

7.9.3 The classical GCN is a tied first-order filter

The familiar GCN layer follows from a strong truncation of Eq. 7.72 [59]. Take $K = 1$ and approximate $\lambda_{\max} \simeq 2$. Since

$$\widetilde{L}_{\text{sym}} = L_{\text{sym}} - I = -D^{-1/2}AD^{-1/2}, \tag{7.73}$$

the one-channel filter becomes

$$\theta_0 x - \theta_1 D^{-1/2}AD^{-1/2}x. \tag{7.74}$$

Tying $\theta_0 = -\theta_1 = \theta$ gives

$$\theta\left(I + D^{-1/2}AD^{-1/2}\right)x. \tag{7.75}$$

The subsequent renormalization trick adds self-loops before normalization:

$$\widehat{A} = A + I, \qquad \widehat{D}_{ii} = \sum_j \widehat{A}_{ij}, \qquad S = \widehat{D}^{-1/2}\widehat{A}\widehat{D}^{-1/2}. \tag{7.76}$$

For a feature matrix $H^{(\ell)}$, the layer is

$$H^{(\ell+1)} = \sigma\left(SH^{(\ell)}W^{(\ell)}\right). \tag{7.77}$$

Equation 7.77 should be read as two operators with different axes:

$$H \xrightarrow{S} \text{mix node states along the declared graph} \xrightarrow{W} \text{mix feature channels.} \tag{7.78}$$

The propagation matrix S is fixed once the input graph is fixed; the trainable matrix W is shared over nodes. Standard GCN therefore learns channel transformations around a prescribed, degree-normalized node-mixing rule, with edgewise freedom limited by the input graph.

The self-loop retains the node's current state. Degree normalization controls message scale; its rows need not sum to one, and it is not a probability law over molecular configurations.

7.9.4 Depth reveals the diffusion assumption

Ignoring nonlinearities and channel maps for a moment, stacking layers repeatedly applies S:

$$H^{(K)} \approx S^K H^{(0)}. \tag{7.79}$$

If $S = U\,\mathrm{diag}(\mu_r)U^{\mathsf{T}}$, then

$$S^K = U\,\mathrm{diag}(\mu_r^K)U^{\mathsf{T}}. \tag{7.80}$$

Modes with $|\mu_r| < 1$ decay, while dominant smooth modes survive. Under the usual connectivity and aperiodicity conditions, node representations within a component become increasingly aligned with a small stationary subspace. This is the linear skeleton of over-smoothing. Nonlinearities, learned channels, and residual paths alter the exact limit, while the spectral pressure created by repeated propagation remains.

The phenomenon reveals the inductive bias encoded by depth. GCN assumes that neighbors should exchange and often homogenize information. That is useful under homophily. If an edge instead connects complementary types, opposite labels, or directed roles, indiscriminate low-pass smoothing can destroy the signal of interest. Adjacency grants permission to communicate; similarity requires an additional assumption about the signal and edge semantics.

APPNP and GCNII counter repeated smoothing by retaining the initial signal [61, 7]. GraphSAGE instead formulates propagation as a permutation-invariant neighborhood aggregator, with sampling as an additional estimation step; GIN studies which neighborhood multisets survive aggregation [49, 101]. These refinements concern propagation and information retention. The next transition changes the route itself.

7.9.5 GAT makes the route depend on the current state

Graph attention replaces degree-determined coefficients by content-dependent scores [38]:

$$\begin{aligned} e_{ij}^{(\ell)} &= a_\theta\Big(Wh_i^{(\ell)}, Wh_j^{(\ell)}, e_{ij}\Big), \\ \alpha_{ij}^{(\ell)} &= \frac{\exp e_{ij}^{(\ell)}}{\sum_{k\in\mathcal{N}(i)} \exp e_{ik}^{(\ell)}}, \\ h_i^{(\ell+1)} &= \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(\ell)} W h_j^{(\ell)}\right). \end{aligned} \tag{7.81}$$

The support may remain fixed while the weighted route becomes $P_\theta(H, E)$. Its rows normalize over allowed sources, so the route is generally directed and state dependent. Its coefficients have the same computational role as Transformer attention weights; a categorical substitution response requires the full input–output audit.

7.9.6 MPNNs promote the edge from a coefficient to a typed computation

The message-passing neural network abstraction separates message, aggregation, and update [47, 39]:

$$\begin{aligned} m_{ij}^{(\ell)} &= \phi_m^{(\ell)}\left(h_i^{(\ell)}, h_j^{(\ell)}, e_{ij}\right), \\ m_i^{(\ell)} &= \bigoplus_{j\in\mathcal{N}(i)} m_{ij}^{(\ell)}, \\ h_i^{(\ell+1)} &= \phi_u^{(\ell)}\left(h_i^{(\ell)}, m_i^{(\ell)}\right). \end{aligned} \tag{7.82}$$

Classical GCN is recovered by choosing a linear isotropic message such as $m_{ij} = S_{ij} h_j W$. General MPNNs can condition on bond type, distance, orientation, molecular identity, or a hidden edge vector. The edge is no longer exhausted by one scalar coefficient; it may implement a map between typed node fibers.

Depth also changes interaction order. A two-layer update at i can depend on paths $k \to j \to i$ and therefore on configurations of several nodes even when every primitive message is pair-indexed. Pairwise execution support can compose into higher-order output dependence across layers.

7.9.7 Geometric networks add transformation law, not molecular energy

For molecular systems the node state may include coordinates $r_i \in \mathbb{R}^3$. A simple invariant message can depend on squared distance,

$$m_{ij} = \phi_m(h_i, h_j, e_{ij}, \|r_i - r_j\|^2), \tag{7.83}$$

while an equivariant coordinate update has the schematic form

$$r_i' = r_i + \sum_j (r_i - r_j)\phi_x(m_{ij}). \tag{7.84}$$

Under a rigid transformation $r_i \mapsto Qr_i + t$, invariant scalar messages remain unchanged and the displacement in Eq. 7.84 transforms by Q. Consequently

$$F(QR + t) = QF(R) + t \tag{7.85}$$

for the coordinate map under the stated architecture [84]. More general geometric networks carry vectors, tensors, or irreducible rotation representations and type every message by its transformation law.

Equivariance licenses a symmetry statement. Energy, momentum conservation, and physical time evolution additionally require integrability, conservation laws, units, and a dynamical model. The same distinction applies to invariant point attention and conditional diffusion in Chapter 10; Section 10.6 develops the latter transport law. Coordinate generation also need not impose equivariance at every layer. AF3 uses unconstrained coordinate projections and attention, together with rigid-motion augmentation; Section 10.4.5 separates this learned handling of orientation from an exact transformation law. Diffusion and architectural equivariance are independent design choices.

7.9.8 Two diffusions that share a word but not a state space

Repeated GCN propagation and a coordinate diffusion model are easily conflated because both use diffusion language. Their operators act on different objects:

$$\begin{aligned} \text{graph-signal diffusion:} \quad & H^{(k+1)} = SH^{(k)}, \quad S : \mathcal{H}_{\text{node}} \to \mathcal{H}_{\text{node}}, \\ \text{generative diffusion:} \quad & \mathrm{d}R_t = f(R_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \\ & p_t \text{ is a density on coordinate configurations.} \end{aligned} \tag{7.86}$$

The first smooths features relative to a declared graph. The second smooths a probability density in a configuration space and learns its score for reverse transport. A graph Laplacian can appear inside a generative model. The state space and generator determine which diffusion is meant and which physical interpretation is available.

7.9.9 The operator progression

The transitions can be recorded in one line:

$$\boxed{\begin{aligned} g_\theta(L_{\text{sym}}) &\longrightarrow \sum_k \theta_k L_{\text{sym}}^k \longrightarrow SHW, \\ &\longrightarrow P_\theta(H)HW \longrightarrow \phi_m(h_i, h_j, e_{ij}), \\ &\longrightarrow \text{typed equivariant edge maps.} \end{aligned}} \tag{7.87}$$

Spectral filters and GCNs share channel maps around a declared positional operator. Attention makes the route adaptive; MPNNs make the edge computation richer; geometric networks constrain its transformation law. AlphaFold adds persistent pair state and triangle composition. The comparison therefore follows which relations carry state and how that state reaches the output.

Coda: correspondence with a remainder

The model portraits and the unified-framework case reveal a recurring pattern. Two systems often share one exact layer of mathematics and diverge at the next. Potts and Gaussian models share a notion of conditional coupling but inhabit different state spaces. Hopfield retrieval and attention can share a softmax update while differing in memory persistence and value tying. RBMs and low-rank pair models share a weak-coupling component while the marginalized RBM retains higher orders. Frozen attention maps lie in a shared head span, while the full network response also travels through queries, keys, residual paths, and nonlinear blocks.

The productive comparison is an explicit map into common operator coordinates, accompanied by the terms, assumptions, and dynamics left in its remainder. The shared coordinates state the exact correspondence; the remainder states where the systems diverge. Once both are fixed, empirical agreement becomes a meaningful question.

The unified-framework case showed how an exact reconstruction equivalence can transfer an estimator or regularizer while leaving probability semantics plural. Section 7.9 traces a complementary graph-network lineage from functional calculus to local propagation, adaptive routing, typed messages, and geometric transformation laws. Both histories locate where complexity moved and which scientific claims remained open.

Chapter 8 next separates the generative tree on MSA rows from the interaction graph on residue positions. Chapter 9 then declares the observations and estimands used to infer an operator graph, and Chapter 10 applies that ledger to three generations of relational state.

Chapter 8

Phylogeny as Tree-Constrained Transport

Phylogeny is not merely a correction applied after an MSA statistic has been computed. It is a distinct generative graph on the sequence axis. Its vertices include latent ancestors, its edges carry evolutionary transition kernels, and its shared topology makes the observed alignment rows dependent. This graph must be separated from the residue-position graph, whose edges describe statistical or functional relations within a sequence, and from an MSA Transformer's row route, which is generated between observed sequences by a learned computation. The distinction supplies the missing type information behind the informal statement that an MSA contains both constraint and common ancestry.

The proof-obligation ledger in Section 2.9 supplies the standard for this separation. A phylogeny-aware interaction claim needs a generative row-tree law as its reference, a connected residual on the within-sequence position space, and a quantitative error connecting that residual to a closed structural or functional object.

8.1 One aligned site on two four-leaf trees

For one alignment column, a phylogenetic model assigns a categorical state to every vertex of a tree. The leaves A, B, C, D are observed homologous sequences. Internal vertices are latent ancestors whose residue states are summed out. A biological root, when identified, orients descent from an ancestor; under a reversible likelihood model, a computational root may simply orient the recursion. Each edge carries a substitution transition kernel. Its branch length usually records expected substitutions per site under the declared model, which need not equal calendar time or raw percentage difference.

Two four-leaf topologies expose the role of shared history:

$$\begin{aligned} &\text{star:} && r \longrightarrow \{A, B, C, D\}, \\ &\text{binary:} && r \longrightarrow \{u, v\}, \qquad u \longrightarrow \{A, B\}, \quad v \longrightarrow \{C, D\}. \end{aligned} \tag{8.1}$$

The *patristic distance* d_{ab} is the sum of branch lengths along the unique path between leaves a and b. Under a stationary reversible single-site process, let φ be a centered eigenfunction whose signal decays at rate $\lambda > 0$. Then

$$\operatorname{Cov}(\varphi(X_a), \varphi(X_b)) = \operatorname{Var}_\pi(\varphi) e^{-\lambda d_{ab}}. \tag{8.2}$$

This formula is a transparent special case: leaf correlation follows the shared path geometry and the substitution process together.

For a star with root-to-leaf length t, every leaf pair has distance $2t$ and the same covariance in Eq. 8.2. For a balanced binary tree with root-to-internal length a, internal-to-leaf length b, and $a + b = t$, the within-clade distances are $d_{AB} = d_{CD} = 2b$, whereas the cross-clade distances are $2t$. The binary tree therefore creates stronger correlation within recent clades while retaining weaker correlation across them. With unequal branch lengths or another substitution process, the comparison changes; topology alone does not order all correlations.

Correlated leaves contain less independent information than the same number of independent rows. Under matched root-to-tip lengths, a star lacks the additional recent clade structure of the binary example and can yield a larger effective sample size for estimating column frequencies and covariation. This improves the visibility of a fixed structural or functional constraint through reduced phylogenetic redundancy. It does not increase the underlying residue coupling.

Selection pressure is a property of the genotype-to-demography map, population process, and substitution rates. A star or binary topology records the branching history and, together with branch lengths, the dependence among sampled leaves. No ranking of selection pressure follows from topology by itself. The basic site-independent phylogenetic model applies this construction separately to each alignment column on a shared tree; residue-interaction models add dependence between columns, and recombination or gene-tree discordance may make the tree vary along the sequence.

8.2 Why a tree is a special computational structure

Let $T = (V, E)$ be a tree with observed leaves $\mathcal{L}$, latent internal vertices $\mathcal{I} = V \setminus \mathcal{L}$, and a root chosen to orient the computation. A tree is not only a sparse graph. It has no cycles, every pair of vertices is joined by one path, and deleting any edge separates it into exactly two components. If p is the parent of u, let D_u be the descendant component containing u after (p, u) is deleted. The Markov property on the tree gives

$$X_{D_u} \perp\!\!\!\perp X_{V \setminus (D_u \cup \{p\})} \mid X_p. \tag{8.3}$$

Thus the single categorical variable X_p separates the complete descendant subtree from the rest of the model. In graphical-model language the separator has size one and the tree has treewidth one.

This separator property is the source of the algorithm. All observations below u can be summarized for the rest of the tree by the q numbers

$$m_{u \to p}(a) = P(x_{\mathcal{L} \cap D_u} \mid X_p = a), \qquad a \in [q]. \tag{8.4}$$

The size of this interface does not grow with the number of descendants. Dynamic programming, belief propagation, variable elimination, and contraction of a tree tensor network are four descriptions of this same compression. On a graph with cycles, eliminating a node generally creates larger separators and higher-order intermediate factors; exact cost is then governed by treewidth rather than vertex count alone.

8.3 The classical recursion is exact message passing on a tree

For one aligned site, let Q be a continuous-time substitution generator on q categorical states. An oriented edge $e = (p, u)$ of length t_e carries the Markov operator

$$M_e = \exp(t_e Q), \qquad M_e(a, b) = P(X_u = b \mid X_p = a). \tag{8.5}$$

Let $\phi_u(b)$ be the observation factor: it is one at an unobserved internal vertex and is the indicator or ambiguity distribution of the observed category at a leaf. Conditional independence between child subtrees gives the Felsenstein message

$$m_{u\to p}(a) = \sum_{b=1}^{q} M_{(p,u)}(a,b)\,\phi_u(b) \prod_{v\in \mathrm{ch}(u)} m_{v\to u}(b). \tag{8.6}$$

The site likelihood is

$$p(x_{\mathrm{leaf}} \mid T, Q, t) = \sum_{a=1}^{q} \pi(a)\phi_r(a) \prod_{v\in \mathrm{ch}(r)} m_{v\to r}(a), \tag{8.7}$$

where r is the root and π its state distribution. For a fixed tree this sum–product recursion evaluates the likelihood in $O(|V|q^2)$ time per site rather than summing explicitly over all ancestral assignments [44]. Its age is therefore not evidence of a missing neural replacement: it is already the exact dynamic program for the declared tree model.

Equation 8.6 also locates phylogeny inside the typed framework. The edge map $M_{(p,u)}$ transports a categorical function from the child fiber to the parent fiber, while the product at u combines conditionally independent descendant evidence. This is a Graph Network on an augmented vertex set with latent nodes, but it is not a residue interaction graph and not a message-passing layer over only the observed MSA rows.

8.4 Tree-conditioned inference and tree identification

Two computational problems are often compressed into the phrase "phylogenetic inference":

$$\boxed{\begin{array}{ll} \text{inference on a tree:} & (T,Q,t) \longmapsto p(x_{\mathcal{L}} \mid T,Q,t), \\ \text{inference of a tree:} & x_{\mathcal{L}} \longmapsto \widehat{T},\widehat{Q},\widehat{t} \text{ or } p(T,Q,t \mid x_{\mathcal{L}}). \end{array}} \tag{8.8}$$

The first problem uses Eq. 8.6 and is exact for the declared model. The second contains a discrete search over topologies and continuous estimation of branch lengths and substitution parameters. Its parameter space has the schematic form

$$\mathfrak{M}_R = \coprod_{T\in\mathcal{T}_R} \left(\mathbb{R}_{>0}^{|E(T)|} \times \mathcal{Q}\right), \tag{8.9}$$

a union of topology-specific continuous regions whose boundaries meet when branches contract. Ordinary gradient descent operates within one continuous region; a topology move, relaxation, or sampling rule is needed to cross between them.

The number of unrooted binary topologies on R labeled leaves is

$$N_R = (2R-5)!!, \tag{8.10}$$

so maximum-likelihood and Bayesian procedures must search or sample a vast outer space. UPGMA, neighbor joining, parsimony, local NNI or SPR moves, stochastic search, and tree-space MCMC are different responses to this topology problem. Modern implementations such as IQ-TREE 2 improve search, substitution models, model selection, bootstrap approximations, and parallel evaluation while retaining the inner likelihood recursion [81, 26].

Rooting supplies a further type distinction. If the substitution process is reversible,

$$\pi(a)M_t(a,b) = \pi(b)M_t(b,a), \tag{8.11}$$

the likelihood can be evaluated by placing the computational root anywhere on the same unrooted tree. The chosen root orients the dynamic program but is not thereby statistically identified. A molecular clock, an outgroup, temporal samples, or a declared nonreversible process supplies additional information for evolutionary direction. Computational direction and biological time direction must therefore be audited separately.

8.5 Three graphs occupy three different base spaces

For an MSA $X \in [q]^{N\times L}$, three graph constructions must be kept separate:

$$\boxed{\begin{aligned} T_f &: \text{ancestral vertices and evolutionary branches,} \\ \mathcal{G}_{\text{pos}} &: \text{residue positions and typed categorical interactions,} \\ A_i^{(\ell,h)}(X) &: \text{observed MSA rows and learned attention routes.} \end{aligned}} \tag{8.12}$$

The first usually shares one topology across sites under the basic model, although recombination or gene-tree discordance can require local trees. The second acts within a sequence and may encode structural or functional constraint. The third is dense, layer- and input-dependent, generally directed after normalization, and defined only on observed homologues.

The associated operators are also different. A tree Laplacian acts on ancestral and observed vertices. The generator Q in Eq. 8.5 acts on categorical states. A position Laplacian acts on residue positions after a position graph has been declared. A Kronecker-sum construction can combine such axes under special separability assumptions, but their spectra are not interchangeable. In particular, a leading singular vector across alignment rows does not become a residue interaction mode merely because both are low rank.

An additive tree metric d_{rs} satisfies the four-point condition: for every four leaves, the two largest members of

$$d_{rs} + d_{uv}, \qquad d_{ru} + d_{sv}, \qquad d_{rv} + d_{su} \tag{8.13}$$

are equal. A generic attention-derived dissimilarity need not satisfy this condition, share a topology across positions, compose along paths, or correspond to a Markov transition kernel. Sequence similarity or clade-like clustering in column attention is therefore evidence of useful row structure, not identification of a phylogenetic tree.

8.6 Static constraint and evolutionary dynamics are compatible

Calling a Potts model static and phylogeny dynamic refers to two different coordinates. A fitted Potts Hamiltonian

$$E_{\Theta_f}(x) = -\sum_i h_i(x_i) - \sum_{i<j} J_{ij}(x_i, x_j) \tag{8.14}$$

is one family-level constraint field shared across MSA rows. Evolution is a stochastic process that transports sequences along the branches of T_f. A hierarchical generative statement is therefore

$$\Theta_f \sim P_\star(\Theta), \qquad (T_f, t_f) \sim P_{\text{tree}}, \qquad X_f \sim p(X \mid T_f, t_f, \Theta_f). \tag{8.15}$$

The static field and the dynamic genealogy are not competitors: the first specifies which sequence configurations are favored, while the second specifies how correlated observations were produced.

There is nevertheless a missing arrow in common practice. A Gibbs distribution proportional to $\exp[-E_{\Theta_f}(x)]$ is an equilibrium law, not an evolutionary transition kernel. To evolve complete sequences on a tree one must additionally specify rates

$$\mathcal{Q}_{\Theta_f}(x, x') \quad \text{with, for example,} \quad e^{-E_{\Theta_f}(x)}\mathcal{Q}_{\Theta_f}(x, x') = e^{-E_{\Theta_f}(x')}\mathcal{Q}_{\Theta_f}(x', x), \tag{8.16}$$

if reversible evolution with the Potts equilibrium is intended. This generator acts on a q^L-state sequence space and is generally intractable. Classical site-independent phylogenetics regains tractability by factorizing the edge law across sites; a Potts model regains tractability by describing an equilibrium or using pseudolikelihood rather than exact evolutionary histories. Neither approximation contains the other.

This separation exposes the central identifiability problem. Correlated residues can arise from shared constraint, while correlated rows arise from shared ancestry; finite and taxonomically biased sampling couples the two in the empirical MSA. Sequence weighting changes an estimator but does not identify the evolutionary process, and a high-scoring Potts block need not be free of phylogenetic contribution [73, 22].

8.7 What modern phylogenetic learning actually replaces

Recent learning methods can be classified by the coordinate they change rather than by the broad label "deep phylogenetics":

1. *Posterior representation.* Variational Bayesian phylogenetic inference uses subsplit Bayesian networks to represent distributions over topologies; normalizing flows enrich the branch-length posterior. The phylogenetic likelihood remains the target [104, 103].

2. *Tree-space parameterization.* GeoPhy embeds topology distributions into continuous geometric coordinates, while Dodonaphy combines hyperbolic embeddings with a differentiable soft-NJ decoder. These methods make optimization differentiable without turning a generic dense matrix into a tree [71, 66].

3. *Combinatorial sampling.* PhyloGFN trains a generative flow network to sample diverse high-reward or posterior-weighted trees, replacing part of topology exploration rather than the biological definition of likelihood [105].

4. *Likelihood-free amortization.* Phyloformer learns evolutionary distance estimation and reconstruction from simulated alignments. It can be faster and can absorb richer simulated dependencies, but its conclusions inherit the training simulator and distribution-shift problem [72].

Thus the modern frontier is not one universal successor to neighbor joining or pruning. It is a collection of learned proposal distributions, posterior approximations, continuous relaxations, and simulator-trained estimators attached to different stages of the classical pipeline.

8.8 MSA attention is an amortized mixed signal

An MSA Transformer is trained to predict masked residues, not to recover (T_f, Q, t_f). Its column attention may exploit close homologues, clades, subfamilies, gaps, or taxonomic redundancy whenever these improve reconstruction. Its tied row attention may simultaneously collect conservation

and coevolutionary evidence across the same rows. Consequently the learned representation is more accurately described as

$$H_f = \mathcal{A}_\theta\big(X_f;\ \text{constraint}, \text{ancestry}, \text{sampling}, \text{gaps}\big), \tag{8.17}$$

where the semicolon lists latent causes, not supplied labels. Successful masked reconstruction does not by itself identify their decomposition.

The useful comparison with classical phylogenetics is therefore not "old tree versus modern attention." Classical phylogenetics declares a tree and substitution law, then performs exact latent-state marginalization conditional on them. MSA Transformer learns a cross-family rule that compresses several latent causes into hidden states and dynamic routes. It is more flexible but less identified. An explicit hybrid would supply a tree-conditioned row operator and reserve a learned residual route for dependence not explained by that tree:

$$H^{(\ell+1)} = \Phi_\theta\Big(H^{(\ell)}, \mathcal{K}_{T_f,Q,t_f} H^{(\ell)}, A^{(\ell)}_{\theta,\mathrm{res}}(X_f) H^{(\ell)}\Big). \tag{8.18}$$

Equation 8.18 is an architectural hypothesis, not a uniquely identified additive decomposition. It becomes scientific only when the tree component and residual are tested by interventions or matched nulls.

8.9 A phylogeny-aware operator audit

A defensible audit should preserve the distinction in Eq. 8.12:

1. infer or sample trees with uncertainty rather than naming leading row principal components "phylogeny" from magnitude alone;
2. simulate an independent-site or declared epistatic process along the fitted tree while matching depth, gaps, composition, and taxonomic sampling;
3. pass real and null MSAs through the identical operator-extraction pipeline and compare full typed blocks before scalar compression;
4. evaluate leave-clade-out transfer and clade-preserving interventions, because arbitrary row shuffling destroys both ancestry and the sampling structure one intends to control;
5. report sensitivity across plausible topologies, branch lengths, substitution models, and product versus empirical background measures.

The resulting null-relative quantity

$$\Delta G_{i\leftarrow j} = G^{\mathrm{real}}_{i\leftarrow j} - \mathbb{E}_{\widetilde{X}\sim Q_{T_f}} G_{i\leftarrow j}(\widetilde{X}) \tag{8.19}$$

asks which operator structure exceeds that generated by the declared genealogy. It does not prove physical contact or causal epistasis; those remain separate validation coordinates. Its advantage is narrower and more rigorous: common ancestry is represented as a generative row-graph rather than removed by an unlabeled spectral heuristic.

Chapter 9

From Observations to Operator Graphs

An interaction graph is never observed in the same way that a sequence is observed. An MSA is a tensor of residues indexed by sequence, position, and category. A structure is an ensemble of coordinates under an experimental and modeling pipeline. A pLM response is produced by a chosen intervention and a chosen set of sequence backgrounds. Each supplies observations from which a particular operator graph can be estimated.

This distinction becomes critical in an MSA because its three axes support different notions of variation. Modes across rows can reflect ancestry, taxonomy, or subfamilies. Modes across positions can describe collective organization along the sequence. Categorical constant modes are removed by the local gauge at each position. Operations on these three axes generally survive one another because estimation, inversion, pair projection, scalarization, and spectral decomposition have a noncommuting order.

The chapter compares two principal estimation routes. A family Potts model uses many homologous sequences to estimate one static pair field. A pretrained pLM uses amortized information to produce a background-dependent response from one query sequence, after which a family or another declared background law can be used to estimate its mean and variance. Common reference-centered coordinates make the results comparable while the estimand ledger records their distinct targets.

The difference between the two estimates can be decomposed into Potts estimation error, pairwise model misspecification, pLM transfer or response gap, and response-sampling error. This ledger is more informative than one correlation coefficient. It shows where disagreement can enter and which source of disagreement an additional dataset could resolve.

Finally, evidence must be relative to a null that preserves the nuisance structure under discussion. Sequence reweighting, phylogenetic projection, APC, and spectral mode removal each encode a specific nuisance hypothesis. Their meaning comes from a declared estimand and a null ensemble that justifies what is being removed.

9.1 MSA observations and interaction operators occupy different spaces

The one-hot encoding of an alignment with N rows belongs to an observation space with three axes,

$$X \in \mathcal{H}_{\text{seq}} \otimes \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{cat}}. \tag{9.1}$$

This tensor belongs to the observation space, while $\mathcal{H}^0 = \bigoplus_i \mathcal{H}_i^0$ is the operator domain. The sequence axis indexes correlated samples; position and category index the object estimated from those samples. A phylogenetic mode across MSA rows, a graph eigenvector across residue positions, and the constant categorical mode removed by P_i therefore live in three different spaces.

Let $w_n \geq 0$ with $\sum_n w_n = 1$, and define the weighted site marginal

$$p_i(a) = \sum_{n=1}^{N} w_n \mathbf{1}[X_i^{(n)} = a]. \tag{9.2}$$

With smoothing for unsupported states, the centered whitened one-hot tensor is

$$Y_{nia} = \sqrt{w_n}\, \frac{\mathbf{1}[X_i^{(n)} = a] - p_i(a)}{\sqrt{p_i(a)}}. \tag{9.3}$$

Singular vectors of an unfolding of Y along the sequence axis describe between-sequence variation that may reflect ancestry, subfamilies, taxonomic imbalance, or function. Position interaction modes arise only after an interaction operator has been estimated.

If U_{phy} has orthonormal columns in the square-root-weighted row coordinates of Y and contains sequence directions declared nuisance by an external phylogeny, replicate null, or preregistered rule, define

$$R_{\mathrm{phy}} = I - U_{\mathrm{phy}} U_{\mathrm{phy}}^{\mathsf{T}}, \qquad Y_{\mathrm{res}} = R_{\mathrm{phy}} Y. \tag{9.4}$$

Projection is exact relative to the declared subspace. External evidence licenses the phylogeny label; magnitude alone identifies leading principal components.

9.2 Two estimators in one coordinate system

The weighted marginal in Eq. 9.2, after declared smoothing, provides a family reference. A Potts fit estimates one $\widehat{J}_{ij}$ and hence one static block $\widehat{G}^{\mathrm{P}}_{i\leftarrow j} = P_i \widehat{J}_{ij} P_j^{\mathsf{T}}$ per direction [28, 36, 9]. This estimate depends on the fitting objective, regularization, sequence weights, and finite MSA sample. Gauge projection addresses parameter redundancy after those estimation choices have been made.

A pretrained pLM takes a single query sequence at inference and directly yields the conditional logits needed for Eq. 4.6. It has amortized information across its training distribution and can therefore produce a dynamic response without an inference-time MSA. An MSA remains useful for specifying the family references p_i, the family background distribution μ_{ij}, the static mean $\overline{G}_{i\leftarrow j}$, and uncertainty under finite family sampling. The two estimators answer different questions:

$$\begin{aligned} &\text{MSA route:} \quad \{x^{(n)}\}_{n=1}^{N} \longrightarrow \widehat{G}^{\mathrm{P}}_{i\leftarrow j}, \\ &\text{pLM route:} \quad x \longrightarrow G^{\mathrm{pLM}}_{i\leftarrow j}(x_{-ij}). \end{aligned} \tag{9.5}$$

Their comparison contains several estimands. Let $G^{\mathrm{P},*}_{i\leftarrow j}$ be the block of the population minimizer of the declared Potts fitting risk; let $G^{F,(2)}_{i\leftarrow j}$ be the pair component of the true family conditional log score under the same anchor reference; and let $\overline{G}^{\mathrm{pLM}}_{i\leftarrow j} = \mathbb{E}_{C\sim\mu_{ij}} G^{\mathrm{pLM}}_{i\leftarrow j}(C)$. If $\widehat{\overline{G}}^{\mathrm{pLM}}$ is a sampled

response mean, then

$$\widehat{G}^{\mathrm{P}}_{i\leftarrow j} - \widehat{\overline{G}}^{\mathrm{pLM}}_{i\leftarrow j} = \underbrace{\widehat{G}^{\mathrm{P}}_{i\leftarrow j} - G^{\mathrm{P},*}_{i\leftarrow j}}_{\text{Potts estimation}} + \underbrace{G^{\mathrm{P},*}_{i\leftarrow j} - G^{F,(2)}_{i\leftarrow j}}_{\text{pairwise objective or misspecification}} + \underbrace{G^{F,(2)}_{i\leftarrow j} - \overline{G}^{\mathrm{pLM}}_{i\leftarrow j}}_{\text{pLM transfer and response gap}} + \underbrace{\overline{G}^{\mathrm{pLM}}_{i\leftarrow j} - \widehat{\overline{G}}^{\mathrm{pLM}}_{i\leftarrow j}}_{\text{response sampling}}. \tag{9.6}$$

Common reference-centered coordinates make all four terms comparable. On synthetic families with a known generating law, each term can be evaluated separately. In an observed family, $G^{F,(2)}_{i\leftarrow j}$ is unknown, so the two middle terms are not separately identified by the MSA fit and pLM audit alone. Their sum,

$$\left(G^{\mathrm{P},*}_{i\leftarrow j} - G^{F,(2)}_{i\leftarrow j}\right) + \left(G^{F,(2)}_{i\leftarrow j} - \overline{G}^{\mathrm{pLM}}_{i\leftarrow j}\right) = G^{\mathrm{P},*}_{i\leftarrow j} - \overline{G}^{\mathrm{pLM}}_{i\leftarrow j},$$

is the population model–response gap; separating it requires a declared estimator or structural assumption for the true family conditional, together with sensitivity analysis. Agreement between a pseudolikelihood Potts optimum and a pLM response mean remains a testable transfer statement of the framework.

An estimand ledger with exact cancellation

The four-term decomposition turns an aggregate comparison into a ledger of discrepancies. It also exposes agreement produced by cancellation. Let U be one unit-norm direction in the common reference-centered operator space and suppose the four gaps along that direction are

$$\begin{aligned}
\widehat{G}^{\mathrm{P}} - G^{\mathrm{P},*} &= +0.10U, && \text{finite Potts estimation,}\\
G^{\mathrm{P},*} - G^{F,(2)} &= +0.25U, && \text{pairwise objective or misspecification,}\\
G^{F,(2)} - \overline{G}^{\mathrm{pLM}} &= -0.30U, && \text{pLM transfer and response gap,}\\
\overline{G}^{\mathrm{pLM}} - \widehat{\overline{G}}^{\mathrm{pLM}} &= -0.05U, && \text{response sampling.}
\end{aligned} \tag{9.7}$$

Their sum is zero. The estimated Potts block and estimated pLM mean agree exactly along U, even though neither estimator equals the family pair component and the pLM has a substantial transfer gap. Agreement arose by cancellation.

The reverse situation is also possible. If the four terms point in orthogonal operator directions, the total discrepancy has squared norm equal to the sum of their squared norms. A single scalar correlation then reports the aggregate disagreement but cannot say whether more MSA depth, a richer statistical model, a different pLM, or more response backgrounds would reduce it.

The ledger suggests an end-to-end comparison protocol.

Fix one anchor geometry. Estimate site marginals from a declared training partition, smooth unsupported categories, and hold those references fixed for both estimators. Transport every block into these fibers before averaging or subtraction.

Separate family sampling from response sampling. Fit the Potts model on weighted or clade-partitioned MSA rows. Estimate the pLM response mean on a separate set of sequence backgrounds. Resample the two axes independently so that each uncertainty term remains identifiable.

Preserve the operator before scalarization. Report block alignment, singular-direction agreement, reciprocity, and dynamic variance before taking norms or contact scores. Two methods can agree in magnitude while disagreeing on which substitutions compensate.

Use a synthetic oracle to identify all four terms. In real protein families the true pair component $G^{F,(2)}$ and Potts population optimum $G^{\mathrm{P},*}$ are unknown. Resampling can estimate the first and fourth finite-sample terms, but misspecification and transfer remain partly confounded. A controlled generative family with a known conditional score is needed to separate all four contributions exactly.

Use biological data only for the claim it can test. Contacts can validate scalar positional association. Double-mutant matrices can test categorical operator alignment. The same mutations across multiple backgrounds can test the dynamic field. No one endpoint identifies every term in Eq. 9.6.

The practical lesson is that a high Potts–pLM correlation is an observation, not an explanation. The estimand ledger identifies which stronger conclusions require additional sampling, a richer family model, a synthetic oracle, or an external intervention.

Finite-sample reporting should include the smoothing convention, regularization path, and uncertainty of full blocks as well as scalar scores. The descriptive quantity $N_{\mathrm{eff}} = (\sum_n w_n^2)^{-1}$ records weight concentration. Independent-sample uncertainty under phylogeny instead requires clade-aware or tree-based resampling rather than an ordinary row bootstrap when estimating confidence intervals for the estimable contrasts derived from Eq. 9.6 and for $\rho^{\mathrm{dyn}}_{i\leftarrow j}$.

Protein language models trained on single sequences and MSA Transformer instantiate different amortization regimes [23, 79]. The former transfers a cross-family prior through fixed parameters and one query sequence; the latter combines that learned prior with an explicit current-family sample as in Eq. 7.55. Both can be audited in the same categorical output coordinates. MSA column attention acts on the sequence axis, tied row attention produces a family-conditioned position route, and neither route is itself the audited categorical residue operator.

9.3 Null-relative evidence

Correlated sampling can bias covariance, inverse-coupling, and sector estimates [73, 22]. Let O^{real} be an estimated operator statistic and let Q_{null} preserve declared nuisance structure, such as one-site composition or a phylogenetic tree, while removing the dependence under test. The null-relative contrast is

$$\Delta O = O^{\mathrm{real}} - \mathbb{E}_{\widetilde{X}\sim Q_{\mathrm{null}}} O(\widetilde{X}). \tag{9.8}$$

Both terms must be produced by the same extraction pipeline and transported to the same reference fibers before subtraction. The contrast updates the evidence for an estimated interaction while leaving the node fiber and Potts parameter-gauge class fixed. Likewise, APC and leading-mode deletion are useful position-level corrections only when the removed direction is supported as nuisance by a declared null or downstream validation [100].

The sequence, position, and categorical axes must be kept separate. Phylogenetic residualization on MSA rows and eigenvector deletion on the final scalar graph produce different results because pair counting, inversion, projection, and norms have a noncommuting order.

For a covariance-like null operator Σ_0 that is positive definite after regularization, relative spectral enrichment can be defined by

$$\Sigma_{\mathrm{real}} u_k = \lambda_k \Sigma_0 u_k. \tag{9.9}$$

Large generalized eigenvalues identify directions enriched relative to the null scale. Contact and mechanism interpretations enter through downstream validation.

Learned representations and task estimators. A pretrained representation can improve an estimator without changing the object it estimates. Several protein methods have the form

$$\boxed{x \xrightarrow{\phi_\theta} z \xrightarrow{\mathcal{T}_D} (\widehat{y}, \widehat{\sigma}) \xrightarrow{\mathcal{A}} \text{ranking, retrieval, or experiment.}} \tag{9.10}$$

The encoder ϕ_θ supplies learned coordinates, $\mathcal{T}_D$ fits task data, and $\mathcal{A}$ selects an action. In EVOLVEpro, a protein language model embeds sequences, a random forest regresses measured activity, and successive experimental rounds supply new labels [33]. VIRAL instead uses an ESM3-derived representation with a Gaussian-process surrogate and acquisition rule [31]. PLMSearch uses learned representations to retrieve candidates before sequence comparison supplies the homology ranking [65].

These systems change the observation and estimator maps of the framework. An assay prediction, surrogate uncertainty, or retrieval score retains its own endpoint. Extracting an interaction operator would additionally require the source intervention, target readout, and reference projection developed here. Evaluation can locate the gain by fixing the representation while comparing estimators, or fixing the estimator while comparing representations. For an adaptive experimental loop, the acquisition rule and newly measured data must also be included.

Coda: the estimand survives the pipeline only by declaration

Weighting sequences, fitting a model, centering tables, averaging responses, scalarizing blocks, and subtracting nulls generally do not commute. The estimand fixes which pipeline is relevant. The four-term ledger then separates family-sampling error, model misspecification, pLM transfer, and response-sampling error as far as the available observations permit.

A Potts block describes a fitted family law; a pLM block describes a predictor's response to substitution. Their agreement can establish a shared categorical pattern. Biological interpretation requires the endpoint-specific tests developed in Chapter 14.

The next chapter follows the evidence through the three AlphaFold generations. Spectral analysis comes afterward, once the graph and its retained information have been declared.

Chapter 10

Three AlphaFolds: The Evolution of Relational State

Chapter 9 fixed the observation and estimand ledgers. We now use those ledgers to ask which evidence each AlphaFold generation retains, compresses, and converts into relational state.

AlphaFold did not advance through three versions of one unchanged algorithm. Each generation reorganized the statistical experiment. The first compressed a family alignment into engineered site and pair features, predicted explicit geometric distributions, and optimized coordinates in the resulting learned potential. The second kept the alignment as an evolving neural state, coupled it repeatedly to a persistent pair representation, and realized geometry through an equivariant structure module. The third shortened the lifetime of the alignment state, enlarged the molecular type system, concentrated relational evidence in a pair representation, and replaced the protein-specific coordinate module with conditional all-atom diffusion.

The progression is often summarized as an improvement in accuracy and scope. For the purposes of this book, it is more revealing to read it as the evolution of an edge object:

$$\begin{aligned} &\text{AF1:} && \text{explicit distance distribution and derived pair potential,} \\ &\text{AF2:} && \text{persistent protein pair state coupled to an MSA state,} \\ &\text{AF3:} && \text{persistent multimolecular pair state conditioning a generator.} \end{aligned} \tag{10.1}$$

At every step the pair representation becomes more capable and less self-interpreting. That tradeoff makes the three systems an unusually strong test of the operator-graph framework.

This chapter uses the theory developed here as a retrospective coordinate system. Its conclusions are therefore audit claims: they classify which sample spaces, information bottlenecks, graph states, symmetries, consistency operations, and probabilistic outputs are present in each published architecture. They are not derivations of AlphaFold from the operator framework, nor equivalence theorems among the three generations. The historical question of the designers' explicit theoretical influences would require different evidence. Within this scope, the framework makes the architectural choices comparable in one language and identifies the additional assumptions needed for physical or biological interpretation.

10.1 One audit, three generations

We will track six coordinates through the three systems.

Evidence. What query-specific information is observed at inference: a current MSA, templates, chemical identity, covalent structure, or recycled predictions?

State. Which axes remain explicit inside the network: alignment row, residue or molecular token, pair, rigid frame, or atom coordinate?

Edge carrier. Is a pair represented by a categorical coupling table, a distance distribution, a hidden vector, an attention route, or an audited response operator?

Composition. How do paths and triples update pair information? Is the operation a linear graph diffusion, a nonlinear triangle computation, or a coordinate-equivariant message?

Normalization and dynamics. Which variable is normalized by softmax or diffusion, and is an iterative update guaranteed to decrease any declared energy?

Licensed interpretation. What can be concluded directly from the architecture, and what would require a gauge, a compatible joint law, a null model, or a biological intervention?

The resulting comparison is summarized before it is derived:

System	Main relational state	Irreversible bottleneck	Strongest immediate semantics
AF1	explicit distogram and torsion-related outputs from an $L \times L$ trunk	raw MSA rows removed before the neural trunk	learned pairwise geometric evidence
AF2	evolving z_{ij} coupled bidirectionally to m_{ni}	alignment information gradually projected into pair and first-row states	joint MSA–pair computation with equivariant coordinate realization
AF3	heterogeneous token-pair state conditioning all-atom denoising	MSA rows discarded after a short MSA module; provenance deposited into z_{ij}	multimolecular relational conditioning and a conditional coordinate generator

None of the final-column statements is yet a claim of microscopic energy, direct evolutionary coupling, or biological mechanism.

10.2 AlphaFold 1: an explicit geometric edge

10.2.1 The alignment disappears before the network begins

AlphaFold 1 constructs a query-specific MSA and transforms it into engineered one-site and pair features before the main neural trunk [25]. Site features include profiles, deletion statistics, and effective alignment counts. Pair features include gap covariation, pseudolikelihood coupling tensors, and scalar summaries of those tensors. Schematically,

$$X_{sia} \longmapsto \{\phi_i(X), \Psi_{ij}(X)\} \longmapsto F_\theta(\phi_i, \phi_j, \Psi_{ij}), \tag{10.2}$$

where the MSA row index n is no longer present after the first arrow.

This is an irreversible statistical decision. The two-dimensional network can learn how to combine profiles, couplings, templates, and positional context. Individual homologous sequences and new representations of row-level phylogeny are no longer addressable after this compression.

Any correction for redundancy, subfamily structure, alignment errors, or finite-depth noise must be performed during feature construction or inferred indirectly from the summaries that remain.

The compression has an advantage that later systems partly surrender: the provenance of the input features is comparatively visible. There is a named profile, a named coupling tensor, and a named gap statistic. What is lost in adaptivity is gained in inspectable interfaces.

10.2.2 Distogram softmax is local geometry, not a global conformation law

The two-dimensional residual network predicts a distribution over distance bins for each residue pair,

$$P_{ij}(d = r_k \mid X) = \mathrm{softmax}_k\, s_{ij,k}(X). \tag{10.3}$$

The normalization axis is the set of possible distances for one fixed pair. This is already much richer than a binary contact: it distinguishes short, intermediate, and long separations and can encode multimodal geometric uncertainty.

This is a collection of pairwise marginals, while a joint distribution over atomic coordinates requires mutual realizability in three dimensions. Triangle inequalities, chain connectivity, chirality, excluded volume, and stereochemistry couple different pairs. The product

$$\prod_{i<j} P_{ij}(d_{ij} \mid X) \tag{10.4}$$

therefore requires a separate compatibility argument before it can be read as the marginalization of one normalized conformation ensemble.

AF1 addresses realization algorithmically. The predicted distance distribution is compared with a learned background distribution and converted into an effective potential of mean force, schematically

$$U_{ij}(r; X) = -\log \frac{P_{ij}(r \mid X)}{P_{ij}^{\mathrm{ref}}(r)}, \tag{10.5}$$

then combined with torsional and stereochemical terms during coordinate optimization. This is an effective learned geometric objective. The logarithm gives it a legitimate energy coordinate; the background ratio gives it a useful statistical reference. Its dependence on the complete input sequence and learned structural database makes it an effective statistical potential whose pair terms can absorb collective constraints beyond a physical two-body force decomposition.

The distinction is subtle but important. AF1 uses energy language at the level needed for optimization: compatible structures should lie in low-objective regions. Calibrated thermodynamics, conformational populations, kinetics, solvent free energies, and transferable atomic forces would require additional physical calibration.

10.2.3 The operator-graph reading

AF1's native pair object is geometric. Each edge carries a distribution over distance states and related orientation information. The input may contain a $q \times q$-like pseudolikelihood tensor, but after learned mixing the output edge no longer says how one amino-acid substitution acts on another residue's categorical distribution.

In the terminology of this book, AF1 builds a dense scalar-state geometric graph from compressed family evidence. It exposes its edge readout clearly; the alignment sample axis is compressed before the network, and the computation has no identified categorical operator. Its great

conceptual move is to replace contact classification with a learned geometric field; its corresponding limitation is that the route from individual evolutionary observations to that field is largely one-way.

10.3 AlphaFold 2: the pair state becomes a dynamical memory

10.3.1 An architectural discontinuity

AlphaFold 2 changes the state space [32]. Persistent single and pair representations replace refinement of a fixed AF1-style potential. The Evoformer maintains an MSA representation

$$m_{ni}^{(\ell)} \in \mathbb{R}^{c_m} \tag{10.6}$$

and a pair representation

$$z_{ij}^{(\ell)} \in \mathbb{R}^{c_z} \tag{10.7}$$

through many blocks. The sequence axis n is now an evolving part of the neural computation rather than a feature-engineering prelude.

The two states communicate repeatedly. Pair features bias row-wise attention in the MSA. The MSA updates the pair state through an outer-product mean,

$$\Delta z_{ij} = W \left[\frac{1}{N} \sum_{n} a(m_{ni}) \otimes b(m_{nj}) \right]. \tag{10.8}$$

The expression is an adaptive, learned second-moment-like statistic. The vectors being multiplied have already passed through attention, pair bias, nonlinear transitions, and earlier blocks. Their row average can carry denoised family signals, deletion structure, phylogenetic patterns, and model priors together.

This explains why retaining raw MSA rows can outperform replacing them by fixed profiles and pair frequencies. The result shows that adaptive access to row-level information matters. Attribution of the gain among higher-order coevolution, improved weighting, denoising, and alignment heterogeneity remains an empirical decomposition.

10.3.2 A graph whose edges have memory

The pair state $z_{ij}^{(\ell)}$ persists and changes across the trunk. It is edge memory on a dense position graph, updated jointly across residue pairs rather than emitted as a final score. Information deposited from the MSA at one layer can influence later MSA routing, later triangle updates, the structure module, and recycling.

This is a major shift in graph semantics. In AF1 the pair grid is primarily the domain of a learned geometric predictor. In AF2 the pair grid is an evolving relational workspace: the model infers and computes with edge state.

The channels of the hidden vector z_{ij} are learned coordinates. A categorical interpretation requires a decoder

$$T_{ij}(a, b) = D_\theta(z_{ij}; a, b) \tag{10.9}$$

followed by reference-gauge projection, while a geometric interpretation requires its own readout. Norms, eigenvectors, and individual channels summarize hidden-state geometry before either typed identification has been performed.

10.3.3 Triangle operations as higher-order computation

AF2 updates an edge through other edges that share an endpoint. A schematic triangle multiplication has the form

$$\Delta z_{ij} \sim \sum_k g_{ijk} \odot A(z_{ik}) \odot B(z_{kj}), \tag{10.10}$$

with learned projections and gates. Triangle attention provides a related content-dependent aggregation. The index k makes relational consistency explicit: the state of (i, j) is judged in the context of two-edge paths through k.

Three claims must be separated.

First, the computation is genuinely triple-indexed. It can compare or compose the relations (i, k) and (k, j) while updating (i, j). Second, the stored output remains a pair state. The architecture can realize the dependence without retaining an explicit hyperedge variable h_{ijk}. Third, a three-body Hamiltonian interpretation additionally requires a compatible energy and global law. The update is gated, channel-dependent, input-dependent, and embedded in a deep residual system. An irreducible three-body statistical component must be established at the audited output after lower orders have been removed.

Triangle operations differ from ordinary graph-Laplacian diffusion. A Laplacian applies a fixed or declared linear edge geometry to node signals; AF2 composes learned edge states nonlinearly and changes the effective operator at every layer. Local Jacobians, singular directions, and intervention responses are more faithful diagnostics than the eigenvectors of one scalarized pair matrix.

10.3.4 Invariant point attention separates symmetry from energy

After the Evoformer, AF2 maps the first-row representation and pair state into three-dimensional structure. Invariant point attention represents residues by rigid frames, forms query and key points in local coordinates, compares them in global space, and uses pair features as additional biases. The construction makes the update equivariant to global rigid motion while allowing the attention weights themselves to depend on invariant distances.

This is a precise use of physical symmetry. Rotating or translating the entire protein changes the coordinates in the required way while preserving the intrinsic prediction. The architecture guarantees this transformation law directly.

Equivariance states how the output changes when the coordinate frame changes. A physical-force or potential-gradient interpretation adds a separate dynamical model. The structure module is a learned coordinate map trained with geometric losses. Its attention softmax normalizes source residues for an update; conformation probabilities live in another space. Symmetry controls transformation law, not thermodynamic semantics.

10.3.5 Recycling as iterative inference

AF2 feeds predicted structure and internal representations back into the network. Recycling lets the model refine a hypothesis after seeing its own geometric consequences. It resembles iterative inference and can correct globally inconsistent early states.

But an iterative map

$$y^{(t+1)} = F_\theta(y^{(t)}, X) \tag{10.11}$$

is an iterative inference dynamics. An energy-descent reading becomes available with a declared state function $E(y)$ and evidence that

$$E(y^{(t+1)}) \leq E(y^{(t)}) \tag{10.12}$$

under fixed memories and update conditions. AF2 provides a powerful learned refinement process; the weaker description is also the accurate one.

10.4 AlphaFold 3: relational conditioning becomes multimolecular

10.4.1 The node and edge types expand

AlphaFold 3 moves from protein structure prediction to complexes containing proteins, nucleic acids, small molecules, ions, modified residues, and covalent links [30]. Standard polymer residues can be represented as tokens, while non-polymer components are tokenized more nearly at heavy-atom resolution. The token-pair representation must therefore relate objects with different internal state spaces and different sources of evidence.

This immediately makes the pair track heterogeneous. Two protein residues may receive family evidence from an MSA, whereas a protein–ligand pair draws on chemical identity, covalent structure, spatial context, templates where applicable, and learned priors. The same tensor index type contains several relation types.

The operator-graph view handles this naturally by allowing typed node fibers and typed edge maps. It also warns against treating the hidden z_{ij} as one homogeneous scalar graph. Any analysis must stratify or transport across entity types before norms and spectra can be compared.

10.4.2 The MSA deposits evidence and then disappears

AF3 still constructs MSAs for polymer chains, including paired rows when cross-chain genetic information is available [40, 30]. But the explicit MSA state has a much shorter lifetime than in AF2. A small MSA module updates both row representations and the pair state through outer-product means, pair-weighted averaging, and triangle operations. After that module, the MSA rows are discarded.

The main Pairformer operates on a single-token state s_i and pair state z_{ij}. Pair features bias token attention, while triangle operations continue to update the pair track. Within the Pairformer, information flow is asymmetric: pair state shapes token updates, while the AF2-style token-to-pair write-back loop is absent. Much of the relational evidence has already been deposited into z.

The change can be written as a shift in bottleneck location:

$$\begin{aligned} &\text{AF2:} \quad m_{ni} \rightleftarrows z_{ij} \quad \text{throughout the trunk,} \\ &\text{AF3:} \quad m_{ni} \longrightarrow z_{ij} \quad \text{near the entrance,} \qquad z_{ij} \longrightarrow s_i \quad \text{in the core.} \end{aligned} \tag{10.13}$$

AF3 gains a common trunk for heterogeneous molecular systems and reduces the cost of maintaining a large alignment axis. It loses direct access to row-level provenance once the compression has occurred.

The pair state becomes the principal relational memory of the model. That makes it more central than in AF2, not more interpretable. Evolutionary evidence, templates, chemical type, covalent relations, recycling information, interface geometry, and confidence-relevant signals can coexist inside the same vector.

10.4.3 OpenDDE: lifting pair memory to structural roles

OpenDDE inserts an intermediate relational scale between residue context and all-atom generation [76]. A residue Pairformer supplies single states s_i^r and pair states z_{ij}^r. A structural refiner expands

each residue into role-labelled tokens, updates their single and pair tracks, and supplies the refined condition to an atom-coordinate denoiser:

$$\text{residue context} \longrightarrow \text{structural-role context} \longrightarrow \text{all-atom coordinates.} \tag{10.14}$$

For residue i, let $\mathcal{R}_i$ contain the instantiated roles, such as backbone and side-chain roles. The expanded node set $V_s = \{(i, \rho) : \rho \in \mathcal{R}_i\}$ has maps

$$\pi : V_s \to [L], \quad \pi(i, \rho) = i, \qquad \varrho : V_s \to \mathcal{R}, \quad \varrho(i, \rho) = \rho. \tag{10.15}$$

An atom-owner map $\omega : [N_{\text{atom}}] \to V_s$ completes the hierarchy. In the vocabulary of Chapter 3, the position set $[L]$ is the base of the categorical graph and the centered contrast spaces $\mathcal{H}_i^0$ are its fibers. The expansion refines the base: V_s is the total space of a role bundle $\pi : V_s \to [L]$ with fiber $\mathcal{R}_i$ over residue i. Every categorical fiber remains untouched, so the construction does not identify new categorical interaction fibers. What changes is which cells of the pair grid exist, not what a node can encode about residue identity.

Remark 10.1 (Roles are not fragments). The role set $\mathcal{R}$ is small, fixed, and functional: backbone and side-chain roles decompose a parent entity. It is not a fragment vocabulary of reusable chemical building blocks whose identities transfer and recombine across molecules. The atom-owner map ω does partition atoms into physical groups, so a fragment-level reading of the tokenization is partly accurate, and it correctly predicts the complexity motivation: pair memory opens on an M^2 grid of role tokens rather than an N_{atom}^2 grid of atoms. What the reading silently adds—library priors, compositional assembly, transfer of fragment identity—is absent: the same role label recurs at every residue, and transfer enters only through learned role-pair weights. A functional decomposition has been upgraded into a compositional semantics that the architecture does not declare.

The initial structural single state inherits its parent's context and adds a role embedding:

$$c_i = s_i^r + f_{\text{split}}(s_i^r), \tag{10.16}$$

$$s_u^{s,0} = c_{\pi(u)} + e_{\varrho(u)}. \tag{10.17}$$

A schematic pair lift applies the parent map on both axes:

$$z_{uv}^{s,0} = z_{\pi(u)\pi(v)}^r + W_{\varrho(u)\to\varrho(v)}\left(z_{\pi(u)\pi(v)}^r\right) + e_{\phi(u,v)}. \tag{10.18}$$

Each term has a distinct office. The first is the pullback $(\pi \times \pi)^* z^r$ of residue pair memory to the refined base: every role pair inherits its parents' record. This movement differs from reference transport, which changes the gauge of an operator while keeping its base fixed; the pullback keeps the record and changes the base. The second applies a linear map graded by the role pair $(\varrho(u), \varrho(v)) \in \mathcal{R} \times \mathcal{R}$, so the lift acts as an $\mathcal{R} \times \mathcal{R}$-graded operator on the pulled-back state. The third is a named metadata section: $\phi(u, v)$ records declared relations such as shared parentage, backbone–side-chain pairing, or polymer adjacency. The lifted pair state is thus a section of a role-blocked bundle $\bigoplus_{(\rho,\rho')} Z_{\rho\rho'}$ with blocks $Z_{\rho\rho'} \cong \mathbb{R}^{d_p}$ over $V_s \times V_s$, and the typed node fibers and typed edge maps that the operator-graph view allows become constructive architecture here, once, at the lift. They are also last guaranteed here: the triangle updates that follow compose paths through all structural tokens regardless of role, so block structure in the refined state is a property to be audited rather than a constraint the architecture enforces.

Triangle multiplication, triangle attention, and pair transitions refine the lifted track. A representative contribution is

$$\Delta z_{uv}^s \propto \sum_{k \in V_s} A(z_{uk}^s) \odot B(z_{vk}^s). \tag{10.19}$$

As in AF2, the state is pair-indexed while the update composes paths through a third node. The statistical order of an output still requires the ANOVA audit.

Let $(s^{s,*}, z^{s,*}) = F_{\text{refine}}(s^{s,0}, z^{s,0})$, and let $\widetilde{z}^s$ combine the refined pair projection with structural relative-position features. One denoising step has the form

$$x_{t-1} = D_\theta(x_t, t;\, s^{s,*}, \widetilde{z}^s). \tag{10.20}$$

In the reported schematic, the refined condition is held across the sampled trajectory. Atom processing queries pair information through ω, global communication acts on $M = |V_s|$ structural tokens, and decoded corrections return to their owned atoms. Persistent pair storage therefore costs $O(M^2 d_p)$ rather than using a global atom-pair grid; direct triangle work remains cubic in M before kernel-specific reductions.

The hierarchy sharpens the bottleneck comparison of this chapter into a one-line information statement.

Proposition 10.2 (Refinement reorganizes; it does not recover)**.** *Let E denote the input evidence (alignment rows, templates, chemical identity) and let (s^r, z^r, ϕ) be the condition deposited from it by the residue trunk and the token construction. If the refined states are computed from these alone, in the sense that $(s^{s,0}, z^{s,0})$ is a function of (s^r, z^r, ϕ) and $(s^{s,*}, z^{s,*}) = F_{\text{refine}}(s^{s,0}, z^{s,0})$, then $E \to (s^r, z^r, \phi) \to (s^{s,*}, z^{s,*})$ is a Markov chain and*

$$I\big(E;\, (s^{s,*}, z^{s,*})\big) \;\le\; I\big(E;\, (s^r, z^r, \phi)\big). \tag{10.21}$$

Proof. $(s^{s,*}, z^{s,*})$ is a deterministic function of $(s^{s,0}, z^{s,0})$, which is by assumption a function of (s^r, z^r, ϕ); the chain is therefore Markov, and the displayed inequality is the data-processing inequality applied to that chain. □

A machine-checked certificate accompanies this proposition (`lean/OperatorGraph/NoRecovery.lean`): the operative core—processing cannot increase entropy, applied to the deterministic-composition setting of the proposition—is formalized over arbitrary finite types, kernel-checked.

In words: the role tier can redistribute relational memory across the refined base, but alignment rows, template provenance, or family evidence removed by the residue-level compression cannot reappear there. The AF3 bottleneck of this chapter survives the refinement unchanged; what refinement buys is resolution and reorganization, not recovery. The same statement applies to any future module that conditions on a deposited state alone.

The distinctive choice is where the long-range bottleneck lies: residue relations are inherited by finer roles, refined there, and cached for coordinate generation. Pair memory and its categorical output response remain separate objects. Role typing at the lift is an architectural guarantee; whether the role blocks carry distinct information is a statistical hypothesis. It is testable by nulling the role embeddings e_ϱ, by severing the metadata section e_ϕ and asking whether the learned updates reconstruct an equivalent blocking, and by auditing decoded response operators within and across (ρ, ρ') blocks (Chapter 14). The next subsection identifies the probability law generated from this condition.

10.4.4 Diffusion changes the output from a coordinate to a law

AF3 replaces the AF2 structure module with a diffusion model over all-atom coordinates. Let R_0 denote a clean structure and R_t a noised version. A denoiser or score-like network is trained under a forward corruption process and conditioned on the trunk representations. Its learned field approximates the score $\nabla_{R_t} \log p_t(R_t \mid s, z)$ of a noised coordinate distribution.

Three operations define the resulting generator:

$$\boxed{\text{multiscale density smoothing} \longrightarrow \text{conditional score estimation} \longrightarrow \text{reverse probability transport.}} \tag{10.22}$$

The forward noise schedule sets a resolution scale. The denoising objective estimates a population conditional mean or score. A reverse SDE or probability-flow ODE then transports a simple terminal law toward the learned conditional structure law. The stochastic and deterministic samplers can share all one-time marginals under an exact score while retaining different path laws.

This construction changes the epistemic status of the output. Repeated samples can express multimodality and sampling variability under the learned model. Their frequencies still depend on training data, conditioning information, the noise schedule, numerical solver, and any ranking applied after generation.

The diffusion clock indexes artificial corruption scale. Molecular time requires a calibrated forward dynamics. Likewise, the learned score is a log-density gradient; a thermodynamic mean force requires the learned density, configuration-space measure, temperature, and physical ensemble to match the intended system. Section 10.6 develops the forward SDE, Fokker–Planck equation, score identities, reverse SDE, and probability-flow ODE in full.

Return to AlphaFold 3. The interlude identifies the coordinate generator, while AF3's protein-specific content enters through the conditioning fields s_i and z_{ij}. The score network is conditional: it reads a token state and a persistent pair state into which MSA, template, chemical, and geometric evidence has already been compressed. Diffusion changes the output law; the pair track still determines which relational evidence is available to shape that law.

10.4.5 All-atom denoising without architectural equivariance

AF3 removes the requirement that each coordinate-processing layer be equivariant to rigid motion [30]. AF2's invariant point attention enforces geometric transformation rules through residue frames and point-based attention. AF3 instead projects noisy Cartesian coordinates into ordinary hidden channels, processes them with atom-level attention and a token-level Transformer conditioned on the trunk states, and decodes atom features back into three coordinate components. Noise-dependent input and output scalings and a coordinate skip connection complete the denoiser. These learned projections do not constrain hidden channels to transform as scalar, vector, or higher-order rotation representations. Direct all-atom output also avoids requiring a residue-frame and side-chain-torsion parametrization for every molecular species.

Rigid-motion augmentation exposes the denoiser to different orientations and translations during training. It also remains active during generation. In the public sampler, every denoising step first centers the current coordinates using the atom mask, applies a random proper rotation and a random translation, and then adds noise, evaluates the denoiser, and advances the sample [48]. The changing coordinate frame leaves the current structure's internal distances intact. It discourages dependence on one fixed orientation without making each network evaluation exactly equivariant.

For comparison, an exactly equivariant coordinate denoiser would satisfy

$$\mathcal{D}_\theta(g\cdot R_t, t; g\cdot c) = g\cdot\mathcal{D}_\theta(R_t, t; c), \qquad g \in SE(3), \tag{10.23}$$

where c contains all conditioning inputs, g acts on any coordinate-valued components, and scalar token and pair conditions remain unchanged. AF3's architecture does not enforce this identity. Augmentation can encourage the corresponding behavior, but its accuracy must be measured. For a denoiser audit, transform the input and geometric conditions, undo that transform on the prediction,

and compare it with the original prediction at the same noise level. A comparison of final sample distributions is a different test and does not establish equality of individual denoising maps.

The diffusion task and the symmetry constraint therefore enter separately. Small-noise denoising trains local stereochemistry; larger noise levels require reconstruction over longer distances, with the token and pair states supplying molecular context [30]. This training task can be implemented with either an equivariant network or an unconstrained coordinate network. In the operator description, both retain routes and feature maps; what changes is whether their geometric transformation law is guaranteed by construction or assessed after training. Neither choice alone establishes a molecular force law.

10.4.6 Input chemical graph, relational graph, and coordinate graph

AF3 also makes three graphs especially easy to conflate. The input chemical graph records known covalent bonds and molecular identity. The Pairformer graph is a dense or broadly connected token-pair relational workspace. The generated coordinate graph is obtained only after distances, contacts, and interactions are defined on a sampled structure. A covalent edge can condition the pair state without being inferred; a noncovalent pair state can be important without becoming a bond; a spatial contact can emerge in a sample without corresponding to one persistent internal edge channel.

The broader molecular scope therefore validates the need for typed graphs. Once proteins, ligands, nucleic acids, ions, and modifications coexist, "one graph over tokens" is too weak a description. Node type, edge provenance, allowed context, and output interpretation become part of the mathematical object.

10.5 What changes across the three generations

AF1 removes MSA rows before the trunk and exposes geometric distributions. AF2 retains the rows through repeated exchange with pair state and realizes coordinates through an equivariant module. AF3 deposits alignment evidence earlier, expands the molecular types, and conditions coordinate diffusion on the resulting single and pair states. The changing bottleneck is the central comparison: later pair tracks integrate more evidence, while its original provenance becomes harder to recover.

This comparison leaves specific identification problems. In AF1, which geometric predictions come from categorical coupling rather than profiles, gaps, or templates? In AF2, can a decoder recover a reference-centered categorical field from z_{ij}, and does its background dependence track triangle computation? In AF3, which relations originate in MSA evidence and which in learned chemical priors, and how should response fibers be typed across molecular species?

These questions call for interventions that preserve the alternative explanations under test: null the MSA while matching composition and depth, alter one evidence track at a time, and audit substitutions through the complete output. Hidden pair channels mix evidence, so a separation into MSA, pair, single, and coordinate tracks does not by itself disentangle ancestry, chemical constraint, and function.

Diffusion adds a further question: what law do repeated coordinate samples represent, and how do their paths relate to molecular dynamics? The following derivation separates the corruption process, learned score, and sampling transport. The MSA Pairformer case then returns the same relational architecture to categorical sequence prediction.

10.6 One probability flow, two path laws

AF3 replaces the AF2 structure module with a diffusion model over all-atom coordinates. This section isolates the mathematical object introduced by this replacement and the scope of its physical interpretation. Let R_0 denote a clean structure and R_t a noised version. A denoiser or score-like network is trained under a forward corruption process and conditioned on the trunk representations. Here (s, z) denotes the fixed single and pair conditioning state introduced in Chapter 10. In score notation, the learned field is associated with

$$s_\theta(R_t, t \mid s, z) \approx \nabla_{R_t} \log p_t(R_t \mid s, z). \tag{10.24}$$

This probability space consists of noisy coordinate configurations conditional on the model inputs and trunk state. Attention, distograms, and masked-token heads normalize over three other sample spaces: source positions, distance bins, and amino-acid identities.

The compact formula contains three different objects: a deliberately chosen corruption process, a vector field estimated from corrupted data, and a numerical transport procedure. Their separation locates what diffusion contributes.

10.6.1 The forward SDE smooths information

A continuous-time idealization of Gaussian corruption is the Itô stochastic differential equation (SDE)

$$\mathrm{d}R_t = f(R_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \qquad 0 \le t \le T, \tag{10.25}$$

where f is a prescribed drift, g is a noise amplitude, and W_t is Brownian motion. An SDE specifies infinitesimal transition laws for an ensemble of nonsmooth paths: over a short interval Δt, the deterministic displacement is of order Δt, while the random displacement is of order $\sqrt{\Delta t}$. Its ensemble density obeys the Fokker–Planck equation

$$\partial_t p_t(R) = -\nabla_R \cdot \big[f(R, t) p_t(R)\big] + \frac{1}{2} g(t)^2 \Delta_R p_t(R). \tag{10.26}$$

The first term transports probability; the second smooths it. In the drift-free case this is the heat equation. If the added variance is σ_t^2, then

$$p_t = p_0 * \mathcal{N}(0, \sigma_t^2 I) = \exp\left(\frac{\sigma_t^2}{2}\Delta_R\right) p_0. \tag{10.27}$$

Thus forward diffusion is Gaussian convolution, or heat-semigroup smoothing, viewed at a sequence of resolutions. Fine features of the data density disappear first; at sufficiently large noise, the terminal law is chosen to be close to a simple reference distribution.

The common finite-time parameterization

$$R_t = \alpha_t R_0 + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{10.28}$$

is the transition law of an appropriate linear SDE. The forward schedule is a design choice, and its clock records noise scale. A physical-time interpretation requires an independently identified molecular dynamics.

10.6.2 Denoising estimates the score of each smoothed density

At a fixed noise level, the score is the spatial log-density gradient

$$s_t(R \mid s, z) = \nabla_R \log p_t(R \mid s, z). \tag{10.29}$$

It points toward locally increasing probability density and contains no normalization constant. For the Gaussian corruption in Eq. 10.28, least-squares noise prediction estimates a conditional mean,

$$\epsilon_\theta(R_t, t, s, z) \longrightarrow \mathbb{E}[\epsilon \mid R_t, s, z], \qquad s_t(R_t \mid s, z) = -\frac{1}{\sigma_t}\mathbb{E}[\epsilon \mid R_t, s, z]. \tag{10.30}$$

Equivalently, a minimum-square denoiser estimates the posterior mean of the clean coordinates. Tweedie's identity gives

$$\mathbb{E}[R_0 \mid R_t, s, z] = \frac{R_t + \sigma_t^2 s_t(R_t \mid s, z)}{\alpha_t}. \tag{10.31}$$

Noise prediction, clean-coordinate denoising, and score estimation are different parameterizations of the same conditional object when the Gaussian kernel and scaling convention are fixed. The essential learned quantity is a multiscale family of density-gradient fields from which clean structures are generated.

The posterior over the particular ϵ used for one corrupted example retains uncertainty given R_t. Squared-error training returns its conditional expectation, which is precisely the quantity needed to reconstruct the score of the noised population.

10.6.3 The reverse SDE transports noise toward data

For the forward process in Eq. 10.25, the time-reversed diffusion has drift

$$\mathrm{d}R_t = \left[f(R_t, t) - g(t)^2 \nabla_R \log p_t(R_t \mid s, z)\right] \mathrm{d}t + g(t)\, \mathrm{d}\overline{W}_t, \qquad t : T \longrightarrow 0, \tag{10.32}$$

where $\overline{W}_t$ is Brownian motion in the reverse-time filtration [2, 93]. The displayed differential is integrated with decreasing t; this convention matters when interpreting its signs. The score correction turns a generic noisy reference law back toward the learned conditional data law. Replacing the unknown score by s_θ and discretizing the equation yields a sampler.

The reverse SDE is a probability-generating dynamics. Its individual paths are algorithmic transport paths; Brownian increments are introduced by the model, the drift depends on a noise-indexed training density, and time runs backward along an artificial corruption schedule. An exact score recovers the intended data distribution. Molecular kinetics additionally requires a physically calibrated forward-time process.

10.6.4 The probability-flow ODE exposes the deterministic core

The same forward marginals p_t also satisfy the continuity equation generated by the ordinary differential equation (ODE)

$$\frac{\mathrm{d}R_t}{\mathrm{d}t} = f(R_t, t) - \frac{1}{2} g(t)^2 \nabla_R \log p_t(R_t \mid s, z). \tag{10.33}$$

Indeed,

$$-\nabla_R \cdot \left[\left(f - \frac{1}{2} g^2 \nabla_R \log p_t\right) p_t\right] = -\nabla_R \cdot (f p_t) + \frac{1}{2} g^2 \Delta_R p_t, \tag{10.34}$$

which is exactly Eq. 10.26. Run from the terminal reference law at T back to time zero, this probability-flow ODE transports the same one-time marginal distributions as the reverse SDE when the score is exact [93].

An ODE assigns a smooth deterministic path once its initial point is fixed. An SDE continues to inject Brownian randomness along the path. Their one-time marginal densities agree, while their path measures, couplings between times, numerical errors, and exploration behavior remain different. In the usual sampling experiment the ODE still begins from a random terminal point drawn from the reference law. The shared mathematical core is

$$\boxed{\text{multiscale density smoothing} \longrightarrow \text{conditional score estimation} \longrightarrow \text{reverse probability transport.}} \tag{10.35}$$

SDE and ODE samplers are two realizations of the last arrow. With an approximate neural score and a finite step size, each incurs a characteristic law-approximation error; sampler choice is a numerical and statistical decision.

Diffusion changes the epistemic status of the output. Multiple coordinate samples can be drawn from the same conditioning state, allowing the generator to express some multimodality and sampling variability. Calibration of sample diversity additionally depends on the learned score, the terminal law, the numerical solver, and any selection or ranking imposed after generation.

10.6.5 Statistical energy and molecular energy

The score is the gradient of a learned log density. A physical equilibrium force additionally requires a calibrated density with physical units and temperature, an appropriate measure on configuration space, correct treatment of solvent and chemical state, and a relation such as

$$F(R) = -\nabla_R U(R) \tag{10.36}$$

for the intended physical potential. The diffusion score follows the noised training distribution and depends on the conditioning network. Its native interpretation is a generative vector field; molecular force-field or kinetic interpretations enter through the additional physical structure.

One may always define a noise-dependent statistical energy

$$E_t^{\mathrm{stat}}(R \mid s, z) = -\log p_t(R \mid s, z) + C_t, \qquad s_t = -\nabla_R E_t^{\mathrm{stat}}. \tag{10.37}$$

To identify $k_{\mathrm{B}}T\, s_0$ with a thermodynamic mean force requires the learned density to equal the relevant equilibrium distribution under the correct configuration-space measure and experimental conditions. A kinetic interpretation further requires a physically valid mobility, noise law, and forward-time dynamics.

Repeated diffusion samples reflect the learned structural distribution, training data, conditioning information, noise schedule, and sampling procedure. Comparing sample frequencies with experimental populations requires separate calibration.

10.7 Gauges and selection on the sampled law

Generative design models add two questions to the transport distinction above. Which statistics of a condition survive when the condition is delivered in a randomly drawn frame, and which statistics of the clean state are exactly recoverable from a noisy interpolant. Both are gauge questions, and both have short exact answers. A third question composes with them: training-time candidate selection, field combination, score rescaling, and trajectory search all replace the law that a sampler actually realizes. The RFD4 case in Chapter 12 instantiates all three.

10.7.1 Randomized gauge representatives

Let G be a group acting measurably on one component x_S of the state, with the group of rigid motions on a coordinate group as the running instance. Assume the action is free on the orbit of x_S, so that a measurable canonical representative $r(x_S)$ and a gauge coordinate $g(x_S) \in G$ decompose $x_S = g(x_S)\, r(x_S)$; for a symmetric condition the gauge coordinate is defined up to its stabilizer. Let X denote the remaining state, jointly distributed with x_S. Delivering x_S transmits both factors. Delivering the randomized gauge representative

$$c = g_0\, r(x_S), \qquad g_0 \sim \nu \text{ independent of } (X, x_S), \tag{10.38}$$

transmits them differently.

Proposition 10.3 (Randomized gauge representation)**.** *With $c = g_0\, r(x_S)$ as in Eq. 10.38:*

1. *$r(c) = r(x_S)$ almost surely, so the invariant part is exactly preserved;*
2. *conditionally on $r(x_S)$, the delivered condition c is independent of $\big(X, g(x_S)\big)$;*
3. *the conditional law of the remaining state satisfies $p(X \mid c) = p(X \mid r(x_S))$: conditioning on the delivered representative is conditioning on the orbit invariant.*

Proof. (1) The representative r is constant on orbits, and c lies on the orbit of x_S. (2) For bounded measurable φ, ψ, first condition on x_S: $\mathbb{E}[\varphi(c)\psi(X) \mid x_S] = \big(\int \varphi(g\, r(x_S))\, d\nu(g)\big)\, \mathbb{E}[\psi(X) \mid x_S]$, because g_0 is independent of (X, x_S). The first factor depends on x_S only through $r(x_S)$; averaging over x_S given $r(x_S)$ therefore factors the joint conditional expectation, which is the claim. (3) By (2), adding c to the conditioning leaves the conditional law of X unchanged once $r(x_S)$ is given, and $r(x_S) = r(c)$ is c-measurable by (1). □

Under absolute-coordinate conditioning the deposited frame is transmitted with the geometry. A deposited frame has no physical content: it constrains the output to one representative of the condition's orbit without adding information about the system. Under a randomized gauge the conditional support over placements is the orbit, placement information must arrive through other inputs or the prior, and any dependence between shape and frame in the data reaches the model only through the joint law of the invariant part. This is the continuous analogue of the support automorphisms in Proposition 7.3: position is carried by whichever inputs break the symmetry, not by the order or frame in which a condition happens to be deposited.

10.7.2 Deterministic statistics of a linear interpolant

The second gauge question concerns the noise rather than the condition.

Proposition 10.4 (Interpolant statistics)**.** *Let $x_t = \alpha_t x_1 + \sigma_t \varepsilon$ with ε independent of x_1, and let S be a linear statistic. Then $S(x_t) = \alpha_t S(x_1) + \sigma_t S(\varepsilon)$. If $S(\varepsilon) = \mu$ almost surely, then*

$$S(x_1) = \frac{S(x_t) - \sigma_t \mu}{\alpha_t} \tag{10.39}$$

is exactly recoverable from the noisy state for every $t \in (0, 1)$.

Proof. Linearity of S and independence give the identity; the constant case solves for $S(x_1)$. □

The per-group mean is such a statistic. With zero-mean noise, the clean placement of every unconditioned group is an identity of the interpolant: available to the network without learning anything, and inherited by outputs at sampling time. Randomizing the statistic, as a per-group offset does, replaces the identity in Eq. 10.39 by a sum of two unknown terms and closes the shortcut. Two scope clauses matter. The randomization must be withheld from every input that could reveal it: an absolutely conditioned region pins the offset, so offsets may enter only groups without coordinate conditioning. And the same operation exists on the angular gauge—aligning the noise frame to a declared local frame is a deterministic gauge fixing, which makes orientation statistics identifiable along the interpolant, while an unaligned frame randomizes them. A design declares, for each gauge component of the noise, whether it is fixed, transmitted, or randomized.

10.7.3 Selection layers and the realized law

Selection acts at four places in a modern design pipeline: on training candidates, on the field at sampling time, on trajectories during search, and on outputs after evaluation. The first three are one operation on the realized law.

Definition 10.5 (Selection layer)**.** Let P be the law the sampler realizes on trajectories. A selection layer is a declared modification of the realized law, of one of two forms:

1. a field substitution: the integrated field v is replaced by a declared field $\tilde{v}$;
2. a trajectory reweighting: $dQ/dP \propto \mathbf{1}_C \prod_i w_i$, with measurable weights evaluated at declared checkpoints and a continuation event C; retention of a fixed number of candidates composes the tilt with a rank truncation.

Proposition 10.6 (Audits run on the composite law)**.** *For a reweighting layer with $W = \mathbf{1}_C \prod_i w_i$, every readout f satisfies $\mathbb{E}_Q[f] = \mathbb{E}_P[fW]/\mathbb{E}_P[W]$. A field substitution instead changes the flow inside $\mathbb{E}_P$. In both cases the audit object is the composite: derivatives and finite perturbations must pass through the sampler actually executed, including branch decisions as discrete operations.*

Proof. The first identity is a change of measure; the second is the definition of the realized law. □

Best-of-K training selection keeps, among K independently noised candidates at one sampled time, the lowest-loss candidate: the regression measure is the base measure transformed by rank selection, concentrating on low-loss noise draws as K grows. Classifier-free interpolation replaces the conditional field by $(1+w)v_c - wv_u$; for $w \notin \{0, 1\}$ this is in general not the conditional field of any single conditioning value, and the realized law is the flow of the declared combination. Score rescaling by a time-dependent factor substitutes a tempered field under matched noise injection. Checkpoint search with continuation criteria reweights trajectories. No layer changes the trained checkpoint, and every layer changes the law that a comparison must match. Selection yields and compute-matched comparisons are therefore separate measurements: a layer moves the law and the cost together, and only the composed statement is well posed.

10.8 Case study: MSA Pairformer and the paired-alignment estimand

MSA Pairformer supplies a compact empirical case for the progression developed in this chapter. It keeps an alignment state m_{ni}, maintains a persistent pair state z_{ij}, writes from the alignment into the pair track through an outer product, and updates pair state through triangular paths.

The model has 111 million parameters and was trained by masked-token prediction on 269,455 single-chain UniClust30 alignments. Its protein–protein interaction behavior appears when a paired alignment is supplied at inference time, even though multichain alignments and complex structures were absent from pretraining [42].

The case contains two statistical experiments. Single-chain alignments support masked-token and intrachain contact evaluations. Protein–protein interaction evaluations first construct a paired alignment, thereby declaring which homologue of chain A and which homologue of chain B form one joint observation. Parameter count describes the learned maps in the middle of these pipelines; it does not specify the information in the input sample or the endpoint used to score the output.

The mathematical audit will therefore ask five questions in order. What empirical measure over MSA rows is induced by the query? What statistic of that measure is written into pair state? How do row-space transport and triangular path composition close the recurrent update? Which output law is identified by query-only masking, and which internal factors remain nonidentifiable? For a protein–protein interaction, what joint sample is created by the pairing rule? The reported 111M-versus-15B comparison can then be interpreted as a consequence of the full estimator rather than as its starting explanation.

The finite estimator and its identification conditions come first. Rank bounds, residual Jacobians, pairing uncertainty, and the adaptive-route audit follow in Section 10.9.

10.8.1 The mathematical object: a measure–edge dynamical system

The coupled state and its three contractions

Each MSA Pairformer layer performs three coupled operations. Pair-weighted averaging lets the pair track shape communication along each sequence. A query-biased outer product writes alignment evidence back into the pair track. Triangle multiplicative updates refine (i, j) by paths through third residues k. Schematically,

$$\begin{aligned} m_{ni}^{(\ell+1/3)} &= \mathsf{PWA}_\theta\Big(m^{(\ell)}, z^{(\ell)}\Big)_{ni}, \\ z_{ij}^{(\ell+2/3)} &= z_{ij}^{(\ell)} + W_\theta\left[\sum_n w_n(q; m^{(\ell+1/3)}) a(m_{ni}) \otimes b(m_{nj})\right], \\ z_{ij}^{(\ell+1)} &= z_{ij}^{(\ell+2/3)} + \sum_k \Phi_\theta\Big(z_{ik}^{(\ell+2/3)}, z_{jk}^{(\ell+2/3)}\Big). \end{aligned} \tag{10.40}$$

The sequence weights w_n depend on the query. Two queries embedded in the same broad family can therefore induce different effective family samples. The pair track stores the resulting relational evidence, while triangle paths give the model a direct computational route for combining two-hop paths. The equation is still only a schematic until the normalization spaces and the missing-data denominator are made explicit. That algebra is the mathematical core of the model.

What the outer-product write estimates

Let $M \in \mathbb{R}^{N \times L \times d_m}$ denote one encoded MSA, with n indexing sampled homologues and i indexing aligned residue columns. These axes have different meanings: n is an empirical-sample axis, while (i, j) is the edge index of a complete residue-position graph. For two learned maps $A, B : \mathbb{R}^{d_m} \to \mathbb{R}^c$,

define

$$\begin{aligned} u_{ni} &= A(\mathrm{LN}\, M_{ni}), \qquad v_{nj} = B(\mathrm{LN}\, M_{nj}), \\ \mathcal{O}_{q,ij} &= \frac{\sum_n \omega_n(q)\, \chi_{ni}\chi_{nj}\, u_{ni} \otimes v_{nj}}{d_{q,ij}}, \\ d_{q,ij} &= \sum_n \omega_n(q)\chi_{ni}\chi_{nj}, \end{aligned} \tag{10.41}$$

where χ_{ni} is the observed-token mask. The released implementation computes this quantity by multiplying both factors by $\sqrt{\omega_n}$ and dividing by $d_{q,ij}$ after the contraction. With complete alignments $d_{q,ij} = 1$; with gaps, every residue pair has its own conditional empirical measure

$$\mu_{q,ij} = \frac{1}{d_{q,ij}} \sum_n \omega_n(q)\chi_{ni}\chi_{nj}\, \delta_n, \qquad \mathcal{O}_{q,ij} = \mathbb{E}_{\mu_{q,ij}}[u_{ni} \otimes v_{nj}]. \tag{10.42}$$

Thus the outer product is a weighted cross-moment. It becomes a covariance only after centering u and v under the same pair-specific measure. A shared linear map and a small hidden width compress this moment before a SwiGLU and a projection write it into z_{ij}.

The query update is an exponential tilt of an empirical MSA measure. Its exact covariance identity and the distinction between feature, router, and denominator terms are collected in Section 10.9.

Alternating contractions close two index spaces

The MSA update has the complementary contraction

$$\beta^h_{ij} = w_z^{h\mathsf{T}}\, \mathrm{LN}(z_{ij}), \qquad A^h_{ij} = \mathrm{softmax}_j(\beta^h_{ij}), \qquad \Delta m_{ni} = W_o\, \mathrm{concat}_h \left[g^h_{ni} \odot \sum_j A^h_{ij} v^h_{nj} \right], \tag{10.43}$$

with v^h_{nj} and g^h_{ni} obtained from normalized MSA features. The pair-to-MSA map is a row-wise transport on the position graph; the MSA-to-pair map in Eq. 10.41 contracts the sample axis and expands a position edge. The model's basic algebra is this alternation of three index operations: contraction over homologues n, expansion to (i, j), and path composition over mediators k.

10.8.2 Adaptive row measure and identification

Query bias as a maximum-entropy row measure

The released model uses two pre-softmax query–key fields. Suppressing layer and batch indices,

$$r_n(q; M) = \frac{1}{\sqrt{d}} \frac{1}{L_n} \sum_i \chi_{ni} \left(q_{1i}^{\mathsf{T}} k_{1ni} - \lambda\, q_{2i}^{\mathsf{T}} k_{2ni} \right), \qquad \omega_n = \frac{\rho_n e^{r_n}}{\sum_t \rho_t e^{r_t}}, \tag{10.44}$$

where λ is a learned differential scale, ρ_n is the row-validity mask, and $L_n = \sum_i \chi_{ni}$. The score averages over observed columns, and the final softmax normalizes over valid MSA rows. Query bias therefore changes the empirical measure used by the outer-product write.

What query-only masking identifies

Let I be a random set of query positions and let $M_{0,-I}$ denote the visible query residues. The second training stage optimizes the denoising risk

$$\mathcal{L}_{\text{query}} = \mathbb{E}_{M,I}\left[-\sum_{i\in I}\log p_\theta(M_{0i} \mid M_{0,-I}, M_{1:N})\right]. \tag{10.45}$$

This is a query-conditional pseudolikelihood/denoising objective. It does not define or maximize a joint Potts likelihood unless additional compatibility conditions are imposed on all conditional logits. Randomly changing the query row trains a row-exchangeable rule: a permutation of the non-query rows leaves the weighted moment unchanged up to the same permutation of weights.

Under a mixture model with latent subfamily H_n, a Bayes-optimal relevance score would contain a log density ratio such as $\log p(M_n \mid q, H_n)/p(M_n \mid q)$, but the paper does not prove that the learned score has this form. The defensible statement is therefore query-conditioned empirical reweighting, with posterior interpretation treated as a hypothesis to test.

Conditional training and downstream estimands

The first training stage masks tokens throughout the MSA; query-only masking instead leaves homologues visible and changes the information available to the target row. At the infinite-capacity population optimum, each risk identifies its conditional output law. Neither risk identifies a unique decomposition into query and key maps, row weights, pair features, and triangle messages.

Contact evaluation adds a supervised logistic head on a frozen pair representation. MP-EM and MP-PDP add assignment-dependent objectives through Potts pseudolikelihood or its differentiable relaxation. Contact calibration and partner assignment can therefore improve without establishing that the base language model identifies a general partner posterior.

10.8.3 Pair-state path algebra and residual depth

Triangle multiplication is a length-two path closure

The triangle block has a precise tensor contraction. After input normalization and learned gates, let $a_{ik}, b_{ik} \in \mathbb{R}^c$ be the two projected versions of the pair edge. The outgoing and incoming updates are

$$\begin{aligned}\mathsf{T}_{\text{out}}(z)_{ij} &= \sum_k a_{ik} \odot b_{jk},\\ \mathsf{T}_{\text{in}}(z)_{ij} &= \sum_k a_{ki} \odot b_{kj}.\end{aligned} \tag{10.46}$$

If projections, gates, and normalization are suppressed and the pair state is scalar, these become ZZ^T and $Z^\mathsf{T}Z$. Each output edge therefore aggregates all length-two walks $i\to k\to j$ or $k\to i, k\to j$ in a shared feature space. Under a factor model $z_{ik} \approx r_i^\mathsf{T} h_k$ and $z_{jk} \approx r_j^\mathsf{T} h_k$, the contraction measures overlap of the two endpoint–mediator factor sets. This gives a path-consistency interpretation for the operation.

The interpretation has a strict boundary. A triangle contraction is generally neither an inverse covariance nor a direct-coupling estimator: it need not be positive semidefinite, symmetric, energy-decreasing, or gauge-fixed. The ablation in the paper supports a task-specific benefit of the path channel; it does not identify the channel with C^{-1} or prove that every selected false positive is an indirect statistical correlation. In the language of cavity methods, the operation is

a representation-level surrogate for combining a mediator. It does not perform cavity deletion, variable marginalization, or normalization over a probabilistic factor graph.

Residual depth is a nonlinear transport process

Residual depth transports perturbations through a coupled MSA–pair state. The full transport Jacobian, singular-value diagnostics, and independent scalar readouts are collected in Section 10.9; residual form alone carries no Lyapunov or free-energy interpretation.

10.8.4 The architectural transfer from AlphaFold

MSA Pairformer reuses AlphaFold 3's MSA-module organization: alignment features write into a persistent pair track, and triangular paths refine that track [32, 30]. Where AlphaFold 3 discards the MSA axis before coordinate diffusion, MSA Pairformer repeatedly updates both tracks and reads amino-acid probabilities. MSA Transformer instead retains the MSA tensor without an equivalent persistent pair track.

The query-weighted cross-moment in Eq. 10.41 replaces a uniform outer-product mean. During the second training stage, homologues stay visible and only the query is masked; changing the designated query trains a conditional relevance rule. This is the specific transfer of structural computation to family-conditioned token prediction. The three AlphaFold generations were compared earlier in this chapter.

10.8.5 Paired MSAs define the protein–protein interaction sample

What joint sample does the pairing rule create?

Suppose two interacting protein families provide separate alignments

$$\mathcal{M}_A = \{a_u\}_{u\in U}, \qquad \mathcal{M}_B = \{b_v\}_{v\in V}. \tag{10.47}$$

These collections do not yet define cross-chain covariance. Within organism or genomic context g, there may be several paralogues $U_g \subset U$ and $V_g \subset V$. A pairing rule chooses a partial assignment

$$\widehat{\pi}_g : U_g \longrightarrow V_g \tag{10.48}$$

and constructs concatenated rows

$$\mathcal{M}_{AB}^{\widehat{\pi}} = \left\{ a_u \| b_{\widehat{\pi}_g(u)} : u \in U_g,\ g \in \mathcal{G} \right\}. \tag{10.49}$$

Only Eq. 10.49 turns residues in the two chains into jointly observed variables. Different assignments preserve the single-chain alignments while changing the cross-chain joint table. For positions i in A and j in B, a weighted covariance block has the form

$$\widehat{C}_{ij}^{\widehat{\pi}}(x,y) = \sum_u w_u \mathbf{1}[a_{u,i} = x]\mathbf{1}[b_{\widehat{\pi}_g(u),j} = y] - \widehat{p}_i^A(x)\widehat{p}_j^B(y). \tag{10.50}$$

The marginals in the second term can remain unchanged while the first term moves from a strong coevolutionary pattern to noise. Pairing error can erase a real interface signal, and a pairing rule correlated with taxonomy or genomic organization can introduce structure that the downstream model will exploit.

The assignment $\Pi = (\pi_g)_g$ is therefore a latent combinatorial variable. The attenuation and total-variance formulas are collected in Section 10.9.

Query bias changes the empirical reference law

The query-biased outer product produces one normalized weight for each concatenated MSA row. For query q and pairing $\widehat{\pi}$, it defines the empirical measure

$$\widehat{\mu}_{q,\widehat{\pi}} = \sum_u w_u(q; \mathcal{M}_{AB}^{\widehat{\pi}}) \delta_{a_u \| b_{\widehat{\pi}_B(u)}}. \tag{10.51}$$

This is an adaptive reference law over observed evolutionary backgrounds. It changes with the query and with the pairing rule. Calling the weights "attention" describes the operation; Eq. 10.51 states their statistical effect. In the presence of gaps, the exact pairwise law is the conditional, pair-specific measure in Eq. 10.42, rather than one global row distribution.

The response-regulator experiment makes that effect visible. OmpR sequences formed only about 12% of a 4,096-sequence mixed-family alignment. Uniform averaging recovered one of 19 subfamily-specific interface contacts, whereas query bias recovered eight. Some strongly weighted sequences had low raw sequence identity to the query. The estimator had learned a relevance rule richer than nearest-neighbor selection.

The benefit is conditional on family heterogeneity. When a property is shared globally or every row supplies useful evidence, query weighting can reduce effective alignment depth and need not improve the estimate. The broader estimator also depends on a deep, trustworthy alignment: evolutionarily recent and de novo proteins, disordered or distantly alignable regions, and sequence rearrangements that violate one-to-one column correspondence remain weakly supported regimes.

This adaptivity also creates a dependency that an operator audit must retain:

$$G_{i \leftarrow j} = G_{i \leftarrow j}\left(C; \mathcal{M}_A, \mathcal{M}_B, \widehat{\pi}, \widehat{\mu}_{q,\widehat{\pi}}\right). \tag{10.52}$$

A reference-centered categorical Jacobian extracted from the model is closest to the edge operator developed in this book. The hidden z_{ij} is a computational edge state, and the scalar contact head is a readout from that state. Neither object alone identifies the directed categorical response in Eq. 10.52.

The externally comparable object is a finite categorical response

Fix an MSA, a pairing, a masked target i, and an audit source j. For source residue b and target logit category a, define

$$T_{i \leftarrow j}^{\mathcal{M}}(a, b) = \ell_i(a \mid M_{0j} = b, M_{0i} = \text{mask}, \mathcal{M}_{-0}), \qquad G_{i \leftarrow j}^{\mathcal{M}} = P_i T_{i \leftarrow j}^{\mathcal{M}} P_j^{\mathsf{T}}. \tag{10.53}$$

This finite table, after reference centering, lives in the same categorical operator space as the Potts and pLM responses developed earlier in the book. It includes every path from the perturbed residue through the MSA state, adaptive row weights, pair state, triangle blocks, and language-model head.

10.8.6 Task-conditional efficiency after the estimand is fixed

On 45 CASP15 monomer targets shorter than 1,024 residues, MSA Pairformer achieved median long-range contact precision $P@L = 0.66$, compared with 0.50 for both MSA Transformer and ESM2-15B. Its 111M parameters are approximately 0.74% of 15B. The comparison licenses a task-specific claim: explicit family evidence, persistent pair state, and triangle computation can dominate parameter scale for long-range contact extraction. ProteinGym variant-effect performance remained strong, with an average Spearman correlation of 0.45 over 217 substitution datasets. The result

does not establish a universal ordering across generation, remote homology, structure prediction, binding affinity, or low-homology proteins.

The contact benchmark also contains a small supervised readout. The pretrained representation was frozen and a logistic contact head was fitted on the same 20 trRosetta proteins used in prior MSA Transformer and ESM evaluations. "Unsupervised contact prediction" describes the language-model objective; the end-to-end contact score still contains labeled structural calibration.

The efficiency result is therefore a statement about an estimator-resource pair. The model receives a structured family sample, compresses it through a finite bilinear feature bottleneck, retains a persistent relational workspace, and uses path closure before a task-specific readout. The 111M-versus-15B contrast follows from that complete statistical and computational pipeline.

10.8.7 What the experiments identify

The paper supplies several useful interventions, each with a different scope.

Interface contacts. On 27 bacterial complexes that fit the MSA Transformer length limit, median interface $P@K$ was 0.52 for MSA Pairformer and 0.19 for MSA Transformer. Among 519 predictions with reported probability above 0.75, 83% were within 8 Angstrom and 94% within 12 Angstrom in the experimental structure. These results evaluate the complete proximity-pairing, representation, and contact-readout pipeline.

Partner removal. In the ParD3–ParE3 toxin–antitoxin landscape, 9,194 antitoxin variants altered four interface positions. MSA Pairformer attained precision 0.74 when ranking the 465 designated binders. Removing the toxin sequences from the paired MSA reduced precision to 0.50. This intervention shows that cross-chain rows carry information used by the score; it does not isolate whether the gain comes from query weighting, pair state, triangle updates, or another learned component.

Pair assignment. For MALG–MALK, the reported pairing precision was 0.70 and 0.67 for the two MSA Pairformer procedures, compared with 0.31 for MSA Transformer DiffPALM. This is evidence that the learned family estimator can help infer partners. Evaluation on two interactions leaves the general paralogue-pairing problem open.

Triangle paths. In a separate supplemental ablation, replacing triangle updates by parameter-matched pair updates reduced CASP15 long-range contact precision by roughly 16–18 percentage points, depending on the training comparison. About 41% of the ablated model's selected false positives could be linked to a third residue contacting both endpoints. The experiment supports the hypothesis that triangle paths provide a useful consistency channel and can suppress some mediated false positives on this task. Equation 10.46 shows why the channel is sensitive to common mediators; it does not establish an algorithmic equivalence with inverse covariance or direct coupling analysis.

10.8.8 A paired-MSA audit

A protein–protein interaction claim should report the paired alignment as part of the estimator, not as an incidental preprocessing file. A minimal audit contains:

1. the sequence databases, search dates, coverage and identity thresholds for both families;

2. the taxonomic units, paralogue counts, genomic-neighborhood rule, and assignment constraints;
3. paired depth, effective depth, discarded candidates, fusion handling, and uncertainty over ties;
4. a within-taxon random-pairing null, which preserves species composition and single-chain marginals while destroying the proposed partner assignment;
5. a cross-taxon random-pairing null, which tests how much broader taxonomic structure drives the score;
6. a partner-removal ablation and a depth-matched unpaired or single-chain baseline;
7. sensitivity of full categorical Jacobian blocks, query weights, and contact readouts to the pairing rule;
8. separate reporting for bacterial operon-linked, eukaryotic paralogue-rich, and cross-species systems.

The strongest factorial experiment crosses pairing and architecture. Run the same model on curated, within-taxon-shuffled, cross-taxon-shuffled, and partner-removed alignments; then run a Potts model, MSA Transformer, and MSA Pairformer on those same inputs. The pairing contrast estimates the value of the joint sample. The model contrast estimates how well each method extracts information from that sample. Their interaction asks whether one architecture is especially robust to pairing noise.

MSA Pairformer therefore connects the three levels of this book in one pipeline. Mathematics specifies a maximum-entropy row tilt, a pair-specific weighted cross-moment, a path-closure dynamics, a latent permutation problem, and a finite categorical response operator. Physics supplies testable hypotheses about when compensatory substitutions and path consistency reflect an interface geometry; no free energy is implied by the architecture alone. AI supplies an amortized estimator that converts a current family sample into persistent relational state. This returns to the observation principle of Chapter 9: the pairing rule is part of the estimator. The next chapter takes spectra only after this provenance has been recorded.

10.9 The MSA Pairformer mathematical interlude

This section collects derivations that support the main Pairformer case but are optional for its architectural and estimand-level argument. The preceding case uses the finite cross-moment, the query conditioned row measure, the triangle path, and the categorical response as its primary objects.

10.9.1 Measure derivatives and rank bounds

Writing $\phi_n(q) = r_n(q; M)$ and $Y_n^{ij} = \chi_{ni}\chi_{nj}(u_{ni} \otimes v_{nj})$, the complete-mask derivative is

$$\mathcal{O}_{q,ij} = \mathbb{E}_{\mu_q} Y_n^{ij}, \qquad \nabla_q \mathcal{O}_{q,ij} = \mathrm{Cov}_{\mu_q}\big(Y_n^{ij}, \nabla_q \phi_n(q)\big)\,, \tag{10.54}$$

when the feature map has no direct query argument. Gaps or query-dependent features add feature and denominator terms. Higher derivatives contain higher cumulants of the tilted empirical law.

For one feature channel, let $U_{in}^a = u_{ni}^a$, $V_{jn}^b = v_{nj}^b$, and $D = \mathrm{diag}(\omega)$. Before nonlinear mixing,

$$\mathcal{O}_{ij}^{ab} = U_{i:}^a D V_{:j}^{b\mathsf{T}}, \qquad \mathrm{rank}(U^a D V^{b\mathsf{T}}) \le \min(N, L). \tag{10.55}$$

For a linearly combined output channel,

$$\mathcal{O}^d = \sum_{n=1}^{N} \omega_n A_n W_d B_n^{\mathsf{T}}, \qquad \operatorname{rank}(\mathcal{O}^d) \le \min(L, Sc). \tag{10.56}$$

Masks make coefficients pair dependent, and gates, residual depth, and triangle nonlinearities remove any global low-rank conclusion about the final contact map.

The linear OPM write has a parameterization gauge. For invertible $R, T \in GL(c)$, $A \mapsto RA$, $B \mapsto TB$, and $W_{\text{opm}} \mapsto W_{\text{opm}}(R \otimes T)^{-1}$ preserve the write up to vectorisation. The masked loss therefore identifies an output law up to output-equivalent parameters, not a unique factorization into row weights and feature maps.

10.9.2 Router derivatives and effective depth

Before LayerNorm, the two query–key fields combine into the bilinear kernel

$$h_{1,ni} - \lambda h_{2,ni} = x_{0i}^{\mathsf{T}} \underbrace{(W_{Q1}^{\mathsf{T}} W_{K1} - \lambda W_{Q2}^{\mathsf{T}} W_{K2})}_{B_\lambda} x_{ni}, \qquad \operatorname{rank}(B_\lambda) \le 2d_q. \tag{10.57}$$

The row softmax has the entropy-variational form of Eq. 2.3, with MSA rows as its sample space. The derivative below records how that reweighting enters the written moment.

The differential of the cross-moment separates feature and router terms:

$$d\mathcal{O}_{q,ij} = \mathbb{E}_{\mu_{q,ij}}[d(u_i \otimes v_j)] + \operatorname{Cov}_{\mu_{q,ij}}(u_i \otimes v_j, dr) + d_{q,ij}^{-1} \text{ mask/denominator terms.} \tag{10.58}$$

The corresponding pair-specific effective depth is

$$N_{\text{eff}}^{ij} = \frac{d_{q,ij}^2}{\sum_n (\omega_n \chi_{ni} \chi_{nj})^2}. \tag{10.59}$$

For dependent rows, the variance has the covariance form

$$\operatorname{Var}(\mathcal{O}_{ij}) = \sum_{n,t} \alpha_{n|ij} \alpha_{t|ij} \operatorname{Cov}(Y_n^{ij}, Y_t^{ij}), \tag{10.60}$$

with a declared tree or resampling scheme.

10.9.3 Symmetry, residual transport, and readouts

For a fixed query row and a permutation π of the remaining rows,

$$M \mapsto \pi M, \qquad \omega(\pi M) = \pi \omega(M), \qquad \mathcal{O}_q(\pi M) = \mathcal{O}_q(M). \tag{10.61}$$

Changing the query is an intervention because it changes the reference law. Independent permutations of two chains preserve their marginals while generally changing their paired joint table.

Collect the MSA and pair states into $x_\ell = (M_\ell, Z_\ell)$. The residual transport is

$$x_{\ell+1} = x_\ell + F_\ell(x_\ell), \qquad Dx_D = \left[\prod_{\ell=D-1}^{0} (I + DF_\ell(x_\ell)) \right] Dx_0. \tag{10.62}$$

The local Jacobian has the two-track form

$$J_\ell = \begin{pmatrix} I + J_{MM}^{(\ell)} & J_{MZ}^{(\ell)} \\ J_{ZM}^{(\ell)} & I + J_{ZZ}^{(\ell)} \end{pmatrix}. \tag{10.63}$$

Singular values of the ordered product diagnose amplification and forgetting; residual form alone does not give a Lyapunov or free-energy interpretation.

Separate supervised scalar heads read contacts from selected pair layers. Their symmetrization does not identify the hidden directed pair field with the categorical response.

10.9.4 Pair assignment and response audits

If a fraction ρ of rows is correctly paired and the wrong pairs are conditionally independent, the centered cross-chain moment obeys the attenuation approximation

$$\mathbb{E}\widehat{C}_{ij}^{\Pi} = \rho C_{ij}^{\star} + (1-\rho) C_{ij}^{\text{wrong}} \approx \rho C_{ij}^{\star}. \tag{10.64}$$

Sampling and assignment uncertainty separate as

$$\operatorname{Var}_{X,\Pi}(\widehat{C}) = \mathbb{E}_\Pi \operatorname{Var}_X(\widehat{C} \mid \Pi) + \operatorname{Var}_\Pi \mathbb{E}_X(\widehat{C} \mid \Pi). \tag{10.65}$$

Gumbel–Sinkhorn relaxation and EM-like hard reassignment optimize moving, noisy pairing objectives; the released procedure has no general exact-coordinate-ascent monotonicity guarantee.

For a source perturbation δM, the difference between adaptive and frozen routing is

$$D\mathcal{O}_{ij}^{\text{adaptive}}[\delta M] - D\mathcal{O}_{ij}^{\text{frozen}}[\delta M] = \sum_n D\omega_n[\delta M] \frac{\chi_{ni}\chi_{nj}}{d_{q,ij}} \left(u_{ni} \otimes v_{nj} - \mathcal{O}_{q,ij}\right). \tag{10.66}$$

A static pairwise Hamiltonian requires the additional reciprocity test

$$G_{i\leftarrow j}^{\mathcal{M}} = G_{j\leftarrow i}^{\mathcal{M}\mathsf{T}}. \tag{10.67}$$

under a common endpoint-excluding context. Query-only training and scalar symmetrization do not impose this condition on the hidden response.

Chapter 11

Spectra After the Graph Has Been Declared

Spectral analysis begins after the graph has been defined, not before. This ordering sounds obvious, yet protein sequence analysis offers many matrices with plausible eigenvectors: an MSA covariance, a Potts norm matrix, an attention route, a dynamic response shadow, a signed coupling graph, or a connection operator acting on categorical fibers. Their spectra live in different spaces and answer different questions.

The first decision is what has been compressed. A scalar Laplacian describes smoothness of functions over residue positions relative to nonnegative edge weights. A signed Laplacian retains one oriented scalar polarity. A connection Laplacian transports vectors between local categorical spaces and can test whether substitution directions remain coherent around the graph. No one spectrum contains the others.

Directionality creates an earlier branch. A row-softmax route is naturally a Markov operator. Its Euclidean asymmetry need not imply irreversible circulation: a symmetric score kernel can produce an entrywise asymmetric but reversible route under its stationary measure. Conversely, a doubly stochastic route can remain nonreciprocal. Reversibility, detailed-balance defect, and directed singular modes should be measured before projecting the route to an undirected shadow.

Context dependence adds another noncommutativity. The spectrum of a mean graph, the mean spectrum of context-specific graphs, and the distribution of spectral projectors are different objects. Near degeneracy makes individual eigenvectors unstable even when their invariant subspace is stable. Spectral claims should therefore be phrased at the level of the appropriate projector and calibrated against replicate null graphs rather than inferred from rank alone.

The guiding question of the chapter is not "what does the leading eigenvector mean?" but "which operator, on which space, relative to which null, produced this eigenvector?" Only then can a spectral mode be connected responsibly to conservation, sectors, communication, allostery, or another biological interpretation.

11.1 A three-node spectral primer

Before applying spectral transforms to protein operators, consider the path on three positions,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad D = \operatorname{diag}(1, 2, 1), \qquad L_G = D - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}. \tag{11.1}$$

For a scalar signal $u = (u_1, u_2, u_3)^\mathsf{T}$ on the positions,

$$u^\mathsf{T} L_G u = (u_1 - u_2)^2 + (u_2 - u_3)^2. \tag{11.2}$$

The quadratic form is small when neighboring positions carry similar values. The eigenvalues are $0, 1, 3$. The zero mode is constant, the middle mode contrasts the two endpoints, and the highest mode alternates across both edges. Low eigenvalue therefore means smoothness relative to this declared graph and this declared node signal.

Three choices precede every biological interpretation of these modes. First, the adjacency must be supplied: contact, covariance, attention, and an operator norm produce different A. Second, the node signal must be typed: a scalar position score, an amino-acid contrast, and an atomic displacement live in different spaces. Third, any null subtraction, symmetrization, or normalization changes the operator being diagonalized.

This primer uses the unnormalized Laplacian. Later sections introduce normalized variants; they rescale node coordinates while preserving the same edgewise smoothness principle. For non-symmetric routes, eigenvectors alone are insufficient, so stationary geometry and left–right structure enter the audit.

The rest of the chapter generalizes this calculation in stages. Directed routes require a stationary geometry; signed edges retain a declared polarity; connection Laplacians transport categorical vectors between sites; wavelets localize selected spectral bands. The three-node path is the baseline against which each additional structure should be read.

11.2 A decision rule for spectral analysis

The word "spectrum" hides several different questions. The operator type determines which decomposition is informative.

For a symmetric matrix $L_G = L_G^\mathsf{T}$, the eigendecomposition

$$L_G = V \Lambda V^\mathsf{T} \tag{11.3}$$

uses an orthonormal basis. Its Rayleigh quotient $u^\mathsf{T} L_G u / \|u\|_2^2$ measures the quadratic energy per unit signal, so low Laplacian eigenvalues identify smooth directions on the declared graph.

For a rectangular map, or for a directed map whose left and right spaces play different roles, use the singular-value decomposition

$$O = U \Sigma V^\mathsf{T}. \tag{11.4}$$

A right singular vector is an input direction, the matching left singular vector is its output direction, and the singular value is the amplification. A square map is nonnormal when $OO^\mathsf{T} \neq O^\mathsf{T} O$. Its eigenvectors can be nonorthogonal, so its SVD and its eigendecomposition answer different questions.

A row-stochastic route P adds a probabilistic question. Its stationary law satisfies

$$\pi^\mathsf{T} P = \pi^\mathsf{T}, \qquad \sum_i \pi_i = 1, \tag{11.5}$$

and describes mass preserved by repeated routing. Reversibility asks whether every stationary edge flux balances its reverse:

$$\pi_i P_{ij} = \pi_j P_{ji}. \tag{11.6}$$

Entrywise symmetry is stronger than this flux condition.

For example, a lazy random walk on the three-node path has

$$P_{\mathrm{rev}} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}, \qquad \pi = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix}. \tag{11.7}$$

The entries $P_{12} = 1/2$ and $P_{21} = 1/4$ differ, while their stationary fluxes both equal $1/8$. The route is asymmetric in Euclidean coordinates and reversible in the π-weighted geometry.

By contrast, the biased three-cycle

$$P_{\circlearrowleft} = \begin{pmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \\ 1/2 & 1/4 & 1/4 \end{pmatrix} \tag{11.8}$$

has the uniform stationary law because it is doubly stochastic. Its clockwise flux exceeds its counterclockwise flux on every edge, so detailed balance fails and a nonzero circulation remains. With the entropy-production convention used later, this example has $\mathrm{EP}(P_{\circlearrowleft}) = \frac{1}{4}\log 2 > 0$.

Finally, an individual eigenvector is not an invariant object inside a repeated or nearly repeated spectral band. If the columns of V_B span such a band, another orthonormal basis has the form $V_B Q$ with $Q^{\mathsf{T}}Q = I$, while

$$(V_B Q)(V_B Q)^{\mathsf{T}} = V_B V_B^{\mathsf{T}}. \tag{11.9}$$

The projector records the stable subspace even when the displayed eigenvectors rotate. Thus the minimal decision rule is: use eigenpairs for self-adjoint quadratic geometry, singular vectors for directed input–output amplification, stationary flux for repeated stochastic routing, and subspace projectors for clustered spectral modes.

11.3 Reference-projected spectral transforms

Let $O : \mathcal{H}_R \to \mathcal{H}_L$ be an observed operator, O_0 a null expectation, and R_L, R_R residualizers away from declared nuisance subspaces in the left and right geometries. A general corrected operator has the form

$$\boxed{\mathcal{D}_{R,\Phi}(O) = R_L \Phi(O - O_0) R_R.} \tag{11.10}$$

Here Φ must be typed for the operator space. For a rectangular $A = U\Sigma V^{\mathsf{T}}$, a singular-value transform is $\Phi_f^{\mathrm{sv}}(A) = U f(\Sigma) V^{\mathsf{T}}$. For a square self-adjoint $A = U\Lambda U^{\mathsf{T}}$, spectral functional calculus instead gives $\Phi_f^{\mathrm{eig}}(A) = U f(\Lambda) U^{\mathsf{T}}$ and preserves eigenvalue signs when f does. Equation 11.10 is an audit template, not an identification theorem. The meaning of Φ depends on a separate model: covariance inversion, graph diffusion, shrinkage, and deletion of an empirical mode make different assumptions.

Operation	Space and observed object	Transform	Identified object
Potts table redundancy	pair table $T_{ij}(a,b)$	$P_i T_{ij} P_j^{\mathsf{T}}$	reference-centered bivariate contrast
Sequence residualization	weighted MSA rows	$R_{\mathrm{phy}} Y$	residual after a declared sequence nuisance model
Position nuisance removal	scalar graph M	$R_Z(M - M_0)R_Z^*$	residual orthogonal to declared position covariates
Gaussian inversion	covariance Σ	Σ^{-1}	conditional graph under a Gaussian law
Graph diffusion	Laplacian L_G	e^{-tL_G}	smoothing relative to a specified graph
Mode deletion	symmetric statistic M	selected $f(\lambda_k) = 0$	residual after a declared spectral choice

For independently specified position nuisance coordinates $Z \in \mathbb{R}^{L \times r}$ and a positive position weight W, let

$$\Pi_Z = Z(Z^{\mathsf{T}} W Z)^{-1} Z^{\mathsf{T}} W, \qquad R_Z = I - \Pi_Z. \tag{11.11}$$

Assume Z has full column rank and let R_Z^* denote the adjoint in the W-weighted position geometry; then $R_Z^* = R_Z$. The corrected scalar position operator is

$$M_\perp = R_Z(M - M_0)R_Z^*. \tag{11.12}$$

This removes exactly the declared linear nuisance space. APC and leading-mode deletion are special rank-one approximations; improvement in contact precision validates their utility for that task, not a universal entropy or phylogeny label.

11.4 Graph Laplacians and multiscale operator spectra

11.4.1 Directed routes and stationary geometry

Graph spectral analysis begins only after the edge object and its compression have been declared. This first branch concerns a route matrix whose rows already sum to one. It asks where routing mass settles after repeated application and whether the stationary flow between every pair is balanced. This branch analyzes an iterated computational route. Categorical substitution effects and thermodynamic equilibrium belong to other constructions. For a nonnegative row-stochastic attention route P, the native object is directed. Assume first that P is irreducible, let $\pi^{\mathsf{T}} P = \pi^{\mathsf{T}}$ be its positive stationary law, and write $\Pi_\pi = \mathrm{diag}(\pi)$. Here P_{ij} uses the query-to-source convention $i \to j$; the value message in Eq. 7.6 travels along the oppositely oriented edge $j \to i$. The convention must remain fixed when cycles are interpreted. The adjoint in the stationary inner product is

$$P_\pi^* = \Pi_\pi^{-1} P^{\mathsf{T}} \Pi_\pi, \qquad P^{\mathrm{rec}} = \frac{1}{2}(P + P_\pi^*), \qquad P^{\mathrm{circ}} = \frac{1}{2}(P - P_\pi^*). \tag{11.13}$$

Both P and P_π^* are row-stochastic. The first term is reversible, while the second has zero row sum and records stationary directed circulation. A dimensionless detailed-balance defect is

$$\delta_{\mathrm{db}}(P) = \frac{\|\Pi_\pi P - P^{\mathsf{T}} \Pi_\pi\|_F}{\|\Pi_\pi P\|_F}. \tag{11.14}$$

It vanishes exactly when $P = P_\pi^*$. Equation 7.9 shows why the criterion uses stationary flux: unequal stationary masses can make a reversible route entrywise asymmetric. Doubly stochastic normalization fixes π to the uniform law while still allowing asymmetric directed cycles, so bistochasticity and reversibility remain different constraints [82, 88].

When both directional fluxes are positive, the same departure can be summarized by the entropy production functional

$$\mathrm{EP}(P) = \frac{1}{2}\sum_{i,j}(\pi_i P_{ij} - \pi_j P_{ji}) \log \frac{\pi_i P_{ij}}{\pi_j P_{ji}} \geq 0. \tag{11.15}$$

One-way support, as in a causal mask, requires a separate support diagnostic or an explicit smoothing convention; otherwise Eq. 11.15 can diverge. Reducible routes likewise require restriction to recurrent classes or a declared regularization. These are properties of a computational Markov route. A thermodynamic entropy-production interpretation additionally requires an independently identified physical stochastic dynamics [86]. Context-conditioned Markov-chain and interacting-particle analyses motivate studying route dynamics; output response and biological causation remain separate audits [55, 46].

The directed random-walk Laplacian

$$L_{\mathrm{rw}} = I - P \tag{11.16}$$

is generally non-self-adjoint and can have nonorthogonal modes. The reversible component admits the symmetric representative

$$L_{\mathrm{rec}} = I - \Pi_\pi^{1/2} P^{\mathrm{rec}} \Pi_\pi^{-1/2}, \tag{11.17}$$

which is the additive reversibilization used in a directed-graph Laplacian [8], whereas P^{circ} must be retained separately. Left/right singular subspaces, stationary fluxes, and nonnormality in the π-geometry retain directed transport omitted by an ordinary eigenbasis. For a frozen attention-only rollout across layers, the route is the ordered product $P^{(L)}(x) \cdots P^{(1)}(x)$; the input-dependent factors form a noncommuting product in general. A spectrum of the mean route, a mean of layer spectra, and a spectrum of the product are therefore different statistics.

The practical output of this branch is therefore a three-part report: stationary mass π, reversible routing P^{rec}, and residual circulation P^{circ}. A single symmetric spectrum sees only the second part. A causal residue-to-residue response is tested by a separate intervention on model outputs.

11.4.2 Scalar, signed, and connection Laplacians

The next branch starts from interaction estimates. Its Laplacian depends on what the signal carried by a node is allowed to remember. A scalar Laplacian retains only strength, a signed Laplacian also retains one declared polarity, and a connection Laplacian retains a transported categorical direction. Moving down this list preserves more structure but also requires more modeling choices.

An arbitrary signed or block-valued response has a native operator geometry; a Markov law arises only after an additional nonnegative normalization. Its directionality should be audited through the block reciprocity split in Eq. 4.36 and through typed left/right singular geometry before any nonnegative scalar normalization is imposed. Only if the scientific question deliberately discards edge direction should one transport all directed blocks to common anchor fibers and define a nonnegative symmetric static shadow, for example

$$A_{ij}^{\mathrm{stat}} = \frac{1}{2}\left(M_{i\leftarrow j}^{\mathrm{static}} + M_{j\leftarrow i}^{\mathrm{static}}\right), \qquad A_{ii}^{\mathrm{stat}} = 0. \tag{11.18}$$

Assuming positive degrees, let $D_{\mathrm{deg}} = \mathrm{diag}(A^{\mathrm{stat}}\mathbf{1})$ and

$$L_{\mathrm{sym}}^{\mathrm{stat}} = I - D_{\mathrm{deg}}^{-1/2} A^{\mathrm{stat}} D_{\mathrm{deg}}^{-1/2}. \tag{11.19}$$

Its quadratic form is

$$u^{\mathsf{T}} L_{\mathrm{sym}}^{\mathrm{stat}} u = \frac{1}{2} \sum_{i,j} A_{ij}^{\mathrm{stat}} \left(\frac{u_i}{\sqrt{d_i}} - \frac{u_j}{\sqrt{d_j}} \right)^2. \tag{11.20}$$

Low eigenvectors are therefore position fields that vary slowly across strong static edges, and the zero mode is $D_{\mathrm{deg}}^{1/2}\mathbf{1}$ [89]. This Laplacian is invariant under any replacement of the full blocks that preserves their norms. It can identify positional domains or bottlenecks. Signs, singular directions, substitution-specific coherence, and directed circulation have already been removed by this compression.

A signed positional score requires a different construction. If a declared categorical contraction produces a symmetric $S_{ij} \in \mathbb{R}$, define

$$D_{|S|} = \mathrm{diag}\left(\sum_j |S_{ij}| \right), \qquad L_{\pm} = D_{|S|} - S. \tag{11.21}$$

Then

$$u^{\mathsf{T}} L_{\pm} u = \frac{1}{2} \sum_{i,j} |S_{ij}| \left(u_i - \mathrm{sign}(S_{ij}) u_j \right)^2 \geq 0. \tag{11.22}$$

Positive edges favor aligned position signals and negative edges favor opposing signals [62]. The sign must come from a declared projection of the categorical operator or from a signed routing variable. A block norm has discarded that information. A negative spectral edge encodes opposition in the declared contraction; probability and biological antagonism have their own meanings.

To retain categorical contrast orientation on an undirected support, work in the anchor-whitened fibers $\widetilde{\mathcal{H}}_i^0 = Q_i \mathbb{R}^q$. First form the reciprocal block

$$\widetilde{G}_{ij}^{\mathrm{rec}} = \frac{1}{2} \left(\widetilde{G}_{i \leftarrow j} + \widetilde{G}_{j \leftarrow i}^{\mathsf{T}} \right). \tag{11.23}$$

For each undirected edge, let $U_{i \leftarrow j} : \widetilde{\mathcal{H}}_j^0 \to \widetilde{\mathcal{H}}_i^0$ be the orthogonal polar factor of $\widetilde{G}_{ij}^{\mathrm{rec}}$ restricted to the centered fibers, and set $U_{j \leftarrow i} = U_{i \leftarrow j}^{\mathsf{T}}$. This rule is unique when the restricted block is nonsingular. A rank-deficient block requires a declared reciprocal completion or restriction to its supported singular subspace. A connection built from one directed block instead requires an explicit edge orientation and a reversal-sensitivity report; its nonreciprocal remainder belongs to the directed audit above. For symmetric nonnegative weights $a_{ij} = a_{ji}$, the connection energy is

$$\mathcal{E}_{\mathrm{conn}}(f) = \sum_{i<j} a_{ij} \left\| f_i - U_{i \leftarrow j} f_j \right\|_2^2 = \langle f, \mathbb{L}_{\mathrm{conn}} f \rangle. \tag{11.24}$$

The resulting connection Laplacian is positive semidefinite [90]. Its low modes are categorical contrast fields that remain coherent after edge-specific transport. The scalar Laplacian asks where a mode is smooth; the connection Laplacian additionally asks how a residue contrast must rotate between positions. Singular blocks admit several reciprocal transport completions, so the chosen completion is an audited modeling coordinate.

Thus the connection construction computes smoothness only after neighboring categorical vectors have been aligned by the chosen transports. It can reveal a coherent substitution pattern that a norm graph erases. Uniqueness, allosteric interpretation, and correspondence to microscopic motion remain validation questions.

11.5 Case study: one covariance matrix, many inverse graphs

An essay written in 2022 under the deliberately exaggerated title "Inverse covariance is all you need" placed a Gaussian chain, MSE reconstruction, sparse and ridge precision estimation, network deconvolution, balanced network deconvolution, and eigenmode deletion in one sequence [99]. The visual argument was compelling because the methods repeatedly took the same form: diagonalize a matrix, alter its eigenvalues, and reconstruct a cleaner graph. Read through the operator framework, this family resemblance is real and stronger than a metaphor. For a symmetric observed matrix

$$O = U\Lambda U^{\mathsf{T}}, \qquad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d), \tag{11.25}$$

most of the methods in the essay can be written

$$O \longmapsto U \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_d))\, U^{\mathsf{T}}, \tag{11.26}$$

possibly preceded by centering or scaling and followed by block compression and contact ranking. The eigenvectors are often retained; the scientific decision is concentrated in the scalar function f.

That algebraic statement explains why the works look so similar. Each f nevertheless estimates the graph defined by a different forward story. Covariance inversion assumes a Gaussian probability law. Network deconvolution assumes that indirect paths are matrix powers of a direct network. Balanced deconvolution changes which path lengths exist in that story. Leading-mode deletion treats selected spectral directions as nuisance rather than as indirect paths. A deep contact predictor may use the transformed matrix only as an input feature, so its final edge is produced by the subsequent network rather than by f alone.

The common family can therefore be typed by the chain

$$\boxed{\begin{array}{c}\text{observed statistic} \longrightarrow \text{propagation law} \longrightarrow \text{inverse or spectral transform} \\ \longrightarrow \text{regularization} \longrightarrow \text{edge object} \longrightarrow \text{validation endpoint}\end{array}} \tag{11.27}$$

Two methods can be nearly identical in numerical linear algebra and different at any later arrow. This section rewrites the historical sequence in those coordinates.

11.5.1 The Gaussian chain: when the inverse really is the graph

Let independent unit Gaussian noises generate

$$A = \epsilon_A, \qquad B = A + \epsilon_B, \qquad C = B + \epsilon_C, \qquad D = C + \epsilon_D. \tag{11.28}$$

The covariance matrix is dense because an upstream fluctuation propagates to every downstream variable. For example, A covaries with C and D even though the structural equations contain no A–C or A–D term. The joint negative log density, however, is

$$-\log p(a, b, c, d) = \frac{1}{2}\left[a^2 + (b-a)^2 + (c-b)^2 + (d-c)^2\right] + \text{constant}. \tag{11.29}$$

Its Hessian, and hence its precision matrix $\Theta = \Sigma^{-1}$, is

$$\Theta = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \tag{11.30}$$

The tridiagonal support recovers the conditional-independence graph: for a nonsingular Gaussian law, $\Theta_{ij} = 0$ if and only if X_i and X_j are conditionally independent given all other coordinates [20]. Here "direct" has an exact meaning because the probability family, conditioning set, and node variables have all been declared.

This example also gives precision a limited physical interpretation. In a Gaussian equilibrium field, covariance measures fluctuations or susceptibility, whereas precision is the quadratic restoring operator: it is the Hessian of the dimensionless Gaussian free energy. A large eigenvalue of Σ is a soft, strongly fluctuating direction; inversion makes it a small stiffness. A small covariance eigenvalue is a stiff direction and becomes a large precision eigenvalue. This language becomes literal molecular physics only if the variables are physical coordinates, the distribution is an equilibrium ensemble, and hidden degrees of freedom have been handled. For one-hot MSA columns, Θ is instead a Gaussian surrogate for categorical sequence statistics. Its graph is probability-exact inside that surrogate; forces and contacts require an additional correspondence.

11.5.2 MSE reconstruction: the same optimum through another objective

The MSE equivalence in Eq. 7.68 supplies a reconstruction route to a regularized precision-like matrix. Chapter 7 derived the shared optimum and the regularization transfer [17]. In the present spectral comparison, it is another route to the same inverse transform; the fitted reconstruction map retains its own objective and constraints.

11.5.3 Sparse and ridge precision: regularization chooses the graph

Given an empirical covariance S, regularized Gaussian precision estimation takes the generic form

$$\widehat{\Theta} = \arg\min_{\Theta \succ 0} \{\operatorname{tr}(S\Theta) - \log\det\Theta + R(\Theta)\}. \tag{11.31}$$

PSICOV used sparse inverse covariance estimation for protein contact prediction [57]. An L_1 penalty makes many fitted entries exactly zero, so it performs two jobs: it stabilizes an underdetermined estimate and declares a sparse Gaussian conditional graph. Its zeros belong to that regularized model and express its selected conditional graph. Experimental influence remains a separate endpoint.

ResPRE instead used the Frobenius penalty $R(\Theta) = \rho\|\Theta\|_F^2$ and then supplied the raw 21×21 precision blocks to a deep residual contact predictor [64]. Its stationarity equation is

$$S - \Theta^{-1} + 2\rho\Theta = 0. \tag{11.32}$$

If $S = V\Lambda V^{\mathsf{T}}$, the solution shares the eigenvectors of S and transforms each eigenvalue by

$$k_\rho(\lambda) = \frac{-\lambda + \sqrt{\lambda^2 + 8\rho}}{4\rho}, \qquad \widehat{\Theta} = V\,\operatorname{diag}(k_\rho(\lambda_i))V^{\mathsf{T}}. \tag{11.33}$$

Thus the L_2 estimator is literally a smooth spectral filter. It usually produces graded shrinkage, while L_1 fitting can select an exact support. ResPRE then continued beyond the precision graph. Its final edge was a contact probability learned by a residual convolutional network from the full categorical blocks. The precision matrix was a representation that made indirect correlations easier for a finite neural predictor to use. Indeed, covariance and invertible precision contain the same information in principle; their empirical difference arose because the downstream learner, sample size, and regularization were finite.

The two papers therefore differ less in their observed statistic than in the composite estimator. PSICOV couples Gaussian fitting to sparse support and a contact score. ResPRE couples ridge spectral shrinkage to a nonlinear, structurally supervised decoder. Calling both "inverse covariance methods" is correct but incomplete in the same way that calling AlphaFold 1 and AlphaFold 2 "distance predictors" is correct but incomplete.

11.5.4 Network deconvolution: inversion of a walk algebra

Network deconvolution begins from a different forward model. It assumes that an observed similarity network contains direct edges and all of their transitive walks:

$$G_{\text{obs}} = G_{\text{dir}} + G_{\text{dir}}^2 + G_{\text{dir}}^3 + \cdots = G_{\text{dir}}(I - G_{\text{dir}})^{-1}. \tag{11.34}$$

Under the convergence and diagonalizability assumptions, inversion gives

$$G_{\text{dir}} = G_{\text{obs}}(I + G_{\text{obs}})^{-1}, \qquad f_{\text{ND}}(\lambda) = \frac{\lambda}{1+\lambda} \tag{11.35}$$

after the paper's required spectral scaling [43]. The expression resembles a precision transform because both contain a matrix inverse. The inverse is solving a different equation. Gaussian precision inverts a fluctuation covariance under a joint probability law; network deconvolution inverts a geometric series of walks under a propagation model. Its off-diagonal entry means "direct relative to the assumed walk closure," not "conditionally independent in a Gaussian law."

The distinction is more than philosophical. Matrix multiplication counts weighted walks only after a node basis, edge composition rule, and treatment of signs have been fixed. A categorical operator graph requires compatible fiber maps before G^2 even has a meaning. For an ordinary scalar symmetric network the composition is automatic, but its biological validity remains an empirical assumption. Network deconvolution improved several biological networks because that assumption was useful; the algebra alone cannot prove that an observed covariance was generated by the full walk series.

11.5.5 Balanced deconvolution: changing which paths exist

Balanced network deconvolution changed the forward model by retaining only odd path lengths:

$$G_{\text{obs}} = G_{\text{dir}} + G_{\text{dir}}^3 + G_{\text{dir}}^5 + \cdots = G_{\text{dir}}(I - G_{\text{dir}}^2)^{-1}. \tag{11.36}$$

Consequently its spectral inverse is

$$f_{\text{BND}}(\lambda) = \begin{cases} \dfrac{-1+\sqrt{1+4\lambda^2}}{2\lambda}, & \lambda \neq 0, \\ 0, & \lambda = 0. \end{cases} \tag{11.37}$$

Removing even powers preserves the sign balance of eigenvalues and maps every real observed eigenvalue to a direct eigenvalue with magnitude below one, avoiding the network-dependent scaling required by the original transform [97]. Applied as a postprocessor, BND improved contact rankings from several coevolutionary methods, with smaller gains for methods such as DCA and PSICOV that had already attempted to suppress transitive effects.

In the language of this book, BND changes the algebra of admissible paths. It should not be described as the same inverse covariance estimator with a technical fix: even-length paths are

absent from its generative story. Nor does odd-only propagation have an automatic physical law behind it. Its justification is a combination of spectral behavior and downstream validation. This is a legitimate scientific route, but the validation endpoint, rather than the square root in Eq. 11.37, supplies the biological evidence.

11.5.6 Leading-mode deletion: a nuisance model, not an inverse

Qin and Colwell analyzed why phylogenetic relatedness contaminates covariance matrices built from MSAs [77]. Their random-matrix calculation predicts a power-law tail among the largest covariance eigenvalues. On that model, selected leading directions are dominated by shared ancestry rather than by structural coupling. The corresponding correction is a spectral projection such as

$$C_{\text{clean}} = C - \sum_{k \in \mathcal{P}} \lambda_k u_k u_k^{\mathsf{T}}, \tag{11.38}$$

where $\mathcal{P}$ is chosen from the declared phylogenetic-tail criterion. Removing those modes improved contact prediction in their tests.

This method is especially revealing because it acts on the same eigendecomposition as precision estimation to target a different object from C^{-1}. Inversion applies $f(\lambda) = 1/\lambda$ to every retained direction, suppressing large covariance modes while amplifying small ones. Leading-mode deletion applies $f(\lambda) = 0$ to a selected nuisance set and $f(\lambda) = \lambda$ elsewhere. The first transform follows from a Gaussian conditional model; the second follows from a phylogenetic null model. They can improve the same contact benchmark for different reasons.

This corrects an overstatement in the 2022 essay. The Colwell construction belongs in the same spectral family. Its conceptual advance is to identify which large modes should be removed and supply a model for why. Without that null, deleting leading eigenvectors is only rank-relative cleaning and may erase real subfamily or functional structure. With it, the projection is an estimand-level decision of the kind developed in Chapter 9.

11.5.7 What is genuinely the same

The longitudinal sequence can now be summarized without either exaggerating or dissolving the unification:

Method	Forward claim	Spectral action	Output semantics and endpoint
Gaussian precision	covariance of one Gaussian joint law	$1/\lambda$	conditional graph in the fitted Gaussian family
MSE reconstruction	best linear reconstruction under squared loss	regularized inverse, up to an affine identity term	reconstruction weight; Gaussian meaning only under the equivalence conditions
PSICOV / ResPRE	high-dimensional Gaussian surrogate plus regularization	sparse support or ridge filter k_ρ	sparse coupling score, or full precision feature decoded into contacts
Network deconvolution	observation is the sum of all directed walk powers	$\lambda/(1+\lambda)$ after scaling	direct edge relative to the all-walk propagation model
BND	observation is the sum of odd walk powers	$f_{\text{BND}}(\lambda)$	direct edge relative to an odd-walk model; validated as contact postprocessing
Qin–Colwell	leading covariance tail is enriched for phylogenetic nuisance	delete selected leading modes	phylogeny-corrected covariance used for contact ranking

At the level of Eq. 11.26, the methods really are close. They reuse the same matrix, eigenvectors, rational or thresholded scalar functions, and often the same contact benchmark. This is why one

implementation can suggest another and why regularization transfers are fruitful. At the level of scientific objects, their differences are concentrated rather than negligible. The forward law determines what has been inverted; the null determines what has been removed; the regularizer determines what can be identified from finite data; and the downstream decoder determines whether the transformed matrix is the answer or only an input feature.

Formally, let a method be the composite map

$$\mathcal{A}_m = \mathcal{V}_m \circ \mathcal{E}_m \circ \mathcal{R}_m \circ \Phi_m \circ R_m(O - O_{0,m})R_m^{\mathsf{T}}, \tag{11.39}$$

where $O_{0,m}$ is a null expectation, R_m a nuisance residualizer, Φ_m the inverse or spectral transform, $\mathcal{R}_m$ regularization, $\mathcal{E}_m$ edge extraction, and $\mathcal{V}_m$ validation or decoding. Two methods are fully equivalent on a declared class of inputs only when these composite maps agree on that class. Sharing O or even sharing $\Phi_m(O)$ is a weaker correspondence. Agreement of final contact rankings is weaker still, because scalarization can hide different categorical operators.

The durable conclusion is therefore not that inverse covariance is literally everything. It is that a surprisingly large body of protein inference can be organized as competing inverse problems on one second-order statistic. Once written that way, what looked like many unrelated algorithms becomes one design space. The important questions are no longer which paper used an inverse, a square root, or an eigenvalue cutoff. They are: which propagation law was assumed, which modes were declared nuisance, which edge object survived the transform, and which experiment made that declaration credible.

11.6 Spectral filters and localized multiscale analysis

If $L_G = V\Lambda V^{\mathsf{T}}$ denotes either a scalar or connection Laplacian, the graph Fourier coefficients of a compatible signal y are $\widehat{y}_k = v_k^{\mathsf{T}} y$, and a spectral filter is

$$g(L_G)y = V g(\Lambda) V^{\mathsf{T}} y. \tag{11.40}$$

The heat kernel e^{-tL_G} probes progressively coarser interaction neighborhoods as t increases; band-pass filters can separate domain-scale modes from localized edge variation. Low frequency means smoothness relative to the inferred graph; conservation, phylogeny, allostery, and function enter through calibrated correspondences.

typed edge operators $\longrightarrow$ declared scalarization or transport $\longrightarrow$ $L_G \xrightarrow{g(L_G)}$ multiscale coefficients,

$g(\lambda) = e^{-t\lambda}$: heat smoothing $\qquad$ $g(\lambda) = \psi(s\lambda)$: wavelets $\qquad$ $g(\lambda) = \lambda^k$: polynomial propagation.

Figure 11.1: Spectral processing is downstream of edge identification. Different choices of g act on the same declared operator geometry but answer different multiscale questions.

11.6.1 Localized spectral filters: graph wavelets

Fourier modes answer which graph frequencies are present, but each eigenvector is generally spread over the whole graph. A wavelet asks a more localized question: at what scale, and around which node, does a change occur? On a self-adjoint graph Laplacian $L_G = V\Lambda V^{\mathsf{T}}$, choose a band-pass generating function ψ and define

$$\Psi_s = \psi(sL_G) = V\psi(s\Lambda)V^{\mathsf{T}}, \qquad \psi_{s,i} = \Psi_s \delta_i. \tag{11.41}$$

The vector $\psi_{s,i}$ is a graph-wavelet atom centered at node i and scale s [50]. Its spectrum is selected by $\psi(s\lambda)$, while its node-space profile records how that band is expressed around i. Graph wavelets couple spectral selectivity to spatial localization.

The scale convention depends on the chosen generator. For the common dilation $\psi(s\lambda)$, increasing s shifts a fixed band-pass profile toward smaller eigenvalues and usually produces a broader node-space atom. Small scales emphasize finer graph variation; large scales emphasize coarser organization. This is a statement relative to L_G. Atomic-motion and protein-domain interpretations require a separately calibrated correspondence between graph scale and the biological geometry.

Localization follows from the spectral construction. If $\psi(s\lambda)$ is approximated by a degree-K polynomial,

$$\psi(sL_G) \approx \sum_{k=0}^{K} c_k(s) L_G^k, \tag{11.42}$$

then $\psi(sL_G)\delta_i$ is supported, up to approximation error, within K graph steps of i. This is the same spectral-to-spatial passage used by Chebyshev graph filters. The wavelet family organizes several band-pass scales around localized centers; a task-specific graph filter instead learns the propagation profile required by its objective.

A coarsest-scale low-pass component completes the bands. Positive lower and finite upper bounds on the summed squared filter responses over the spectrum give a stable frame [50]; this is the additional condition needed for stable signal reconstruction.

Heat diffusion, wavelets, and graph convolution can now be placed in one functional calculus:

$$\begin{aligned} e^{-tL_G} &: \quad \text{low-pass smoothing at diffusion scale } t, \\ \psi(sL_G) &: \quad \text{localized band-pass analysis at scale } s, \\ \sum_{k=0}^{K} \theta_k L_G^k &: \quad \text{finite-hop polynomial filtering}, \\ S^K H &: \quad \text{repeated propagation with possible over-smoothing.} \end{aligned} \tag{11.43}$$

Their common mathematics is $g(L_G)$; their state, objective, and interpretation remain different. A heat kernel describes smoothing relative to a graph, a wavelet transform analyzes a signal, and a GCN layer is trained as part of a predictor.

For an operator-valued protein graph, the analyzed signal must still be declared. It might be a scalar conservation score, a static interaction norm, a mutational response field, or a categorical section $f \in \bigoplus_i \mathcal{H}_i^0$. Scalar signals can be analyzed with a scalar Laplacian. Categorical sections require the connection Laplacian so that neighboring contrasts are transported between fibers before they are compared:

$$f \longmapsto \psi(s\mathbb{L}_{\text{conn}})f. \tag{11.44}$$

The resulting coefficient measures scale-dependent coherence in the declared transport geometry. Recovery of the original edge operator would require an additional invertible representation and its associated reconstruction map.

Classical spectral graph wavelets also assume a self-adjoint operator with real spectral calculus. A directed attention route P is generally nonnormal and therefore calls for a different spectral construction. One may analyze the reversible Laplacian, a singular-value construction, or a declared symmetrization, but each discards or reorganizes directed circulation differently. Thus the order of reasoning remains

$$\boxed{\begin{array}{c}\text{declare the graph and signal} \longrightarrow \text{choose the operator geometry} \\ \longrightarrow \text{choose the scale filters} \longrightarrow \text{interpret localized coefficients.}\end{array}} \tag{11.45}$$

Low-, band-, and high-frequency coefficients are mathematical facts about that construction. Calling them conservation, allostery, physical vibration, or mechanism requires an external endpoint that licenses the name.

11.7 Context averaging and null-relative spectra

Static and dynamic shadows define different Laplacians. Construct A_{ij}^{dyn} from $M_{i\leftarrow j}^{\mathrm{dynamic}}$ by the same declared symmetrization as Eq. 11.18, and compare the resulting spectra as subspaces. If V_r^{stat} and V_r^{dyn} contain a chosen r-dimensional spectral band, one basis-invariant distance is

$$d_r^{\mathrm{spec}} = \frac{1}{\sqrt{2r}} \left\| V_r^{\mathrm{stat}} V_r^{\mathrm{stat}\mathsf{T}} - V_r^{\mathrm{dyn}} V_r^{\mathrm{dyn}\mathsf{T}} \right\|_F . \tag{11.46}$$

Individual eigenvectors are unstable inside a nearly degenerate band, whereas the spectral projector remains well defined. For context-specific graphs, spectral averaging is nonlinear: in general

$$L_{\mathrm{sym}}(\mathbb{E}_C A(C)) \neq \mathbb{E}_C L_{\mathrm{sym}}(A(C)), \qquad \lambda_k(\mathbb{E}_C L(C)) \neq \mathbb{E}_C \lambda_k(L(C)). \tag{11.47}$$

When averaging graphs creates a mode that no context has

Consider three positions and two equally likely sequence contexts. In context A, only positions 1 and 2 are connected; in context B, only positions 2 and 3 are connected. Their unnormalized Laplacians are

$$L_A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad L_B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}. \tag{11.48}$$

Each graph consists of one edge and one isolated vertex, so both spectra are

$$\mathrm{spec}(L_A) = \mathrm{spec}(L_B) = \{0, 0, 2\}. \tag{11.49}$$

The mean Laplacian is

$$\overline{L} = \frac{1}{2}(L_A + L_B) = \begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1 & -1/2 \\ 0 & -1/2 & 1/2 \end{pmatrix}, \tag{11.50}$$

with spectrum

$$\mathrm{spec}(\overline{L}) = \{0, 1/2, 3/2\}. \tag{11.51}$$

The mean graph is connected even though no context contains both edges simultaneously. Averaging has created a path $1{-}2{-}3$ that exists only across contexts. Correspondingly, the second zero mode present in every individual context disappears from the mean.

This tiny example separates three claims. The mean context spectrum is $\{0, 0, 2\}$. The spectrum of the mean graph is $\{0, 1/2, 3/2\}$. The distribution of zero-mode projectors records which vertex is isolated in each context. None determines the others.

For protein responses, the distinction can be biological. Two alternative sequence backgrounds may support mutually exclusive communication paths. Their mean graph suggests a connected route through all three positions, while no individual protein realizes that route. A spectral analysis of $\mathbb{E}_C A(C)$ therefore answers a population-level aggregation question, not a statement about a

typical sequence. Context-specific projectors or their distribution must be retained when that difference matters.

A machine-checked certificate accompanies this example (`lean/OperatorGraph/MeanSpectrum.lean`): all eigenvalue witnesses for both context spectra $\{0, 0, 2\}$, the mean spectrum $\{0, 1/2, 3/2\}$, and the linear independence that exhausts each spectrum are formalized in Lean 4, kernel-checked. The mean graph, mean Laplacian, mean spectrum, and distribution of spectral projectors answer different questions. Because this construction begins after symmetrization, it compares undirected shadows only; context-dependent detailed-balance defect, circulation, and directed singular subspaces must be compared before that projection.

Finally, a null-relative Laplacian spectrum must account for zero modes. For unnormalized Laplacians one may restrict to $\mathbf{1}^{\perp}$; for normalized graphs with unequal degree modes or unequal null spaces, use an explicitly declared ridge $\epsilon > 0$. Generalized modes then extremize

$$\gamma(u) = \frac{u^{\mathsf{T}} L_{\mathrm{real}} u}{u^{\mathsf{T}} L_0 u + \epsilon \|u\|_2^2}. \tag{11.52}$$

Small and large values identify directions that are respectively smoother or rougher in the real graph than under the null geometry. Significance should be calibrated from replicate null graphs for eigenvalues, gaps, localization, and spectral projectors. The position Laplacian, signed Laplacian, connection Laplacian, categorical projector P_i, and Potts pair-table redundancy therefore occupy distinct coordinates of the framework.

Coda: spectra inherit every earlier choice

Scalar modes describe positional smoothness; signed modes retain a chosen polarity; connection modes compare transported categorical directions; directed decompositions retain circulation. Gauge projection, scalarization, symmetrization, normalization, and context averaging determine which of these objects is available. Near a degenerate band, the subspace projector is the stable comparison.

A physical mode needs a corresponding operator: vibration requires stiffness and a mass metric, while relaxation requires a dynamical generator. In particular, categorical connection modes act on amino-acid contrasts, not atomic displacements. Their correspondence with sectors or allosteric motion must be tested against the relevant observations.

The next part asks whether a trainable architecture can realize the declared interaction under its information and execution constraints.

Part IV

Realizability, Evidence, and Interpretation

Chapter 12

Architectural Consequences and Realizability

Once an interaction object has been defined, it is tempting to design an architecture that emits it directly. The difficulty is that semantic requirements become information constraints. If an edge is claimed to be invariant under substitutions at its endpoints, then the mechanism choosing that edge cannot depend on those endpoint values. If a zero edge is claimed to save computation, the implementation must actually skip its work. If an update is claimed to descend an energy, its memories, state variables, and parameter tying must satisfy the assumptions of the descent proof.

These constraints are easy to blur because neural architectures combine several roles in one tensor. An attention coefficient can select a route, scale a message, and contribute to an output response. A pair representation can store persistent state or merely cache a transient feature. A soft gate can express uncertainty without producing exact support. The operator framework separates these roles before asking which implementation is efficient.

The next two chapters develop complementary results. This chapter treats routing geometry, persistent relational state, operator reuse, ordered composition, sparse execution, and structural alignment. These are capacity and systems questions. Chapter 13 then treats endpoint-removable context, strict targetwise support, affine deletion, and optional MSA supervision as measurability and information questions.

The separation matters because representational capacity does not settle conditional meaning. A router may compute an accurate adaptive graph while reading evidence that a later endpoint intervention is supposed to hold fixed. Conversely, withholding endpoint identities can create an irreducible prediction floor even for a high-capacity architecture.

Realizability marks the boundary between a mathematically well-defined interaction and a trainable mechanism that can infer or execute it under finite information and compute. This chapter begins with capacity and execution; the next chapter isolates the information contract.

The operator coordinates distinguish a semantic interaction object from an implementation that computes it. Support selection, routing normalization, persistent state, endpoint measurability, and update dynamics are independent architectural decisions. Endpoint measurability, support invariance, and state–energy compatibility are semantic requirements. Signed routing, affine deletion, persistent pair tracks, and optional-MSA supervision form a menu of constructive realizations. Other realizations can satisfy the same requirements.

The realizability analysis uses three evidence levels:

Evidence level	Established in this part	Remaining evaluation
Exact result	Algebraic identities, rank and coverage bounds, deletion conditions, and operation counts under their stated assumptions	Assumption checks on each implementation and numerical agreement with the identities
Constructive design	Executable recurrences, routing rules, state updates, and sparse schedules	Optimization stability, approximation error, and compatibility with efficient kernels
Empirical hypothesis	Operator reuse and context-dependent routing may improve quality per stored parameter or per unit of measured compute	Matched-data, matched-parameter, and matched-compute comparisons with confidence intervals and ablations

The labels apply locally: a proposition about a frozen route establishes an identity for that route, while model quality and systems performance remain empirical quantities.

12.1 Routing geometry and persistent relational state

Standard attention chooses a nonnegative row-stochastic route

$$\alpha_{ij} = \frac{e^{z_{ij}}}{\sum_k e^{z_{ik}}}, \qquad y_i = \sum_j \alpha_{ij} V_j. \tag{12.1}$$

For fixed values, y_i lies in $\operatorname{conv}\{V_j\}$. A more general interaction operator may use signed coefficients with bounded total variation. If $\sum_j |a_{ij}| \le 1$, then

$$y_i = \sum_j a_{ij} V_j \in \operatorname{conv}(\{0\} \cup \{V_j, -V_j\}_j), \tag{12.2}$$

which permits source-specific contrasts and cancellation. As one illustrative design, a continuous signed sparse map is

$$w_{ij} = \operatorname{sign}(z_{ij})(|z_{ij}| - \lambda)_+, \qquad a_{ij} = \frac{w_{ij}}{\epsilon + \sum_k |w_{ik}|}. \tag{12.3}$$

It has exact zeros and bounded total variation. Its relative performance against softmax, sparsemax, entmax, and other signed attention rules is an empirical question [67, 75, 102].

Routing geometry is separate from edge-value capacity. A node-only message-passing layer retains $O(Ld)$ persistent states even if it evaluates dense transient connectivity. A dense pair track retains $O(L^2 d_p)$ relational state, as in structure models with explicit pair representations [32]. A degree-k operator graph retains $O(Lkd_p)$ edge state once support is known. Categorical rank and interaction order are independent capacity coordinates.

State dynamics is another coordinate. The same $\mathbb{K}_z$ may define a one-pass readout, an iterated map, or part of a descent dynamics. A Hopfield claim combines a specified state space, fixed memories, update schedule, and Lyapunov decrease in addition to graph support.

The chapter now turns from admissible routing geometries and state carriers to theorem-driven capacity and execution statements. Definitions, rank bounds, and operation counts are exact under their stated assumptions; the architecture assembled from them is a constructive design whose optimization and quality remain empirical.

12.2 Parameter efficiency from operator reuse

Parameter efficiency has several inequivalent meanings. For a width-d Transformer block with feed-forward width d_{ff}, square query, key, value, and output projections give the leading parameter count

$$P_{\mathrm{block}} \simeq 4d^2 + \nu d d_{\mathrm{ff}}, \qquad \nu = \begin{cases} 2, & \text{ordinary two-matrix feed-forward block,} \\ 3, & \text{gated three-matrix feed-forward block,} \end{cases} \tag{12.4}$$

up to biases and normalization parameters. Grouped or multi-query attention changes the key–value part of the first term. A useful resource ledger is the typed tuple

$$\mathscr{R}(B, L, p, \mathcal{H}) = (P_{\mathrm{total}}, P_{\mathrm{active}}, F, T_{\mathrm{latency}}, M_{\mathrm{context}}, M_{\mathrm{train}}, B_{\mathrm{storage}}, C_{\mathrm{comm}}), \tag{12.5}$$

recording total and active parameters, arithmetic work, latency, inference-time context memory, training activations, stored bytes, and communication. These quantities require a declared batch size B, sequence length L, numerical precision p, and hardware–kernel protocol $\mathcal{H}$; P_{active} also requires a per-token, per-example, or per-batch convention. Sparse routing primarily changes sequence-dependent computation and activations. Mixture-of-experts routing can increase P_{total} while controlling tokenwise P_{active} and F, although batchwise expert coverage can be much larger. A recurrent state can make M_{context} constant in L for its recurrent layers, while training activations can still grow with L and a hybrid attention layer retains a length-dependent key–value cache. Quantization changes B_{storage} and usually leaves the number of scalar parameters unchanged. Cross-depth sharing directly changes P_{total} and may retain essentially the same number of sequential update steps. Claims of "efficiency" should therefore name the coordinates and protocol of Eq. 12.5.

Architecture hypothesis and failure criteria. The proposed dictionary tests whether a small set of reusable channel maps, combined through state-dependent coefficients and ordered depth, can preserve task quality while reducing P_{total} or a declared systems cost. A matched evaluation varies dictionary size, factor rank, routing sparsity, and recurrent depth while holding training data, optimization budget, and output head fixed. It reports held-out likelihood and task endpoints together with measured forward and backward latency, peak memory, communication, and support-discovery cost. Static coefficients, untied depth, dense routing, and a conventional Transformer provide the required ablations. The hypothesis fails in a tested regime when the tied dictionary loses held-out quality at matched resources, when state-dependent coefficients add no reproducible gain over the static control, or when routing and kernel overhead erase the predicted systems advantage.

12.2.1 Attention as a separated operator

Let $H \in \mathbb{R}^{L\times d}$ store token states by row. In the column-state convention of Eq. 7.10, absorb the value projection and the corresponding output block of head h into $B_h = W_{O,h}W_V^{(h)} \in \mathbb{R}^{d\times d}$, for which $\mathrm{rank}(B_h) \le d_v$. Write its position-space route as $P_h(H) \in \mathbb{R}^{L\times L}$. For ordinary multihead attention with value bias, dropout, and value-side positional modulation omitted, the value path is

$$Y = \sum_{h=1}^{H_{\mathrm{head}}} P_h(H) H B_h^{\mathsf{T}}. \tag{12.6}$$

For a fixed routing state H_0, define the linear map

$$\mathcal{V}_{H_0}(X) = \sum_{h=1}^{H_{\mathrm{head}}} P_h(H_0) X B_h^{\mathsf{T}}. \tag{12.7}$$

Using column-stacking vectorization,

$$\operatorname{vec}(\mathcal{V}_{H_0}(X)) = \left[\sum_{h=1}^{H_{\text{head}}} B_h \otimes P_h(H_0)\right] \operatorname{vec}(X). \tag{12.8}$$

For a linear map on position–channel arrays, define its operator-Schmidt rank as the smallest s for which it can be written $\mathcal{L}(X) = \sum_{r=1}^{s} A_r X C_r^{\mathsf{T}}$ with separate position maps A_r and channel maps C_r. This is a tensor separation rank, distinct from the ordinary matrix rank of $\mathcal{L}$ on $\mathbb{R}^{Ld}$.

Proposition 12.1 (Frozen-route operator-Schmidt bound)**.** *The position–channel operator-Schmidt rank of the frozen value operator in Eq. 12.8 is at most H_{head}. Its directed edge map is*

$$\mathcal{A}_{i\leftarrow j}(H_0) = \sum_{h=1}^{H_{\text{head}}} [P_h(H_0)]_{ij} B_h \in \operatorname{span}\{B_1, \ldots, B_{H_{\text{head}}}\}. \tag{12.9}$$

Proof. Equation 12.8 is an explicit sum of H_{head} position–channel Kronecker products. Directly expanding row i of Eq. 12.6 gives Eq. 12.9, which is Proposition 7.1. □

Dense $L \times L$ routing therefore need not contain L^2 learned parameter blocks. It evaluates a small shared family of channel operators at many dynamically weighted edges. The dense route still costs $O(L^2 d)$ leading arithmetic. Materializing every head route costs $O(H_{\text{head}} L^2)$ activation memory; exact tiled kernels can avoid storing the full route while retaining quadratic arithmetic [15]. Since each P_h is row-stochastic, Eq. 12.9 also obeys $\sum_j \mathcal{A}_{i\leftarrow j} = \sum_h B_h$ in the dropout-free computation. The separation-rank bound permits an ordinary matrix rank as large as Ld and has no converse under the nonnegative routing constraints.

The frozen map is a stop-gradient value-path audit and agrees with the actual value output at $X = H_0$. The full differential is

$$DY(H_0)[\Delta] = \sum_h P_h(H_0) \Delta B_h^{\mathsf{T}} + \sum_h DP_h(H_0)[\Delta] H_0 B_h^{\mathsf{T}}. \tag{12.10}$$

Only the first term satisfies Proposition 12.1; route derivatives, earlier layers, and nonlinear blocks contribute to the model-level response.

12.2.2 A recurrent operator dictionary

The separated form suggests sharing at a finer scale than complete layers. Let

$$B_r = U_r V_r^{\mathsf{T}}, \qquad U_r, V_r \in \mathbb{R}^{d\times\rho}, \qquad r = 1, \ldots, R, \tag{12.11}$$

be a reusable dictionary of rank-at-most-ρ channel maps. At recurrent step n, a sparse dynamic operator graph computes

$$m_i^{(n)} = \sum_{j\in\mathcal{N}_i^{(n)}} \sum_{r=1}^{R} a_{ijr}^{(n)} B_r h_j^{(n)}, \qquad a_{ijr}^{(n)} = \left[g_\theta\left(h_i^{(n)}, h_j^{(n)}, S^{(n)}\right)\right]_r, \tag{12.12}$$

followed by

$$h_i^{(n+1)} = h_i^{(n)} + \eta_n \left[m_i^{(n)} + f_\theta\left(h_i^{(n)}, S^{(n)}\right)\right], \tag{12.13}$$

$$S^{(n+1)} = \mathcal{M}_\theta\left(H^{(n)}\right) S^{(n)} + \mathcal{U}_\theta\left(H^{(n)}\right). \tag{12.14}$$

Here $S^{(n)}$ declares a compressed memory as a separate coordinate from the route matrix. The dictionary, router, node update, and state update can be tied across recurrent depth. A small number C of phase-specific cells or low-rank step adapters can preserve coarse depth specialization.

For N conventional untied blocks, Eq. 12.4 gives

$$P_{\text{standard}} \simeq N\left(4d^2 + \nu d d_{\text{ff}}\right). \tag{12.15}$$

A conservative tied construction with $C \ll N$ full cells and rank-ρ step adapters has

$$P_{\text{tied}} \simeq C\left(4d^2 + \nu d d_{\text{ff}}\right) + O(N\rho(d + d_{\text{ff}})). \tag{12.16}$$

Replacing full channel maps inside a shared cell by Eq. 12.11 stores $2Rd\rho$ raw factor parameters, together with the parameters of g_θ, f_θ, $\mathcal{M}_\theta$, and $\mathcal{U}_\theta$. Each factorization has the reparameterization $U_r \mapsto U_r Q$, $V_r \mapsto V_r Q^{-\mathsf{T}}$ for invertible Q, so its intrinsic generic rank-ρ dimension is $\rho(2d - \rho)$; $2d\rho$ is its raw storage count. The edge coefficients are activations, not learned edgewise parameters. Each edge map has rank at most $\min\{d, R\rho\}$; retaining all R coefficients on a degree-k graph costs $O(LkR)$ activation memory and arithmetic still carries an R factor. Selecting top-k edges by dense all-pairs scoring can itself cost $O(L^2)$. These counts omit embeddings and output heads. They are accounting hypotheses; realized quality and compute remain empirical quantities [27].

12.2.3 Capacity from ordered operator words

Weight sharing does not force every effective depth map to be identical. Consider the channel subflow obtained after freezing and scalarizing one position-space path:

$$h^{(n+1)} = T_n h^{(n)}, \qquad T_n = I + \eta_n \sum_{r=1}^{R} a_{nr} B_r. \tag{12.17}$$

Let

$$\mathrm{Alg}(B_1, \ldots, B_R) = \mathrm{span}\left\{I,\ B_{r_k} \cdots B_{r_1} : k \geq 1,\ r_1, \ldots, r_k \in \{1, \ldots, R\}\right\} \subseteq \mathbb{R}^{d \times d} \tag{12.18}$$

denote the unital algebra generated by the dictionary.

Proposition 12.2 (Ordered-word expansion of tied depth)**.** *For any frozen coefficient sequence* $\{a_{nr}\}$*, the depth-N transport* $T_{N-1} \cdots T_0$ *belongs to* $\mathrm{Alg}(B_1, \ldots, B_R)$ *and expands as*

$$T_{N-1} \cdots T_0 = I + \sum_{k=1}^{N} \sum_{0 \leq n_1 < \cdots < n_k < N} \sum_{r_1, \ldots, r_k = 1}^{R} \left[\prod_{\ell=1}^{k} \eta_{n_\ell} a_{n_\ell r_\ell}\right] B_{r_k} \cdots B_{r_1}. \tag{12.19}$$

Proof. Expand the ordered product of the N factors in Eq. 12.17. Selecting k nonidentity terms at times $n_1 < \cdots < n_k$ places the latest selected operator on the left and gives the word $B_{r_k} \cdots B_{r_1}$. □

Thus a compact generator set can produce a much larger family of effective maps through ordered composition. With a state-dependent router, the coefficients are evaluated along the trajectory, so the complete input–output map is nonlinear; the realized linear channel transport at each trajectory still has the expansion above as a conditional, stop-gradient transport. The $(R+1)^N$ formal terms are not independent degrees of freedom: words can collapse through commutation, nilpotence, common invariant spaces, Cayley–Hamilton identities, and low-rank overlap, while their coefficients remain coupled products of router outputs. Full algebra generation is a span condition; reachability by one finite-depth product requires an additional argument.

The factorized dictionary gives a sharper ceiling. Define

$$\mathcal{U} = \mathrm{span}_r \,\mathrm{col}(U_r), \qquad \mathcal{V} = \mathrm{span}_r \,\mathrm{col}(V_r), \qquad u = \dim \mathcal{U}, \quad v = \dim \mathcal{V}. \tag{12.20}$$

Proposition 12.3 (Coverage ceiling of a factorized dictionary)**.** *Every nonempty word in the generators of Eq. 12.11 has rank at most ρ, image contained in $\mathcal{U}$, and row space contained in $\mathcal{V}$. Consequently,*

$$\mathrm{Alg}(B_1,\dots,B_R) \subseteq \mathrm{span}\{I\} + \mathcal{U}\otimes\mathcal{V}, \qquad \dim \mathrm{Alg}(B_1,\dots,B_R) \le \min\{d^2, 1+uv\}. \tag{12.21}$$

Generating all of $\mathbb{R}^{d\times d}$ therefore requires $u = v = d$, and in particular $R\rho \ge d$.

Proof. For $k \ge 2$,

$$B_{r_k}\cdots B_{r_1} = U_{r_k}\left(V_{r_k}^{\mathsf{T}}U_{r_{k-1}}\right)\cdots\left(V_{r_2}^{\mathsf{T}}U_{r_1}\right)V_{r_1}^{\mathsf{T}}. \tag{12.22}$$

The stated image, row-space, and rank bounds follow directly; their linear span lies in the space of maps from $\mathcal{V}$ to $\mathcal{U}$, plus the identity. Finally $u, v \le R\rho$. □

For a one-step state-dependent map $F(h) = T(h)h$, the full Jacobian contains

$$DF(h) = T(h) + \eta\sum_{r=1}^{R}(B_r h)\nabla a_r(h)^{\mathsf{T}}, \tag{12.23}$$

with further cross-terms from S and f_θ. The outer-product terms can leave the generated algebra. Proposition 12.2 therefore describes frozen channel transport, while the response audit must include router derivatives and position-space products. The relevant capacity diagnostics are the dimension, conditioning, and reachable subset of the word span. The resulting design principle is

$$\boxed{\text{small operator dictionary} + \text{state-dependent coefficients} + \text{ordered composition}} \tag{12.24}$$

Low rank alone supplies no expressivity guarantee. The ordered-word algebra constrains an internal latent channel subflow. Its externally observable categorical capacity is further limited by the injection and readout maps: Eq. 7.22 bounds every local response by the ranks of $\mathsf{E}_{i\leftarrow j,C}$ and $\mathsf{R}_{i\leftarrow j,C}$ even when the intervening latent algebra is full. Conversely, router derivatives and nonlinear paths may leave the frozen word span, so neither the dictionary algebra nor the categorical audit alone determines the other.

12.2.4 Computational realizability is a separate audit

The algebraic results above declare which maps can be represented. An execution schedule must also specify which intermediate arrays are formed, how support is discovered, which operations are sequential, and whether the resulting kernels use the hardware efficiently. These questions can be analyzed without expanding any operator word.

Suppress the recurrent-step superscript and define $A_r \in \mathbb{R}^{L\times L}$ by

$$[A_r]_{ij} = a_{ijr}\mathbf{1}\{j \in \mathcal{N}_i\}. \tag{12.25}$$

Thus rows index targets, columns index sources, and the declared directed support is $\mathcal{E} = \{(j,i) : j \in \mathcal{N}_i\}$ with $E = |\mathcal{E}|$. With token states stored by row, the complete dictionary message in Eq. 12.12 is exactly

$$M_{\mathrm{msg}} = \sum_{r=1}^{R} A_r H V_r U_r^{\mathsf{T}}. \tag{12.26}$$

Since $B_r = U_r V_r^{\mathsf{T}}$, column-stacking vectorization gives

$$\operatorname{vec}(M_{\mathrm{msg}}) = \left(\sum_{r=1}^{R} B_r \otimes A_r \right) \operatorname{vec}(H). \tag{12.27}$$

With the route matrices held fixed, the parenthesized operator is the dictionary contribution to the token-state block of $\mathbb{J}_C^{\mathrm{lat}}$ in Eq. 7.28. State-dependent routes add derivatives through $A_r(H, S)$; node, state, residual, and nonlinear paths add the remaining Jacobian blocks.

Proposition 12.4 (Conditional factorized execution on supplied sparse support)**.** *Assume the matrices A_r have already been supplied and $E \le Lk$. Equation 12.26 can be evaluated by*

$$Z_r = HV_r, \qquad T_r = A_r Z_r, \qquad M_{\mathrm{msg}} = \sum_{r=1}^{R} T_r U_r^{\mathsf{T}}, \tag{12.28}$$

using the exact multiply–accumulate count

$$\mathrm{MAC}_{\mathrm{message}} = R\rho(2Ld + E) = O(LR\rho(2d + k)) \,. \tag{12.29}$$

Counting a multiplication and addition as separate FLOPs approximately doubles this quantity. Materializing sparse indices, all route values, all Z_r, T_r, and the output uses $O(E + ER + LR\rho + Ld)$ forward storage up to constant factors. Streaming or recomputing both the atom intermediates and route values reduces the forward workspace to $O(E + L\rho + Ld)$ when one atom is processed at a time. Reverse-mode differentiation has a separate saved-tensor ledger.

Proof. For each r, the dense thin products HV_r and $T_r U_r^{\mathsf{T}}$ cost $Ld\rho$ multiply–accumulates each, while sparse–dense multiplication by A_r costs $E\rho$. Sum over R and use $E \le Lk$. The storage bounds follow from the shapes of the support, coefficient, intermediate, and output arrays. □

The factorization replaces an independent $O(LkRd\rho)$ edgewise implementation by shared thin projections and sparse aggregation. Its count is conditional on supplied routes. End-to-end execution has three additional costs.

Finding support. For candidate sets $|\mathcal{C}_i| \le c$, define

$$\mathcal{N}_i = \mathrm{TopK}_{j \in \mathcal{C}_i} \, s_\theta(h_i, h_j, S). \tag{12.30}$$

A width-d_s dot-product scorer costs $O(Ldd_s + Lc\,d_s)$ before selection and coefficient production. All-pairs candidates set $c = L$, so top-k output alone gives no subquadratic search guarantee. Local or block candidates, approximate content routing, and rebuilt indices make different coverage and execution tradeoffs [4, 37, 60, 80]. Periodic global attention also retains a quadratic term when its frequency is fixed.

Repeated execution. With C shared cells evaluated over N recurrent steps,

$$P_{\mathrm{depth}} \simeq CP_{\mathrm{cell}}, \qquad F_{\mathrm{depth}} \simeq NF_{\mathrm{cell}}, \qquad C \ll N. \tag{12.31}$$

Parameter tying saves parameter and optimizer storage. It retains the sequential updates and, without recomputation, their saved training activations. A nonlinear state-dependent router has no general associative-scan guarantee. Node states, sparse support, and route coefficients still scale with L even if a separate recurrent memory has fixed size.

The rest of the cell. For $d_{\text{ff}} = 4d$, an ordinary FFN has approximately $8d^2$ parameters, compared with $4d^2$ for the four attention projections. Compressing messages alone can leave most parameters in the FFN. The complete per-step ledger is

$$F_{\text{step}} = F_{\text{candidate}} + F_{\text{score}} + F_{\text{message}} + F_{\text{state}} + F_{\text{FFN}}. \tag{12.32}$$

Sparse kernel overhead and memory traffic can also erase an arithmetic saving. Benchmarks should time the supplied-route message, route discovery, and complete recurrent cell separately, including backward execution, peak memory, and matched-quality comparisons.

The capacity audit can be conducted outside the training path. Dictionary coverage is obtained from

$$u = \operatorname{rank}[U_1 \ \cdots \ U_R], \qquad v = \operatorname{rank}[V_1 \ \cdots \ V_R]. \tag{12.33}$$

Rank tolerances and singular values matter; at large width, sampled word-span probes and Jacobian–vector products replace explicit algebra bases and full response Jacobians. One sufficient frozen-family stability certificate is

$$T_n^{\mathsf{T}} Q T_n \preceq \gamma^2 Q, \qquad Q \succ 0, \tag{12.34}$$

over a declared update set. Failure to find a common Q does not establish instability. State-dependent and nonlinear response requires the full Jacobian audit.

12.2.5 Protein realization and tests

For proteins, the shared B_r can encode reusable computational primitives for amino-acid exchange, charge, packing, or geometric compatibility. Their scientific interpretation remains an experimental question. The coefficients $a_{ijr}^{(n)}$ supply a sequence- or family-conditioned graph over those shared primitives. Column support can combine fixed chain edges with learned top-k long-range edges. MSA rows can retain permutation equivariance or use an explicitly declared tree geometry, while columns retain chain order. Persistent pair features on selected edges cost $O(Lkd_p)$ memory at degree k, compared with $O(L^2 d_p)$ for a dense pair track. Periodic global attention or latent-memory retrieval can refresh information omitted by a local sparse support.

The construction has identifiable failure modes. A fixed-dimensional recurrent state can collide or forget; inadequate $\mathcal{U}$ or $\mathcal{V}$ coverage creates a hard bottleneck on the dictionary message path; sparse support can miss long-range dependencies; router collapse or top-k route churn can impair optimization; repeated shared updates can oversmooth, explode, vanish, or converge to unhelpful fixed points; and lower parameter count can require more iterative computation. A frozen-family stability audit can use Eq. 12.34; state dependence and nonlinear paths require corresponding Jacobian bounds. External retrieval may still be needed for factual memory. Endpoint-removable conditional semantics also continue to require the measurability conditions developed below.

Cross-layer sharing in ALBERT, recurrent depth in the Universal Transformer, fixed-point depth in deep equilibrium models, and shared key–value heads in grouped-query attention provide precedents for individual parts of this construction [63, 19, 3, 1]. R-GCN basis decomposition is a direct precedent for a shared graph-operator dictionary; dynamic filter networks provide input-conditioned operator generation; and low-rank adapters provide a standard depth-modulation mechanism [85, 56, 27]. KDA and related state-space attention supply precedents for compressed recurrent context [34, 35]. A distinct contribution would require a typed operator-algebra analysis, parameter–expressivity–stability bounds, and matched experiments at equal data and compute. Reports should separate the coordinates of Eq. 12.5 and include prediction quality, latency, route audits, state-propagator audits, and interventional-response audits.

These propositions establish representability and message execution conditional on supplied support and coefficients. End-to-end efficiency additionally requires candidate generation, coefficient routing, sparse aggregation, nonlinear updates, recurrent backward propagation, and communication to satisfy the same resource ledger.

The final block changes evidence level. It uses concrete protein generators to examine whether representation, relational architecture, objective, data curation, and evaluation are aligned with one endpoint. Pallatom examines structural alignment; RFD4-Proteina then makes multiplicative writes, hidden-state refinement, and output response explicit. These cases illustrate the preceding design principles without turning them into a theorem about biological validity.

12.3 Pallatom: structural alignment across scales

Pallatom allows the representation, relational architecture, training distribution, objective, and evaluation endpoint to be examined together [78]. The relevant question is which protein regularities each choice makes easy to represent, and which alternatives the reported tests exclude.

The first bias is introduced before the neural network sees any data. A residue with variable chemical topology is embedded into fourteen coordinate slots. Native heavy atoms occupy their standard slots; absent atoms are represented by virtual points at the C_α coordinate:

$$(x, a) \xmapsto{\Psi_{14}} r_0 \in \mathbb{R}^{N\times 14\times 3}. \tag{12.35}$$

This map converts transdimensional generation into ordinary Euclidean point-cloud diffusion. It asserts that a residue is the stable compositional unit, that fourteen slots suffice for the canonical amino acids, and that the artificial coincidence of unused slots at C_α is a learnable nuisance. These assumptions make the problem computationally regular; their status is representational, while molecular cardinality and atom existence belong to the physical ontology.

The second bias is relational and multiscale in the sense emphasized by Battaglia et al. [39]. Pallatom uses local atom-level attention for fine geometry and a globally communicating residue track with pair features for fold-scale organization. Residue features are broadcast to atoms; atom features are aggregated back to residues; denoised coordinates are recycled into pair features; and triangle updates encourage relational consistency. The architecture therefore assumes

$$\text{local atomic geometry} \longleftrightarrow \text{residue identity and state} \longleftrightarrow \text{global residue–residue organization}. \tag{12.36}$$

This hierarchy is well matched to folded proteins and avoids global attention over every atom pair. Long-range atom–atom effects are therefore mediated through a residue-level bottleneck, which defines the model's relational capacity at fold scale.

Two multiscale bottlenecks. Pallatom recycles partially denoised coordinates into residue-level pair or self-conditioning features. OpenDDE instead lifts residue memory to structural roles and, in the reported refiner schematic, holds the refined condition across atom denoising (Chapter 10). The comparison locates both the global communication bottleneck and the direction of coordinate feedback; exact recycling contracts remain implementation-specific. Under either contract the finer tier inherits, but cannot exceed, the deposited condition (Proposition 10.2).

The slogan "learn P(all-atom)" should be read with care. Pallatom supplies the denoiser with a reduced side-chain labeling, and several amino acids share geometrically degenerate atom14 patterns. Coordinates alone therefore leave sequence ambiguous. The trained object combines coordinate score matching with supervised sequence cross-entropy, residue- and atom-level distogram

losses, smooth-LDDT loss, and intermediate supervision. Schematically,

$$\mathcal{L} = \lambda\mathcal{L}_{\text{atom}} + \alpha_0\mathcal{L}_{\text{seq}} + \alpha_1\mathcal{L}_{\text{LDDT}} + \alpha_2\mathcal{L}_{\text{dist,res}} + \alpha_3\mathcal{L}_{\text{dist,atom}} + \alpha_4\mathcal{L}_{\text{intermediate}}. \tag{12.37}$$

Thus the representation, architecture, and auxiliary targets together make the sequence–structure relation learnable beyond the information in an unsupervised coordinate density alone. Pallatom encourages the $SE(3)$ identity statistically through coordinate centering, random rigid-motion augmentation, and a Kabsch-aligned coordinate loss. The function class still admits departures from exact equivariance. This distinction is substantive: symmetry learned from augmentation is an approximate empirical regularity, whereas symmetry imposed by the architecture holds by construction.

The training set is restricted to short monomers, with crops of at most 128 residues and filtered PDB and AFDB structures selected for confidence, compactness, and redundancy. The learned density is better written as

$$p_\theta(r \mid r \in \mathcal{F}_{\text{compact,designable}}(\text{PDB} \cup \text{AFDB})) \tag{12.38}$$

than as a universal distribution over proteins. This is a legitimate design choice: it aligns the training set with the desired endpoint of compact, prediction-compatible monomers. It also limits its claims about disordered proteins, complexes, alternate states, membrane environments, and functional negative examples.

The ablations support this alignment interpretation. In the reported setup, atom14 achieves 87% all-atom designability, compared with 5% for a sequence-guided hybrid14 representation; removing differentiable recycling lowers it to 21%. These results identify the computational representation that works with this denoiser and objective. A claim about physical ontology would require independent physical evidence. Indeed, changing the sampling step scale raises reported designability while reducing diversity and shifting the fraction of all-helical outputs. The sampler is therefore part of the inductive bias and changes which region of the learned distribution is visited.

The primary designability endpoint requires ESMFold mean pLDDT above 80 and all-atom RMSD below 2 Å between a generated structure and the structure predicted from its generated sequence. This tests predictor-mediated sequence–structure consistency. Stability, activity, specificity, and equilibrium weights require separate measurements.

Pallatom remains context-dependent even though its final sequence head is residuewise. The head pools atomic embeddings and applies a linear classifier after those embeddings have received global residue, pair, triangle, and recycled-coordinate information. The output logit at residue i is therefore a function of the generated molecular context. A local readout can therefore express pairwise and higher-order context already mixed into its input representation.

Pallatom shows how representation, architecture, objective, and evaluation can point toward one endpoint. The next case follows the computation more closely: RFD4-Proteina writes single states into pair memory, repeatedly refines that memory, and reads it through sequence and coordinate outputs. Its explicit update rules let us ask which interaction components a write introduces and which survive the complete input–output path.

12.4 RFD4-Proteina: writing, refining, and reading relational state

RFD4-Proteina writes single states multiplicatively into pair memory, uses that memory to bias token routing, refines both tracks repeatedly within one generation step, and delivers its conditions

through declared gauges [11]. The operator audit isolates four objects: the centered component introduced by a write, the input response of repeated refinement, the gauge content of a delivered condition, and the commitment of a readout along the generation orbit.

12.4.1 Declare the represented objects

Each supported polymer residue has an anchor $x_i \in \mathbb{R}^3$ and a latent $\ell_i \in \mathbb{R}^{32}$. The autoencoder reconstructs identity, relative atom coordinates, and atom presence; other components can remain explicit atom tokens. Its self-attention mixes tokens without a persistent pair track, while the flow trunk maintains single states s_i and pair states z_{ij} (Supplementary Sections S3.1–S3.2).

A per-token latent is therefore not an endpoint-local encoding: changing one source may change several latents. Its width 32 bounds the differential rank of a path passing exclusively through that latent, while anchors, other latents, and pair states supply additional paths. The categorical rank bound in Eq. 7.22 applies only to a specified intervention and readout.

12.4.2 Why multiplicative writes survive a pair audit

In the flow trunk, normalized single states are projected and written into the pair track as

$$u_i = \mathrm{LayerNorm}(s_i), \qquad a_i = W_a u_i, \qquad b_j = W_b u_j, \qquad z_{ij} \leftarrow z_{ij} + a_i \odot b_j. \tag{12.39}$$

Each channel lets one endpoint gate the other's contribution. The reference-centered geometry gives a precise comparison with additive writing.

Proposition 12.5 (Centered additive and multiplicative writes)**.** *Fix an endpoint-excluding background, positive references p_i, p_j, and feature tables $A \in \mathbb{R}^{q_i \times d_p}$ and $B \in \mathbb{R}^{q_j \times d_p}$ that depend only on their respective endpoint categories. Let $w \in \mathbb{R}^{d_p}$ be a fixed linear channel readout. The additive and multiplicative write tables are*

$$T^{+} = (Aw)\mathbf{1}^{\mathsf{T}} + \mathbf{1}(Bw)^{\mathsf{T}}, \tag{12.40}$$

$$T^{\times} = A\,\mathrm{diag}(w)B^{\mathsf{T}}. \tag{12.41}$$

Their centered components satisfy

$$P_i T^{+} P_j^{\mathsf{T}} = 0, \tag{12.42}$$

$$P_i T^{\times} P_j^{\mathsf{T}} = (P_i A)\,\mathrm{diag}(w)(P_j B)^{\mathsf{T}}, \tag{12.43}$$

$$\mathrm{rank}(P_i T^{\times} P_j^{\mathsf{T}}) \le \min\{d_p, q_i - 1, q_j - 1\}. \tag{12.44}$$

Proof. Since $P_i\mathbf{1} = P_j\mathbf{1} = 0$, both endpoint-only summands vanish. Matrix multiplication gives Eq. 12.43, and the rank bound follows from the ranks of its factors. □

One product channel supplies an outer product of centered category profiles. It vanishes if either profile is constant or its readout weight is zero; dependence and cancellation can reduce the rank across channels. In the actual trunk, context-mixed s_i may already depend on both endpoints, so an additive write can inherit an interaction and later nonlinearities can create one. A multiplicative component can also disappear at readout. The proposition isolates the write under endpoint-local features and a fixed linear readout; it does not predict the complete model's ablation performance.

12.4.3 From pair memory through routes to output response

RFD4 projects normalized pair states into head-specific biases:

$$\alpha_{ij}^h = \mathrm{softmax}_j\left(\frac{(q_i^h)^\mathsf{T} k_j^h}{\sqrt{d_h}} + \beta_{ij}^h\right), \qquad \beta_{ij}^h = b_h(\mathrm{Norm}(z_{ij}))\,. \tag{12.45}$$

Thus pair memory affects token routing, and routed token states subsequently write back into pair memory. Outgoing and incoming triangle multiplication further update a pair through pairs sharing a token, schematically

$$\Delta z_{ij} \sim \sum_k A(z_{ik}) \odot B(z_{jk}). \tag{12.46}$$

The third index gives a relational path; its statistical order requires the audit of Chapter 5. All single, pair, coordinate, and latent paths enter $\mathbb{J}^{\mathrm{lat}}$ in Eq. 7.27, followed by the encoder and readout maps. Freezing α_{ij}^h keeps only value transport. The full response also includes the query, key, pair-bias, gate, and upstream derivatives of Eq. 12.10.

The output contract needs equal care. The flow trunk predicts coordinate and latent velocities, with auxiliary heads for sequence identity and motif correspondence. The auxiliary heads are trained on detached single states, but their outputs still depend on those states at inference. Sequence predictions also condition the decoder. Consequently, an audit must include this conditioning path when it is used. Argmax choices, sampling, and Hungarian motif assignment are discrete operations: local derivatives describe a fixed smooth branch, while finite interventions can also change the selected branch.

RFD4 does not automatically supply the masked conditional required by Eq. 4.5. To extract such a categorical object, an investigator must define the target-withholding and source-substitution protocol. The protocol must remove target identity from every input intended to exclude it, including cached latents, side-chain geometry, and atom-presence patterns. Re-encoding the intervened structure and changing one latent while holding the others fixed are different experiments.

12.4.4 Conditions arrive through a gauge

Motif and target geometry enter either as absolute coordinates or as internal coordinates, produced by centering and independently rotating each coordinate group; noise added to unconditioned groups carries independent per-group offsets; and noise aligned to stereochemical frames fixes the corresponding angular gauge. Propositions 10.3 and 10.4 type these choices.

Absolute conditioning transmits the deposited frame with the geometry; the frame is an arbitrary gauge element, and the output placement is constrained by it without gaining information. The randomized internal-coordinate representative carries exactly the invariant geometry, so placement must arrive through other inputs or the prior, and the conditional support over placements is the orbit rather than a point. Replacing a mandatory origin token by optional, noisily specified chain centroids is the same declaration on the translational gauge. Zero-mean noise would expose the clean placement of every unconditioned group as an interpolant identity; the per-group offsets close the identity, and applying them only outside coordinate-conditioned regions withholds the offset from the condition. The reported exchange of absolute for internal target conditioning is consistent with this typing in direction; the propositions fix no magnitude.

The gauge protocol joins the masking protocol of the previous subsection. An intervention on a condition must declare whether its frame is transmitted or redrawn; replicates over redrawn gauges vary input realizations while holding invariant content fixed; and leakage checks on cached conditions must include frame statistics.

12.4.5 Recycling has repeated input injection

The model refines hidden states repeatedly while holding the current noisy coordinates, latents, and molecular conditions fixed. This places two clocks in the computation: generation time and within-step recurrent depth. Hidden single and pair states are reset at each generation step; within that step, each pass adds projected previous states to the original embeddings.

Write u for the fixed-step inputs, including noisy coordinates, latents, times, and conditions; let $e(u)$ be the initial single/pair embeddings and $m^{(r)}$ their recurrent state. A schematic form of the reported recurrence is

$$m^{(0)} = 0, \qquad m^{(r+1)} = F_\theta\Big(e(u) + \mathcal{P}(m^{(r)}); c_r(u)\Big), \qquad 0 \le r < R, \tag{12.47}$$

where $\mathcal{P}$ includes normalization and projection, and c_r includes pass-dependent conditioning. Hold the discrete pass count R fixed for the differential audit. Along one orbit define

$$A_r = D_1 F_\theta \, D\mathcal{P}(m^{(r)}), \qquad B_r = D_1 F_\theta \, De(u) + D_2 F_\theta \, Dc_r(u), \tag{12.48}$$

with both derivatives of F_θ evaluated at the arguments in Eq. 12.47. The complete input sensitivity obeys

$$Z_0 = 0, \qquad Z_{r+1} = A_r Z_r + B_r, \tag{12.49}$$

$$Z_R = \sum_{k=0}^{R-1} A_{R-1} \cdots A_{k+1} B_k, \tag{12.50}$$

where the product is the identity for $k = R-1$. For a smooth head $o = g(m^{(R)})$, its input response is $Do = Dg(m^{(R)})Z_R$.

The sum is essential: every pass reinjects the original evidence. The homogeneous product $A_{R-1} \cdots A_0$ instead describes propagation of a hidden-state perturbation with inputs fixed. Even a stable homogeneous product can coexist with a substantial forced response from the B_k terms. Parameter sharing also does not make the A_r identical, because they are evaluated at different states. More passes can change response directions and path lengths without adding a new set of trunk parameters; improvement, convergence, and Lyapunov descent each require additional evidence.

The reported training recipe propagates gradients through only the final trunk pass. That optimization convention must be separated from the derivative of the executed inference function. An inference response audit must differentiate through, or numerically perturb, all executed passes rather than inherit the training-time detach operations.

12.4.6 A third clock: readout commitment

Generation time and within-step depth are now separate clocks. Temporal search reads a third: the progress at which a readout of an intermediate state predicts the same readout of the final state. Fix a sampling procedure with declared shared randomness and trajectory states $X^{(s)}$, $0 \le s \le 1$. For a readout f, define its commitment by

$$\kappa_f(s) = 1 - \frac{\mathbb{E}\big[\mathrm{Var}\big(f(X^{(1)}) \mid X^{(s)}\big)\big]}{\mathrm{Var}\big(f(X^{(1)})\big)}. \tag{12.51}$$

Commitment is an estimand, not a local object: it is not a derivative along the orbit and not a function of the forced response in Eq. 12.50. No monotonicity in s is guaranteed, and the ordering of commitment across readouts is an empirical property of the trained field. Measurement

requires protocol declarations: how intermediates are evaluated (single-step clean prediction or completed rollout), which randomness is shared (seeds and branch decisions), the full conditioning path through sequence-conditioned decoding, and the treatment of discrete branch selections. Rank correlations between intermediate and final readouts, as reported for interface area, hydrogen bonding, and predictor confidence, estimate a commitment-type quantity under exactly such a protocol.

Checkpoint termination is a stopping decision measurable in the trajectory filtration; it changes the selected ensemble through a selection layer and leaves the field untouched. Both time-varying recycle schedules and time-varying search rewards allocate against this clock. The observed resolution order—global interface properties before local contacts before predictor confidence—is evidence about the trained field, not a consequence of the architecture.

12.4.7 Probability transport, energy supervision, and selection

Flow matching generates coordinates and residue latents jointly, with temporal score rescaling and stochastic noise injection in the released sampler. The transport distinction of Section 10.6 applies in this joint space under the model's time and noise conventions.

Rosetta supervision fine-tunes atomic reconstruction: energy penalties accompany reconstruction losses, the encoder stays fixed, and low-rank adapters modify the decoder (Supplementary Section S4.3.1). This does not identify a Rosetta Boltzmann law for the complete generator. External search further changes the selected law through user metrics, branching, pruning, and evaluation budget; best-of-noise training selection, interpolated conditioning fields, and score rescaling are further instances of the same operation on the realized law (Definition 10.5 and Proposition 10.6). The resulting predictor-based passing rate and an experimental binding or catalytic endpoint remain separate measurements.

12.4.8 Three tests suggested by the translation

Compare additive and Hadamard writes under matched interventions, references, and readouts, measuring both the centered write and the output response. Then fix a noisy input and vary recurrent depth, reporting homogeneous propagation separately from the forced response in Eq. 12.50; full-generation comparisons must match compute. Finally, carry selected source substitutions through sequence and coordinate readouts to structural and mutational endpoints. Two further protocols accompany the gauge and commitment subsections: replicate audits over redrawn condition gauges at fixed invariant content, and commitment curves measured under a declared rollout protocol alongside the search thresholds that read them. Replicate seeds or shared random numbers can control sampling variation, while finite audits must retain changes of discrete branch.

Coda: capacity and execution require separate ledgers

This chapter moved from routing geometry to persistent pair state, reusable operator dictionaries, ordered composition, sparse execution, and structural alignment. These constructions answer how a relational computation can be represented and executed. Their algebraic capacity, stored parameters, arithmetic work, memory traffic, training stability, and biological performance remain separate coordinates.

RFD4-Proteina makes this separation operational. A multiplicative write can retain a centered bivariate component under endpoint-local features; its contribution to output still passes through routes and readouts. Within-step recycling shares parameters while repeatedly injecting the original

evidence, so its input response is a forced recurrence. Neither calculation supplies a physical energy or a biological validation result.

The next chapter imposes a different kind of constraint. A claimed conditional interaction must remain measurable when its endpoint identities are varied. That requirement restricts which information a router may use and creates exact recoverability limits. Architecture design then becomes an information problem as well as a capacity and systems problem.

Chapter 13

Conditional Realizability and Information Limits

An architecture can have enough parameters to represent a relational map and still lack the information contract required by a particular interpretation. A route selected after reading both endpoint residues is a valid adaptive computation. Holding that same route fixed during an endpoint intervention requires a context that can be reconstructed without those endpoint values.

This chapter develops the resulting information boundary. Endpoint-removable state gives one constructive mechanism. Strict targetwise support states the measurable object. A spanning-support theorem limits globally invariant adaptive graphs, and a Bayes floor quantifies the prediction error that remains when endpoint evidence is withheld. These results continue the endpoint-exclusion condition introduced in Chapter 4 and separate semantic realizability from the capacity and execution questions of Chapter 12.

13.1 Endpoint-removable context

Definition 13.1 (Endpoint-removable context)**.** For a deleted set $A \subseteq [L]$, a context map $C_{-A} : [q]^L \to \mathcal{S}$ is endpoint-removable if $x_{-A} = x'_{-A}$ implies $C_{-A}(x) = C_{-A}(x')$.

Endpoint-removable context must be constructed before information propagates from the deleted positions; post hoc subtraction leaves their traces in retained states. The following construction establishes realizability for associative token-local recurrences as one concrete design.

Theorem 13.2 (Associative affine deletion)**.** *Let token k define the affine transition $F_k(h) = A_k(x_k)h + b_k(x_k)$ and augmented matrix*

$$M_k(x_k) = \begin{bmatrix} A_k(x_k) & b_k(x_k) \\ 0 & 1 \end{bmatrix}. \tag{13.1}$$

For a deleted set A, replace M_k by the identity when $k \in A$. Then

$$M_{-A} = M_L^{(A)} M_{L-1}^{(A)} \cdots M_1^{(A)} \tag{13.2}$$

is endpoint-removable and preserves the order of every retained transition.

Proof. Every deleted factor is the identity and is independent of the deleted category. All remaining token-local factors occur in their original order, while associativity permits arbitrary parenthesization. □

The theorem applies exactly to the affine recurrence core. Surrounding convolutions, normalizations, and context-dependent parameter generators require their own deletion laws. A segment-product data structure can evaluate an endpoint-deleted context with $|A| = m$ by composing at most $m + 1$ retained intervals.

13.2 Strict targetwise conditional operators

For a target i and selected source set N_i, the shared strict context is the whole-neighborhood endpoint-excluding context

$$\mathcal{C}_i^E = \sigma\{X_k : k \notin \{i\} \cup N_i\}. \tag{13.3}$$

Every reference, route decision, coefficient, and pair table directed into i must be measurable with respect to this same context. Only the final categorical lookup may observe a selected source state x_j.

Definition 13.3 (Strict targetwise support)**.** A directed router is strict for target i if $N_i(x) = N$ implies $N_i(x') = N$ whenever x' differs from x only on $\{i\} \cup N$. For a randomized router the condition is pathwise after fixing a seed independent of the sequence.

Conditioned on c_i^E, define

$$\widehat{G}_{i\leftarrow j}(c_i^E) = P_i(c_i^E) S_{i\leftarrow j}(c_i^E) P_j(c_i^E)^{\mathsf{T}}, \tag{13.4}$$

$$\ell_i(a) = b_i(a; c_i^E) + \sum_{j \in N_i} a_{i\leftarrow j}(c_i^E) \widehat{G}_{i\leftarrow j}(a, x_j; c_i^E). \tag{13.5}$$

Proposition 13.4 (Conditional pairwise operator)**.** *If N_i is strict and every object in Eq. 13.5 is measurable with respect to $\mathcal{C}_i^E$, then the interaction branch is exactly pairwise in the active endpoint variables under the declared product reference. Its only access to a selected source category is the final lookup $\widehat{G}_{i\leftarrow j}(a, x_j; c_i^E)$.*

Proof. Condition on the endpoint-excluding context and the realized support. All tables, references, and coefficients are fixed while (x_i, x_{N_i}) vary. Theorem 3.2 centers each target–source term, and additivity over j excludes components involving two active sources. Theorem 5.1 identifies the branch as the pairwise truncation. □

Projection with an ordinary full-context table remains a useful local gauge. The stricter global functional semantics comes from the shared endpoint-excluding construction above.

13.3 Dynamic support is directed and targetwise

Let a deterministic router $E : [q]^L \to \mathcal{G}_L$ output an undirected graph. Call it globally endpoint-invariant if changing the categories of any active vertices leaves the entire realized graph unchanged.

Theorem 13.5 (Spanning-support no-go)**.** *If a globally endpoint-invariant router has one input for which its output graph has no isolated vertices, then the router is constant on $[q]^L$. Equivalently, every output of a nonconstant globally endpoint-invariant router contains at least one isolated vertex.*

Proof. For a realized graph with no isolated vertices, every sequence position is active. The complement of the active set is empty, so every other input agrees outside the active set. Global endpoint invariance therefore forces the same graph for every input. □

Targetwise routing remains adaptive under the theorem. $N_i(x)$ stays fixed when its own target and realized sources vary, while other residues may still alter routes for targets to which they are unselected. The resulting predictive object is a directed endpoint-excluding operator; an undirected contact graph requires a separate identification step.

13.4 A binary Bayes-error primer

Let $T \in \{0,1\}$ be a desired edge label and let Y contain exactly the information available to a router. The posterior probability of an edge is

$$\eta(y) = \Pr(T = 1 \mid Y = y). \tag{13.6}$$

At a fixed observation y, predicting 1 makes an error with probability $1 - \eta(y)$, while predicting 0 makes an error with probability $\eta(y)$. The optimal binary decision is therefore

$$\widehat{T}^*(y) = \mathbf{1}\{\eta(y) \geq 1/2\}, \qquad \Pr(\widehat{T}^*(Y) \neq T \mid Y = y) = \min\{\eta(y), 1 - \eta(y)\}. \tag{13.7}$$

Averaging over the available observations gives the Bayes error

$$e_Y^* = \mathbb{E}_Y \min\{\eta(Y), 1 - \eta(Y)\}. \tag{13.8}$$

No classifier restricted to Y can have smaller zero–one error. Independent randomization cannot improve the bound because its conditional error is a convex combination of the two deterministic errors.

Two limiting cases fix the intuition. If $\eta(y) = 0.8$, the best prediction is 1 and the unavoidable conditional error is 0.2. If Y carries no information about a balanced label, $\eta(y) = 1/2$ for every y and the Bayes error is $1/2$.

For endpoint-excluding support prediction, the allowed observation is

$$Y_{ij} = X_{[L]\setminus\{i,j\}}. \tag{13.9}$$

A strict router can use any function of Y_{ij}, including a large learned representation, while it cannot recover label information that is absent from the conditional posterior $\Pr(T_{ij} = 1 \mid Y_{ij})$. The Bayes-floor result below applies Eq. 13.8 to this declared information set.

A machine-checked certificate accompanies this primer (`lean/OperatorGraph/Bayes.lean`): the pointwise optimality of the min and the convex-combination bound for randomized rules are formalized in Lean 4, kernel-checked.

13.5 Single-sequence recoverability and optional-MSA supervision

Let $T_{ij} \in \{0,1\}$ be a training-only support label, $Y_{ij} = X_{[L]\setminus\{i,j\}}$, and $Z_{ij}(X) = \mathbf{1}\{j \in N_i(X)\}$ for a strict router. Strictness implies that Z_{ij} is a function of Y_{ij} alone. Indeed, invariance to x_i is immediate. If j is selected, invariance to x_j follows from the definition. If j is not selected but changing x_j made it selected, applying strictness at the modified input while reverting x_j would force the modified support at the original input, a contradiction. Define

$$e_{ij}^{\mathrm{excl}} = \mathbb{E} \min\{\eta_{ij}(Y_{ij}), 1 - \eta_{ij}(Y_{ij})\}, \qquad \eta_{ij}(y) = \Pr(T_{ij} = 1 \mid Y_{ij} = y). \tag{13.10}$$

Proposition 13.6 (Endpoint-exclusion Bayes floor)**.** *Every strict deterministic or independently randomized router satisfies*

$$\Pr(Z_{ij} \neq T_{ij}) \geq e_{ij}^{\mathrm{excl}}. \tag{13.11}$$

Proof. Conditional on $Y_{ij} = y$, endpoint invariance restricts the prediction to a label that observes neither endpoint. The minimum conditional zero-one risk is the smaller posterior mass. Independent randomization has the same or larger linear conditional risk. Taking expectation proves the claim. □

The exact price of endpoint exclusion

The endpoint-exclusion Bayes floor can be maximal even in a two-bit example. Let X_i and X_j be independent uniform binary variables and define a desired edge label

$$T_{ij} = \mathbf{1}[X_i = X_j]. \tag{13.12}$$

An endpoint-visible router predicts the label perfectly: it compares the two bits. A strict endpoint-excluding router observes only

$$Y_{ij} = X_{[L]\setminus\{i,j\}}, \tag{13.13}$$

which is independent of T_{ij} in this construction. Consequently,

$$\Pr(T_{ij} = 1 \mid Y_{ij}) = \frac{1}{2}, \qquad e_{ij}^{\mathrm{excl}} = \frac{1}{2}. \tag{13.14}$$

No training procedure, teacher graph, model scale, or randomized decision rule can reduce the strict inference error below one half without restoring endpoint information or adding a correlated side channel.

The example exposes the semantic tradeoff cleanly. The endpoint-visible router has perfect label accuracy, but its support changes when either endpoint is intervened upon. It is a valid dynamic computational route and an invalid fixed conditional pair selector. The endpoint-excluding router preserves the edge definition during endpoint substitution, but the desired label is not identifiable from its allowed information.

Real proteins lie between these extremes. Neighboring sequence context, predicted secondary structure, templates, MSA-derived family evidence, or other modalities may make an edge partly predictable without direct endpoint access. The endpoint-exclusion posterior then moves away from one half and the Bayes floor falls. That gain should be attributed to the added observable evidence, not to an architecture overcoming an information theorem.

This is why router performance and interaction semantics must be reported together. A useful comparison includes at least three variants:

$$\begin{aligned} q_{ij}^{\mathrm{visible}} &= q(X_i, X_j, Y_{ij}), && \text{adaptive computational support,} \\ q_{ij}^{\mathrm{excl}} &= q(Y_{ij}), && \text{strict endpoint-invariant support,} \\ q_{ij}^{\mathrm{fixed}} &= q(i, j), && \text{static candidate geometry.} \end{aligned} \tag{13.15}$$

The first measures the attainable predictive frontier, the second the price of strict conditional meaning, and the third the value of content-dependent routing itself. No one variant is universally correct; each licenses a different claim.

As an illustrative training design, an optional MSA teacher can estimate family-level edge importance while the inference-time endpoint-exclusion floor remains fixed. Let E_c be a declared candidate set, $q_{ij}^{\theta} \in [0, 1]$ a student support confidence, and $\widehat{G}_{ij}^{\theta}$ its reference-centered block. Teacher and student must use the same detached categorical reference:

$$G_{ij}^{T} = P_i^{[\bar{p}]} T_{ij}^{\mathrm{int}} P_j^{[\bar{p}]\mathsf{T}}, \qquad T_{ij}^{\mathrm{int}} = T_{ij}^{\mathrm{real}} - \mathbb{E}_{\widetilde{X} \sim Q_{\mathrm{null}}} T_{ij}^{\widetilde{X}}. \tag{13.16}$$

A schematic objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_m \sum_i \mathrm{KL}(p_i^{\text{MSA}} \| p_i^\theta) + \lambda_G \sum_{(i,j)\in E_c} \| q_{ij}^\theta \widehat{G}_{ij}^\theta - G_{ij}^T \|^2_{\bar{p}_i, \bar{p}_j} + \lambda_0 \sum_{(i,j)\in E_c} q_{ij}^\theta. \tag{13.17}$$

MSA availability changes the training supervision while the inference contract remains single-sequence. This objective gives one architecture consistent with the framework. Convergence and statistical consistency require separate analysis.

Coda: information constraints survive architecture

Capacity and admissible information impose different limits. A router can predict an edge accurately by reading its endpoints, while strict conditional semantics require its selection to survive changes to those endpoints. The Bayes floor quantifies the resulting inference limit. A teacher can improve training; only additional inference-time evidence or a weaker invariance contract can lower that floor.

The two realizability chapters therefore give separate tests for representability, execution, and measurability. A useful dynamic route can fail the strict conditional test without failing its computational task. The final empirical question is which declared operator corresponds to evolutionary constraint, structure, or a measured assay response. The next chapter specifies experiments that can reject those correspondences individually.

Chapter 14

Falsification and Biological Interpretation

Interpretability claims are often evaluated by accumulation: more correlations, more visual examples, more downstream benchmarks. A typed framework suggests a different standard. Every semantic upgrade should have an observation that could reject it while leaving weaker claims intact.

The first level is mathematical calibration. A gauge projection should satisfy its zero-marginal identities. Reference transport should preserve the underlying equivalence class. A Potts audit should recover the known coupling in the additive limit. A claimed Hopfield realization should show Lyapunov descent under the conditions of the proof. These tests establish whether the named mathematical object has actually been implemented; biological validation begins at a later rung.

The second level compares estimators. Static MSA fields and mean pLM responses should be evaluated as full categorical blocks in a common reference, with their estimand gap and uncertainty exposed. Dynamic residuals should be tested across natural, mutational, and held-out backgrounds. Route reversibility should be separated from response reciprocity. Null ensembles should preserve the nuisance structure that the analysis claims to remove.

The third level is biological and must match the type of the prediction. A scalar position score can be compared with contacts or distances. A typed edge operator calls for a gauge-aligned double-mutant matrix. A background-dependent response calls for the same perturbations across multiple sequence contexts. A higher-order residual calls for combinatorial interventions beyond pairs. Validation becomes stronger when the endpoint retains the structure of the inferred object instead of compressing it prematurely.

Falsification is constructive here. Failure of static agreement can reveal transfer or misspecification. Failure of reciprocity can expose directed computation. Failure of pair truncation can quantify higher-order dependence. A theory earns scope not by surviving every possible test, but by saying precisely what each failure means.

14.1 A four-genotype grammar for experimental interaction

Consider reference residues (a_0, b_0) at sites (i, j), an alternative a at site i, and an alternative b at site j. A linked double-mutant experiment measures all four genotypes in the same molecular construct and assay pipeline:

	$j = b_0$	$j = b$
$i = a_0$	$Y_e(a_0, b_0; c)$	$Y_e(a_0, b; c)$
$i = a$	$Y_e(a, b_0; c)$	$Y_e(a, b; c)$

Here $Y_e(\cdot; c) \in \mathcal{Y}_e$ is the endpoint of assay e under condition and sequence background c. Choose a declared scalar coordinate $g_e : \mathcal{Y}_e \to \mathbb{R}$ and define $F_e = g_e \circ Y_e$. Experimental epistasis in that coordinate is the mixed finite difference

$$\epsilon_{ij}^{(e,g)}(a, b; c) = F_e(a, b; c) - F_e(a, b_0; c) - F_e(a_0, b; c) + F_e(a_0, b_0; c) = \Delta_i \Delta_j F_e. \tag{14.1}$$

The additive null in the F_e coordinate has $\epsilon_{ij}^{(e,g)} = 0$. A complete substitution panel repeats the four-genotype comparison across many choices of a and b, producing a categorical table rather than one scalar pair label.

The coordinate is part of the null. For a positive scalar endpoint Y_e, the identity choice $g_e(y) = y$ tests additive endpoint effects. The logarithmic choice $g_e(y) = \log y$ gives

$$\epsilon_{ij}^{(e,\log)}(a, b; c) = \log \frac{Y_e(a, b; c) Y_e(a_0, b_0; c)}{Y_e(a, b_0; c) Y_e(a_0, b; c)}, \tag{14.2}$$

so zero epistasis tests a multiplicative endpoint model. A nonlinear transformation can change the numerical epistasis and occasionally its sign. The assay, endpoint, units, transformation, reference genotype, condition, and uncertainty must accompany the table. If a competition assay directly identifies the Malthusian rate m_c, choosing $F_e = m_c$ yields Malthusian-growth epistasis with a population-dynamic interpretation. Other assay coordinates retain their own experimental meaning.

The remaining residues form a sequence background C. Repeating the panel in several backgrounds defines $\epsilon_{ij}^{(e,g)}(a, b; C, c)$ and tests whether the interaction is stable or modulated by other sites. Assay conditions supply another background: temperature, binding partner, expression system, time point, and cellular environment can all change the measured relation. Compensation means that the double perturbation restores the chosen endpoint relative to the declared single-mutant null; it does not identify a physical contact by itself.

Replicates, matched controls, and an explicit treatment of missing or low-count variants are needed to separate biological nonadditivity from measurement noise. A scalar structural contact can test the endpoints of a positional prediction. A reference-aligned epistasis table retains the substitution-specific degrees of freedom needed to test a categorical edge operator, with a fitted link function when model scores and experimental measurements use different scales.

A machine-checked certificate accompanies this primer (`lean/OperatorGraph/Epistasis.lean`): the additive null (zero mixed difference), the log-coordinate identity, and the multiplicative null (zero log-epistasis for multiplicative endpoints) are formalized in Lean 4, kernel-checked.

14.2 One residue pair, end to end

Chapter 1 introduced five lines that can be drawn between the same positions (i, j). The following worked audit carries one hypothetical pair through the complete framework. Its deliberately synthetic outcome exposes the logic of interpretation without making an empirical claim about a particular protein.

Assume that one protein family provides a clade-partitioned MSA, a query structure, a pretrained protein language model, and linked double-mutant measurements in two sequence backgrounds. The pair (i, j) is selected before any of these endpoints are compared.

14.2.1 Declare the comparison before fitting it

Training clades define smoothed anchor marginals p_i, p_j. They are then held fixed. A Potts model is fitted on the training MSA, and its block is projected into the anchor fibers. The pLM is audited

on held-out natural and designed backgrounds by changing the source residue at j while measuring the target distribution at i. Every response table is transported to the same fibers before its mean, residual, norm, or singular directions are computed.

This order matters. If each method supplied its own reference, a difference in block norm could come from the reference rather than the relation. If blocks were scalarized first, agreement of strength could conceal disagreement about which amino-acid substitutions compensate.

14.2.2 A hypothetical outcome ledger

Suppose the audit produces four observations. The Potts block and the mean pLM block have similar dominant singular directions. Their scalar norms both rank (i, j) highly, and the query structure places the sites in contact. The pLM nevertheless has a substantial dynamic residual associated with backgrounds that vary a third site k, and its two directed mean responses retain a measurable reciprocity defect. Finally, the double-mutant table agrees with the predicted compensatory direction in one background but changes when k is altered.

Observation	Licensed conclusion	Conclusion still unavailable
Potts and pLM mean blocks align in categorical direction	the pLM contains a reusable family-like response component	equality of training laws or microscopic energies
Both scalar shadows rank the pair highly and the sites contact	the two estimators share a contact-predictive positional signal	equality of the full operators or a direct physical pair force
The pLM residual changes with site k	one static edge is insufficient under the audited backgrounds	one identified three-body physical mechanism
Directed means fail reciprocity	the response is not integrable into one additive pair energy under the tested reference	absence of useful directed computation
Double-mutant shape changes with k	background-dependent biological nonadditivity survives intervention	universal validity outside the tested assay and backgrounds

No single verdict summarizes this ledger. The pair is simultaneously a structural contact, a stable statistical relation to first approximation, a context-dependent model response, and a background-dependent experimental interaction. The failed reciprocity test blocks one stronger claim: these directed responses cannot be compressed into a single additive pair energy without discarding information.

14.2.3 Where the spectral analysis enters

Only after the operator graph has been assembled across many pairs is (i, j) placed in a larger network. A scalar norm may contribute to a contact-weighted Laplacian; a signed contraction may enter a signed Laplacian; the full categorical orientation may enter a connection Laplacian. Heat kernels or wavelets can then ask whether the pair belongs to a localized or domain-scale pattern. Those transforms inherit the categorical resolution retained by the declared compression; any discarded reciprocity defect must be reported separately.

The vertical audit therefore closes the path opened in Chapter 1:

$$\boxed{\begin{array}{l}\text{shared endpoints} \longrightarrow \text{typed observations} \longrightarrow \text{common operator coordinates} \\ \longrightarrow \text{declared compression} \longrightarrow \text{matched structural and mutational evidence.}\end{array}} \tag{14.3}$$

The framework places the five original edges in common coordinates, making their exact level of agreement and the first failed upgrade in interpretation visible in one report.

The worked pair shows how several claims can succeed and fail independently. The following cases turn each proposed upgrade into a rejection test.

14.3 A falsifiable static–dynamic program

The framework is most useful when its objects are forced to fail separately. The following three cases move from a system whose truth is known, through a family comparison with partially known estimands, to biological interventions where the endpoint itself limits the conclusion.

Case I: the additive limit must be boring

Begin with sequences sampled from a known pairwise Potts model under a product or exactly controlled background law. The ground-truth fields, pair blocks, support, and gauge class are available. A correct substitution audit must recover

$$G_{i\leftarrow j}(C) = P_i J_{ij} P_j^{\mathsf{T}} \tag{14.4}$$

for every background C. Dynamic variance must vanish up to numerical and sampling error, and the two directed responses must satisfy reciprocity after transport to the same reference.

This case is intentionally uninteresting biologically. Its purpose is to make implementation errors visible. A nonzero dynamic residual can reveal that endpoint identities leaked into the reference or router, that logits were compared in inconsistent gauges, or that the source intervention changed more than one coordinate. A reciprocity defect can reveal transposed endpoint conventions or incompatible reference transport. A mismatch between tangent norm and exact mutual information at weak coupling can reveal a scaling error.

The same synthetic system can then be perturbed one assumption at a time. Add a pure third-order term while keeping the pair component fixed. The mean pair response should remain the projected second-order component under the product reference, while dynamic variance increases by the exact higher-order mass. This test must assess localization, not merely nonzero variance: rank audited pairs or source–modulator tuples by their recovered order-specific mass and report AUC or enrichment against the known synthetic hyperedges, not against ordinary contact pairs: on a pairwise system, approximation error can scale with the static operator norm and spuriously enrich contacts. If one pair belongs to several planted hyperedges, its truth is the full modulator set; test set recovery or whether the top recovered modulator belongs to that set rather than assigning one arbitrary label. Apply the nontriviality threshold in Eq. 4.23 before ranking ρ^{dyn}. Train the same deep architecture on a column-wise shuffled alignment that preserves one-site marginals while destroying the planted dependence. A large ρ^{dyn} under this null is expected from generic nonlinear context sensitivity; failure to localize the planted hyperedges above the shuffled control rejects the claimed higher-order recovery even when dynamic variance is positive.

Introduce phylogenetic sampling without changing the generative Hamiltonian. The underlying operator stays fixed, but its estimator becomes biased and its uncertainty ceases to follow an independent-row bootstrap. Replace the product background by a correlated one. The mean–residual Pythagorean identity remains exact under the new law, while the decomposition of dynamic variance into order-specific ANOVA masses acquires cross terms. Report both $\rho^{\mathrm{dyn,prod}}$ and $\rho^{\mathrm{dyn,emp}}$ from Eq. 4.24, together with $\Delta\rho^{\mathrm{geom}}$. The product quantity tests the exact ANOVA identity but may probe off-manifold backgrounds; the empirical quantity stays closer to the sampled manifold

but has no orthogonal order decomposition. Also repeat the additive calibration under multiple full-support endpoint references: positive magnitudes may change, whereas the zero-dynamic verdict must obey Eq. 3.30.

Each modification targets a different coordinate. Higher-order structure changes the estimand, phylogeny changes the sampling law, and background correlation changes the geometry of the decomposition. Distinguishing them requires operator and sampling diagnostics alongside contact precision.

Case II: family field and pLM response may agree for the wrong reason

Choose a protein family with sufficient effective depth to fit a range of regularized Potts models. Partition the MSA by clade rather than by random rows. Use the training clades to define an anchor reference and fit the family field; use held-out natural or designed sequences as backgrounds for the pLM substitution audit.

The first comparison is operator-valued:

$$\widehat{G}^{\mathrm{P}}_{i\leftarrow j} \quad \text{versus} \quad \widehat{\widehat{G}}^{\mathrm{pLM}}_{i\leftarrow j}\,. \tag{14.5}$$

Report alignment of full blocks, singular directions, reciprocity, and the pLM dynamic fraction. Only then compress to contact scores. A high norm correlation with poor singular-direction agreement means the models locate strong pairs but disagree about substitution chemistry. Good mean-block agreement with large dynamic variance means the pLM contains a reusable family-like component plus substantial context dependence. Poor mean agreement with useful contact ranking means the scalar benchmark has hidden an operator-level discrepancy.

The second comparison separates routes from responses. Measure attention or another internal route using its own stationary geometry, detailed-balance defect, and directed support. Then replace the route by its stationary reversibilization or a matched doubly stochastic control while holding value and output maps as fixed as the architecture permits. If masked-token loss and the categorical response change, the model functionally uses the removed circulation. If contact scores change while output responses remain stable, the intervention localizes the fragility to the attention interpretation rather than the model behavior.

The result can fail in several informative ways. Potts–pLM disagreement may reflect finite MSA sampling, pairwise misspecification, transfer failure, or response sampling. Off-manifold substitutions may dominate the pLM variance. MSA removal may alter confidence without altering the decoded edge operator. The estimand ledger determines which additional control is relevant; one aggregate correlation does not.

Case III: biological evidence should retain the predicted type

Suppose the model predicts a strong static operator on (i, j), a particular compensatory substitution direction, and a dynamic modulation by a third position k. Three increasingly rich experiments test different parts of that statement.

A structure supplies a distance or contact endpoint. It can test whether the position pair is geometrically associated, but not whether the predicted amino-acid direction is correct. A linked double-mutant panel supplies an epistasis table. After its own one-body terms are removed and its reference is aligned, it can test the categorical shape of the static operator. Repeating that panel in multiple backgrounds that vary k can test the predicted dynamic component.

The experiments need not agree. A spatially distant pair can carry a reproducible functional or compensatory operator. A close contact can show weak epistasis under one phenotype. A static

mean can agree with double-mutant data while the predicted background modulation fails. Together these outcomes identify which edge semantics survive the intervention, replacing an abstract choice of the one "correct" graph with a typed empirical comparison.

Scale and units require their own care. Model logits, evolutionary energies, binding free energies, assay coordinates $g_e(Y_e)$, and Malthusian growth rates occupy different numerical scales. A link function between an inferred operator and an experimental table must be fitted on one subset and evaluated on held-out mutations. The words "energy" and "fitness" acquire technical meaning only after the relevant physical, statistical, or population-dynamic construction has been specified.

What a clean failure looks like

A falsifiable theory permits each layer to succeed or fail independently. The outcomes can be read locally:

- failure in the additive synthetic limit rejects the implementation;
- recovery in the additive limit but failure under controlled third-order terms rejects the claimed higher-order audit;
- nonzero dynamic variance without planted-hyperedge localization above a dependence-destroying null demonstrates generic nonlinearity, not recovered higher-order structure;
- stable estimation with Potts–pLM disagreement rejects transfer, not the common coordinate;
- contact agreement with operator disagreement supports a scalar positional signal, not typed mechanism recovery;
- operator agreement in one background with failed replication rejects background invariance;
- route ablation without response change weakens the proposed internal mechanism while leaving the output association intact.

This locality is the practical value of the claim ladder. Evidence can stop at the strongest rung it supports without forcing the weaker results to disappear.

14.4 Three statistical-mechanics claims and their failure conditions

The three solvable lenses become useful only when their heavier vocabulary can fail. The following ledger records the minimum object, the relevant observation, and a clean rejection condition.

Claim	Required mathematical object	Evidence to report	Clean failure
Glassy or frozen	joint energy over complete sequences, temperature or control parameter, overlap observable	finite-size free energy, entropy, overlap distribution, metastability, shuffled-energy null	transition or overlap structure disappears under size scaling or is matched by the null
Cavity inference	declared factor graph and a specified deleted factor or variable	exact-tree calibration, loop residual, marginal and susceptibility error	message equations fail on trees or loop error is treated as exact without calibration
Chaotic response	dynamical state, perturbation norm, time axis, and trajectory law	finite-time singular values, Lyapunov estimate, width and background uncertainty	positive growth is transient, norm-specific, or absent under longer horizons and matched perturbations

The corresponding protein audits should retain the book's typed coordinates. A random-energy comparison tests whether low-order operator mass exceeds a landscape with no reusable positional structure. A cavity comparison tests a message approximation to a declared sequence law; endpoint exclusion separately tests invariance of support construction. A dynamical-response comparison tests propagation through layers or recurrent time, then asks whether that internal transport predicts the externally audited categorical operator.

Each label combines an object, an ensemble, a limiting or replication protocol, and an observation that a simpler null fails to reproduce. Reporting these four fields turns statistical-mechanics language into a falsifiable extension of the operator audit.

14.5 Biological validation is typed

The scalar graph and the full operator make different predictions. If $F_{e,ij}(a, b; c) = g_e(Y_e(a, b; c))$ denotes the transformed assay endpoint introduced in Eq. 14.1, the resulting table retains a separate measured contrast for each substitution pair. The operator-level comparison is between $G_{ij}(a, b)$ and the corresponding projected epistasis table in a common reference, rather than only between M_{ij} and a binary contact label [41, 74]. Model logits, evolutionary energies, transformed assay coordinates, and Malthusian growth rates generally occupy different units and may be related nonlinearly. Their comparison therefore uses a scale or link function fitted on a training subset and evaluated on held-out mutations, rather than direct numerical equality. A population-selection claim additionally requires a declared genotype-to-demography map.

Inferred quantity	Primary hypothesis	Independent endpoint
Declared symmetrization of $M^{\mathrm{static}}_{i\leftarrow j}$	persistent positional association or contact candidate	held-out structure and distance-stratified contact precision
Full $G_{i\leftarrow j}(a, b)$	substitution-specific compensation	linked, gauge-aligned double-mutant matrix
$M^{\mathrm{dynamic}}_{i\leftarrow j}$ or $\rho^{\mathrm{dyn}}_{i\leftarrow j}$	background-dependent response	the same mutations across multiple sequence backgrounds
P^{circ} or $\delta_{\mathrm{db}}(P)$	nonreciprocal computational routing	masked-loss and response change under stationary reversibilization
Low scalar or connection eigenspace	coherent positional domain or transported substitution sector	held-out domain boundaries and multi-position mutational coherence
Higher-order residual	non-pairwise epistasis	designed triple-mutant or combinatorial assay

Evolutionary constraints can arise from folding, stability, binding, catalysis, regulation, and phylogenetic sampling. A structurally distal operator may therefore encode reproducible function or compensation. Contact prediction supports positional association; causal epistasis requires matched perturbational evidence [95, 94]. The validation endpoint must match the mathematical object.

Coda: evidence should preserve type

Validation is strongest when the endpoint matches the inferred object's degrees of freedom. Binary contact labels are appropriate for a scalar support hypothesis. Substitution-specific action calls for a linked mutational table. One double-mutant panel tests an operator near a reference sequence;

replication across backgrounds tests its invariance. A downstream prediction gain establishes utility, while targeted intervention addresses the internal mechanism that produced it.

This discipline keeps several legitimate results from competing for one label. A model may learn useful evolutionary statistics, reproduce a family interaction field, exploit directed routing, and predict experimental epistasis to different degrees. Reporting those degrees separately is more informative than deciding whether the model has or has not "learned interactions."

The framework is now complete enough to return to the question that opened the book. When five descriptions draw a line between the same two residues, what kind of object, if any, do they share? They share a structured relation whose precise meaning is given by the coordinates and evidence that survive the translation.

Chapter 15

What Kind of Object Is a Protein Interaction?

The contact, Potts block, attention route, substitution response, and experimental epistasis of a residue pair have now been constructed in their respective spaces. The reference-centered operator graph makes Potts blocks and model responses comparable through a common categorical action. Context dependence, scalar compression, probability compatibility, and biological validation determine which further comparisons are available.

The remaining questions concern how to extend those comparisons beyond the settings proved or worked through here.

15.1 The open mathematical problems

The framework leaves several problems deliberately unresolved.

From factorized edges to graph ensembles. Independent edge-state distributions provide local Gibbs coordinates but omit global topology. A coupled theory must determine which degree, reciprocity, loop, geometry, and chemical constraints belong in a graph Hamiltonian, and whether the resulting ensemble remains learnable at protein scale.

From directed responses to integrable dynamics. Reciprocity characterizes additive pair compatibility, but deep models generate context-dependent operators and higher-order residuals. The appropriate integrability conditions for these fields, and the meaning of their nonintegrable circulation, remain open.

From pair operators to higher-order bundles. Functional ANOVA identifies the mass beyond pairs, but an economical representation of typed hyperedges must balance support, categorical rank, gauge, and realizable computation. A pair graph records the pairwise projection of higher-order behavior, while typed hyperedges retain the remaining information.

From family inference to transferable mechanism. Agreement between a family Potts field and a pLM response mean would show transfer in common coordinates. Whether such agreement obeys a cross-family scaling law in MSA depth, alphabet, degree, phylogenetic distortion, or model scale is an empirical question with the shape of a statistical-physics problem.

From predictive operators to biological intervention. The most decisive evidence will preserve categorical type across multiple sequence backgrounds. Large, linked double- and higher-order mutant panels remain rare. Experimental design and operator estimation must therefore be developed together; reducing assay data to a final scalar benchmark would discard the structure the theory aims to test.

15.2 The edge revisited

Return to the pair (i, j) from Chapter 1. The structural contact, Potts block, attention route, substitution response, and experimental epistasis may eventually converge on one coherent account. Such coherence comes from a sequence of explicit correspondences: common categorical coordinates, controlled contexts, compatible directions, matched nulls, and interventions that retain the predicted structure.

The operator graph treats unity among the five edges as a hypothesis of correspondence. It makes agreement precise enough to be discovered and disagreement precise enough to be informative.

Appendix A

Population Selection as an Optional Bridge

A.1 Population selection as an optional bridge

This appendix records one explicit population-genetic construction for the boundary introduced in Chapter 2.9. It is not needed for the operator or MSA arguments. Its role is to show exactly what must be added before an assay endpoint can receive a population-dynamic interpretation.

On the finite sequence space $\mathcal{X} = [q]^L$, let $n_x(t)$ be the expected abundance of genotype x. Let Q_{mut} be an irreducible mutation generator in the column convention: $Q_{xy} \geq 0$ is the rate from y to x for $x \neq y$, and $\sum_x Q_{xy} = 0$. For an infinite asexual population in a fixed environment, with frequency-independent Malthusian rates $m_c(x)$ and no recombination,

$$\frac{dn}{dt} = \mathcal{L}_c n, \qquad \mathcal{L}_c = Q_{\mathrm{mut}} + \mathrm{diag}(m_c). \tag{A.1}$$

After normalization by total abundance, $p = n/(\mathbf{1}^{\mathsf{T}} n)$ obeys the replicator–mutator equation

$$\frac{dp}{dt} = Q_{\mathrm{mut}} p + [\mathrm{diag}(m_c) - \overline{m}_c(p) I]\, p, \qquad \overline{m}_c(p) = \sum_x m_c(x) p_x. \tag{A.2}$$

The irreducible Metzler operator $\mathcal{L}_c$ has a positive principal right eigenvector,

$$\mathcal{L}_c v_c = \lambda_c v_c, \qquad p_c^\star = \frac{v_c}{\mathbf{1}^{\mathsf{T}} v_c}, \qquad \overline{m}_c(p_c^\star) = \lambda_c. \tag{A.3}$$

This equilibrium is an eigenproblem coupling mutation and growth. It follows from the declared population model, not from exponentiating an arbitrary assay value.

A different, reversible stationary construction gives a Gibbs-like law. Suppose a coarse-grained substitution chain has rates $K_c(x \mid y)$ from y to x, the neutral chain has full-support stationary law π_{mut}, and a dimensionless selection potential Ψ_c is derived from a fixation or diffusion model such that

$$\frac{K_c(x \mid y)}{K_c(y \mid x)} = \frac{\pi_{\mathrm{mut}}(x)}{\pi_{\mathrm{mut}}(y)} \exp[\Psi_c(x) - \Psi_c(y)]\,. \tag{A.4}$$

Detailed balance then gives

$$\pi_c(x) = \frac{\pi_{\mathrm{mut}}(x) e^{\Psi_c(x)}}{\sum_{y \in \mathcal{X}} \pi_{\mathrm{mut}}(y) e^{\Psi_c(y)}}, \qquad H_c(x) = -\log \pi_{\mathrm{mut}}(x) - \Psi_c(x) + \log Z_c. \tag{A.5}$$

The potential Ψ_c is a coordinate of this specified construction. It can be identified with Y_e, m_c, or a model score only after a separate derivation and dimensional calibration.

Finite-population drift, recombination, changing environments, frequency dependence, and genealogy alter the state dynamics or the observation law. They belong to the population model and should not be silently absorbed into one scalar called fitness.

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Glossary

This glossary collects the words that travel most easily between statistical physics, graphical modeling, neural computation, and experimental biology. Each entry lists the meanings the word carries in those literatures and states what this book requires before the word is used in a claim. The entries are grouped by the space in which the word is most often anchored; within each group they are ordered alphabetically. The notation chapter fixes symbols; this chapter fixes usage.

Probability and statistical physics

Term	Meanings across literatures	Required declaration
cavity	a site removed from a locally tree-like factor graph; the surrounding fields passed to it; by extension any message-passing approximation to a marginal	the factorization being assumed and the graph to which the cavity belongs (Chapter 6)
coupling	a pair term of a joint Hamiltonian; a conditional-dependence parameter; a message coefficient; a measured epistatic effect	the sample space of the law in which the term appears and its gauge
energy	minus the logarithm of an unnormalized weight; a learned scalar score; a physical potential with units and dynamics	which normalization converts the score into an energy coordinate
field	a one-body term of a Hamiltonian; an external input; a feature map; a gauge degree of freedom	whether the object is a physical field or a reparametrization artifact
Hamiltonian	the generator of a joint Gibbs law; the drift of a stochastic dynamics; a learned energy function	the law or dynamics it generates and its intended domain
joint law	the normalized measure on complete configurations; sometimes claimed from a collection of conditionals or marginals	whether a compatibility condition has actually been checked
marginal	a projection of a joint law onto sites or blocks; an empirical frequency estimate of one	population versus sample, and the estimator that produced it
message	a cavity summary in belief propagation; a feature vector in a neural aggregation; an edge payload in a graph network	the state space it lives in and the operation that consumes it
partition function	the normalizer of a Gibbs law; a sum over exponentially many states; an intractable quantity approximated by variational or cavity methods	which sample space is being summed over
temperature	a physical scale; a softmax parameter; a noise level in sampling dynamics	whether the parameter is a physical temperature or a local normalization constant (Chapter 2)

Computation and architecture

Term	Meanings across literatures	Required declaration
attention	a source-normalized routing coefficient inside a computation; a claimed relational structure; a soft dictionary lookup	the normalization axis and the fact that a route is not yet an interaction (Chapter 10)
Laplacian	the divergence of the gradient on a declared graph; a normalized operator on a position graph; a discretized diffusion generator	which graph, which weights, and which vertex state it acts on (Chapter 11)
memory	a stored pattern in an attractor network; an addressable key–value table; a pair representation updated across layers	the state being stored, retrieved, and confused with others
mode	an eigenvector over positions; a categorical constant direction; a phylogenetic principal component; an attractor	the space in which the mode is a mode (Chapter 11)
normalization	division by a partition function; by a source-position sum; by a memory sum; by a marginal	the axis of the simplex on which the operation projects
route	a one-step computational path; a chosen edge in an adaptive graph; a sparse execution decision	whether the route is a declared part of the model or an observed trajectory
softmax	the map from scores to a probability simplex	which sample space the resulting law lives on (Chapter 2)
spectrum	eigenvalues of a declared operator on a declared state space; sometimes reported without either declaration	the operator and the null model against which it is read
state	a configuration of a physical system; a hidden vector of a network; a relational buffer updated across layers	the state space and its update law
structural role	a fixed functional decomposition of a parent token, such as backbone or side-chain; sometimes read as a reusable chemical fragment	the parent map, the role set, and the fact that a role is not a fragment with transferable identity (Chapter 10)
transport	moving an operator between reference geometries; moving mass in optimal transport; propagating a state through layers or time	the origin, the target, and the Jacobian of the movement

Inference and evidence

Term	Meanings across literatures	Required declaration
background	the surrounding sequence on which a response is conditioned; a null model; a measurement noise floor	the conditioning set and the reference with respect to which residuals are taken (Chapter 4)
commitment	the degree to which a readout of an intermediate generation state predicts the same readout of the final state; by extension, the progress at which a property stops changing under the remaining sampling randomness	the rollout protocol, the shared randomness, and the readout path (Chapter 12)
contact	a geometric finding at a threshold; a predicted edge; a validated interaction	the criterion and the evidence level (Chapter 14)
direct	a one-step route; a term surviving a conditional-independence test; a physical proximity; an effect remaining after a lower-order null	which alternative explanations were excluded (Chapter 2)
endpoint	the measured quantity of an assay; the validation target of a model; the output coordinate of a readout	the physical meaning attached to the number
epistasis	a nonadditive intervention effect relative to a declared scale and null	the measurement scale, the background genotype, and the assay conditions (Chapter 14)
estimand	the population quantity an estimator targets, before any data are seen	the full pipeline from observations to the target (Chapter 9)
falsification	a failed prediction; a rejected identification claim; an audit outcome with a declared alternative	the claim, the test, and the alternative that was preferred
gauge	a reparametrization freedom that leaves a law invariant; a fixing convention; a centering projection	the chosen reference and the transported coordinates (Chapter 3)
realizability	whether an architecture can represent a function class; whether a conditional structure is compatible with any joint law; whether a sparse program computes the same map	which capacity notion is meant (Chapters 12 and 13)
reciprocity	equality of forward and backward responses; symmetry of a pair energy; a compatibility condition	the reference transport under which the two directions are compared
shadow	a scalar compression of a typed operator, such as a contact score or a Frobenius norm	which information the compression provably discards (Chapter 5)

Two habits make this glossary redundant. Before using any term above, name the space in which the object lives and the alternative explanations that have been excluded; after using it, keep the carried object visible until the validation endpoint has been declared. The book's chapters repeat those two requirements in progressively stronger forms.